NxM Templates for SuperCDMS SNOLAB Run 4 using Ge Activation Data

K-line event selection · 1x1 and NxM (PCA) templates

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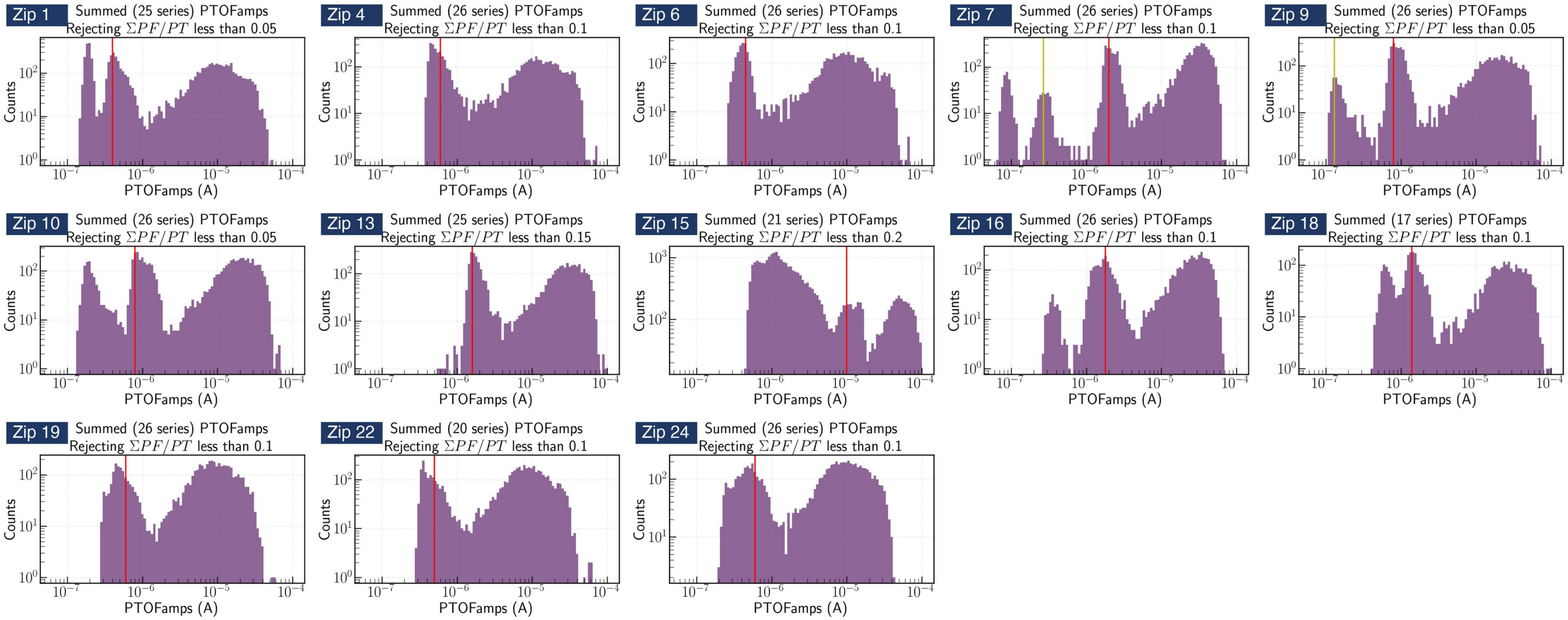
Table of Contents

Ge activation dataset

Credit: Prof. Tarek Saab

- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in **PTOFamps** (the red line in each panel below)
- A **PTOFamps** window bracketing the red line: position ×/÷ 1.35, a rough eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata, saved as per-series pickle files (reusable for any later analysis)
- Events in the window per zip: Z1 10.1k Z4 26.1k Z6 17.0k Z7 2.2k Z9 2.6k Z10 7.6k Z13 9.1k Z15 4.2k Z16 1.6k Z18 22.1k Z19 23.3k Z22 25.0k Z24 22.9k (total 174k)
- **Focus: Z7, the best detector; examples use its PBS1 channel; other detectors in the backup slides**

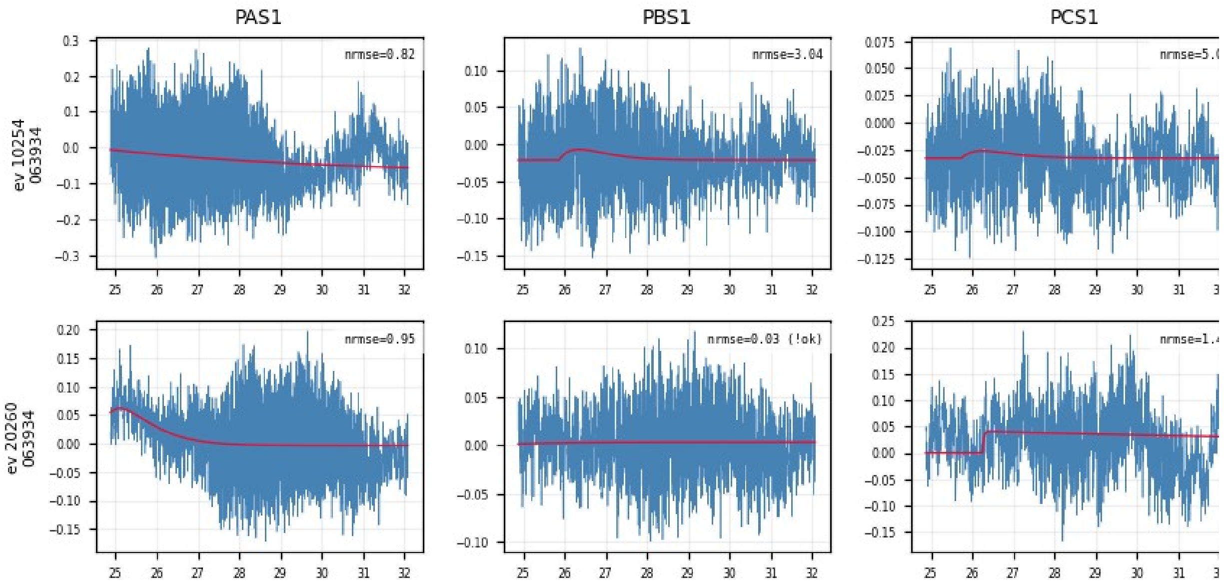


All 13 detectors, PTOFamps spectra (Prof. Saab's study)

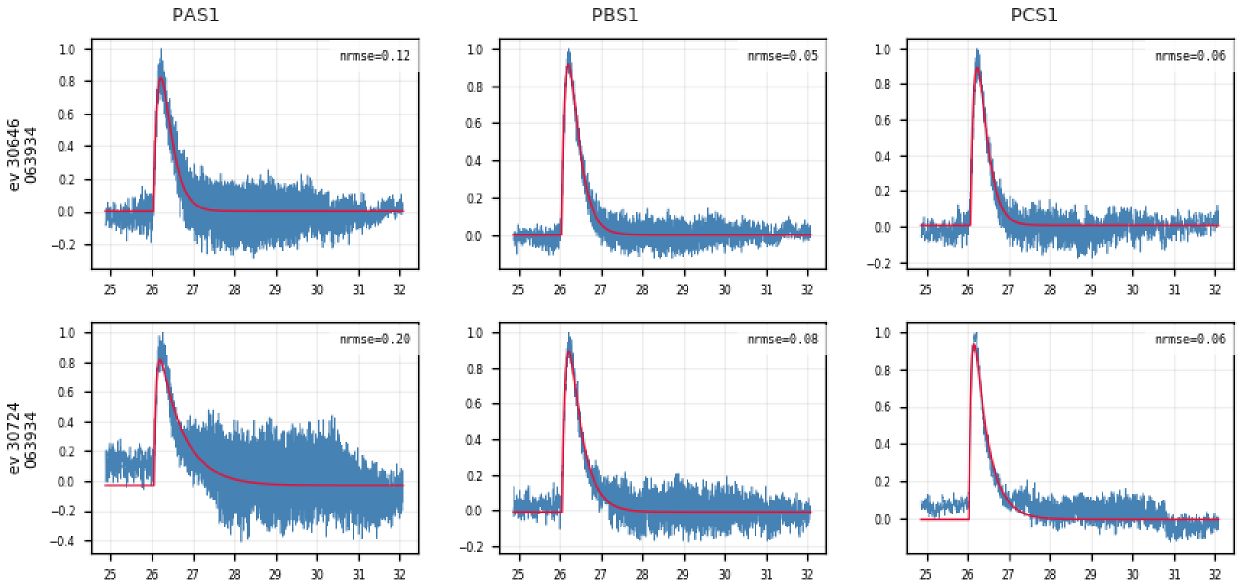
Fit examples

- Real events and noise triggers can be told apart by visual comparison (no cut applied at this stage)

NOISE



REAL PULSE



Per-trace algorithm: low-pass → fit → align

1. **Low-pass**: 100 kHz Butterworth
2. **Normalize**: subtract the baseline (median of the pre-pulse samples), scale the trace to unit peak
3. **Fit**: two-exponential model:

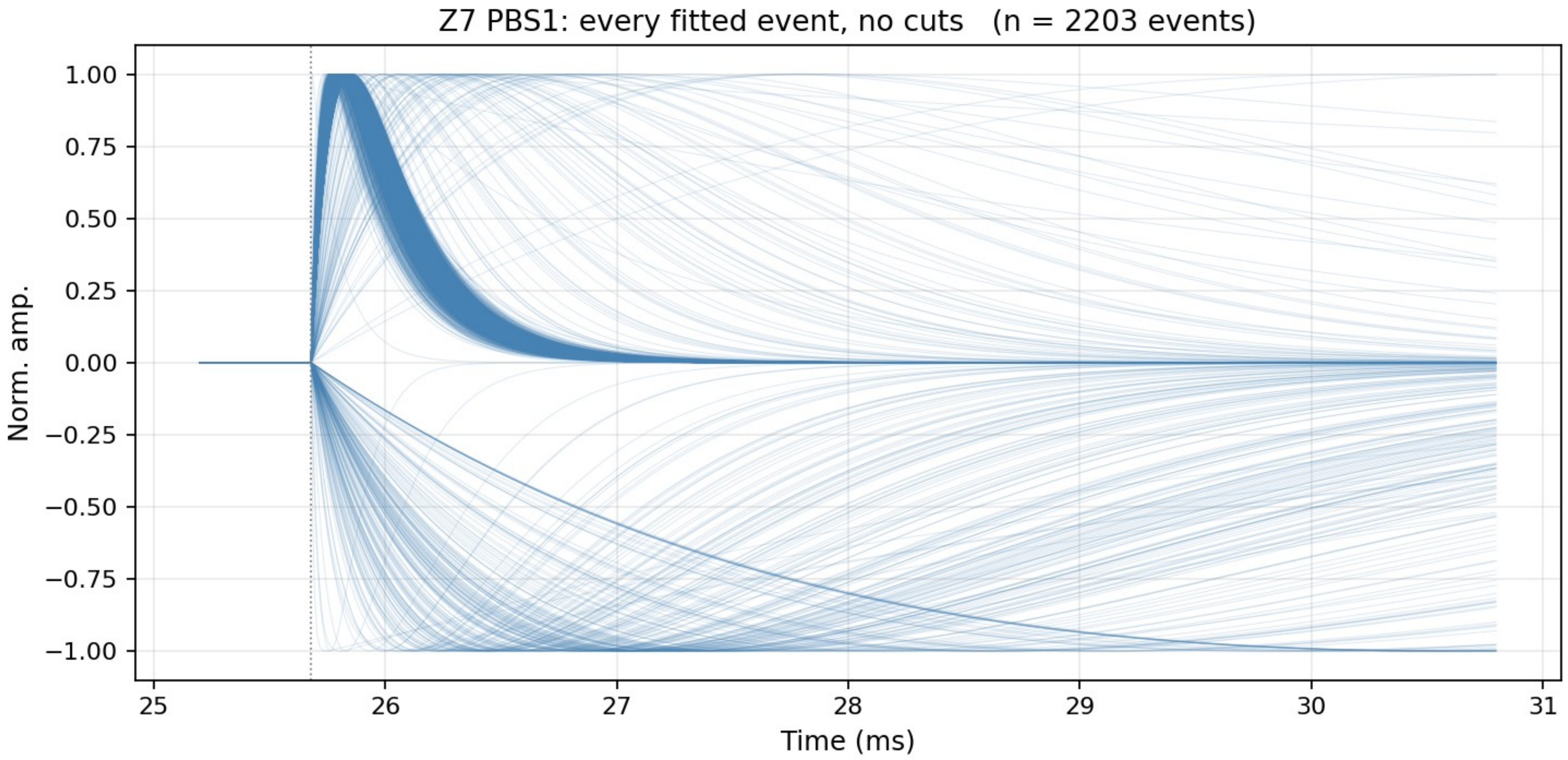
$$y(t) = A\left[\, e^{-(t-t_0)/\tau_{\mathrm{fall}}} - e^{-(t-t_0)/\tau_{\mathrm{rise}}} \,\right] + b$$

$t_0 = $ pretrigger, free within 16050 ± 3000 samples (τ_{rise}, τ_{fall}: rise / fall time)

Fit parameters: A amplitude · τ_rise rise time constant · τ_fall fall time constant · t_0 pulse onset (pretrigger) · b baseline

4. **Two quality checks per trace** (calculated and recorded here, not cut yet):
   - **fit_ok**: amplitude is positive, and the pulse rises faster than it falls
   - **NRMSE**: RMS of the fit residual ÷ pulse peak (smaller = better fit)

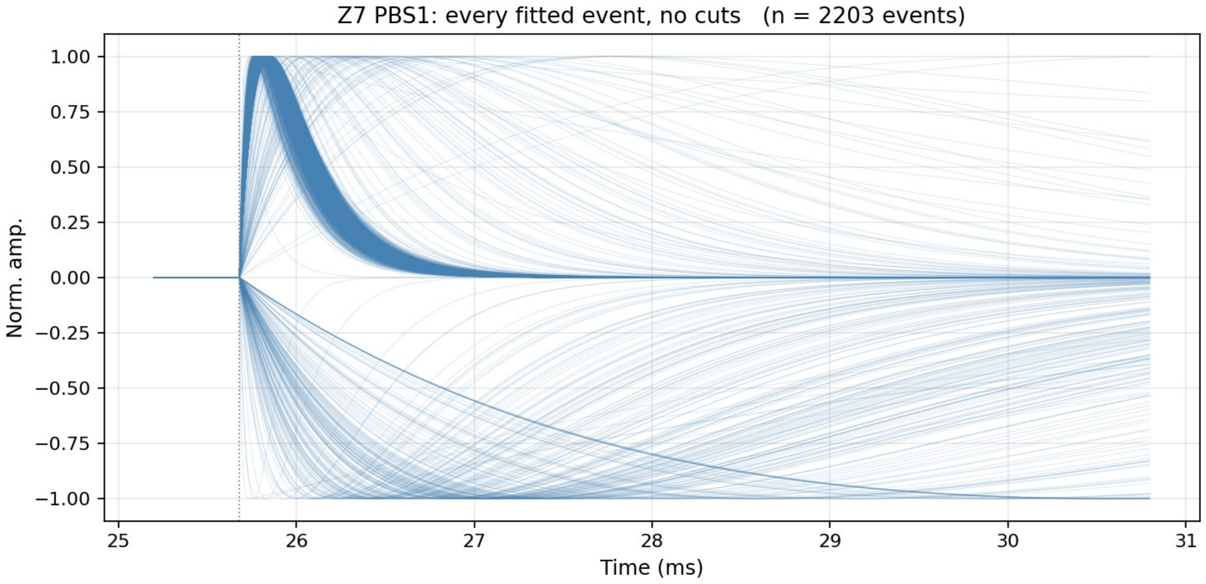
5. Align



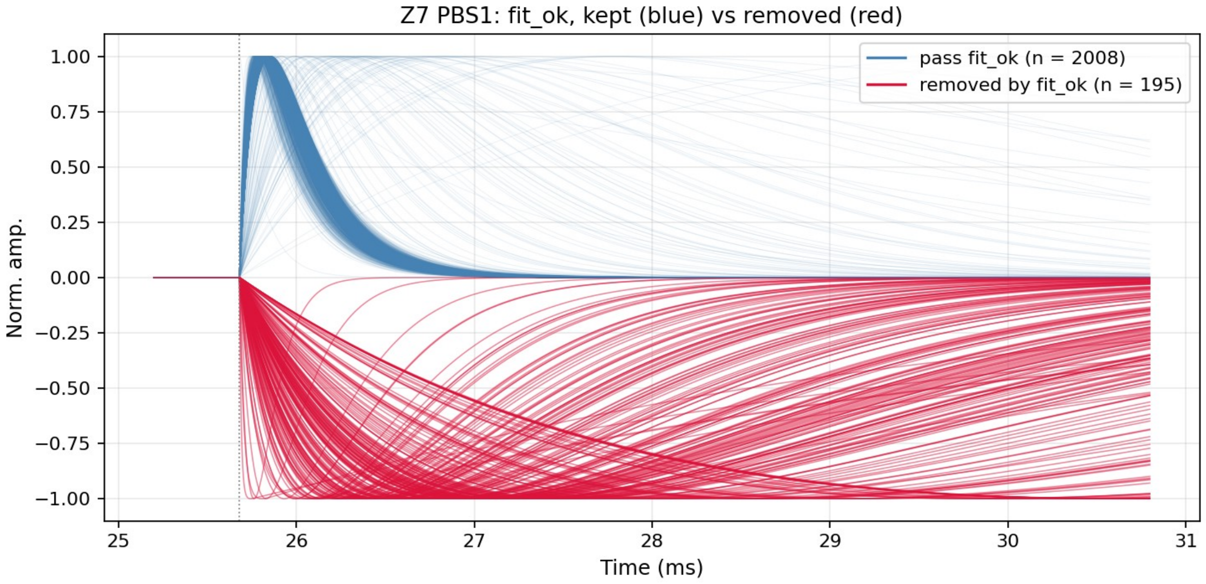
Z7 PBS1: every fitted event drawn at the common pretrigger, no cuts of any kind (n = 2203, inverted fits included)

Quality cut 1: fit_ok

- fit_ok: amplitude > 0 AND rise faster than fall (the fit is a physical pulse)
- Z7 PBS1: 2008 of 2206 events pass; 195 removed (red, inverted or negative shapes), 3 fits did not converge



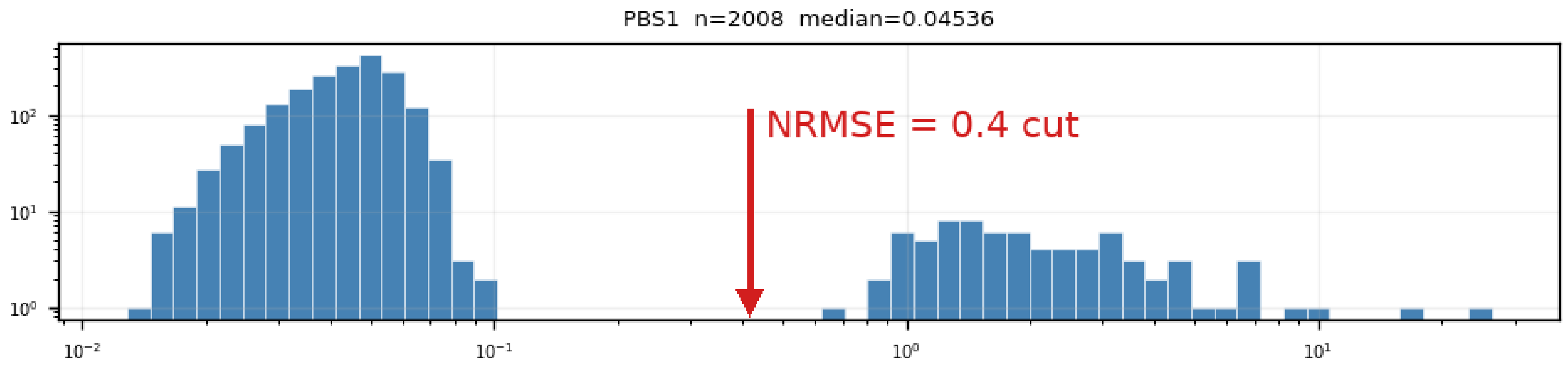
All fitted events, no cuts (n = 2203 drawn, inverted fits included)



fit_ok: kept (blue) vs removed (red, drawn at |peak| normalization)

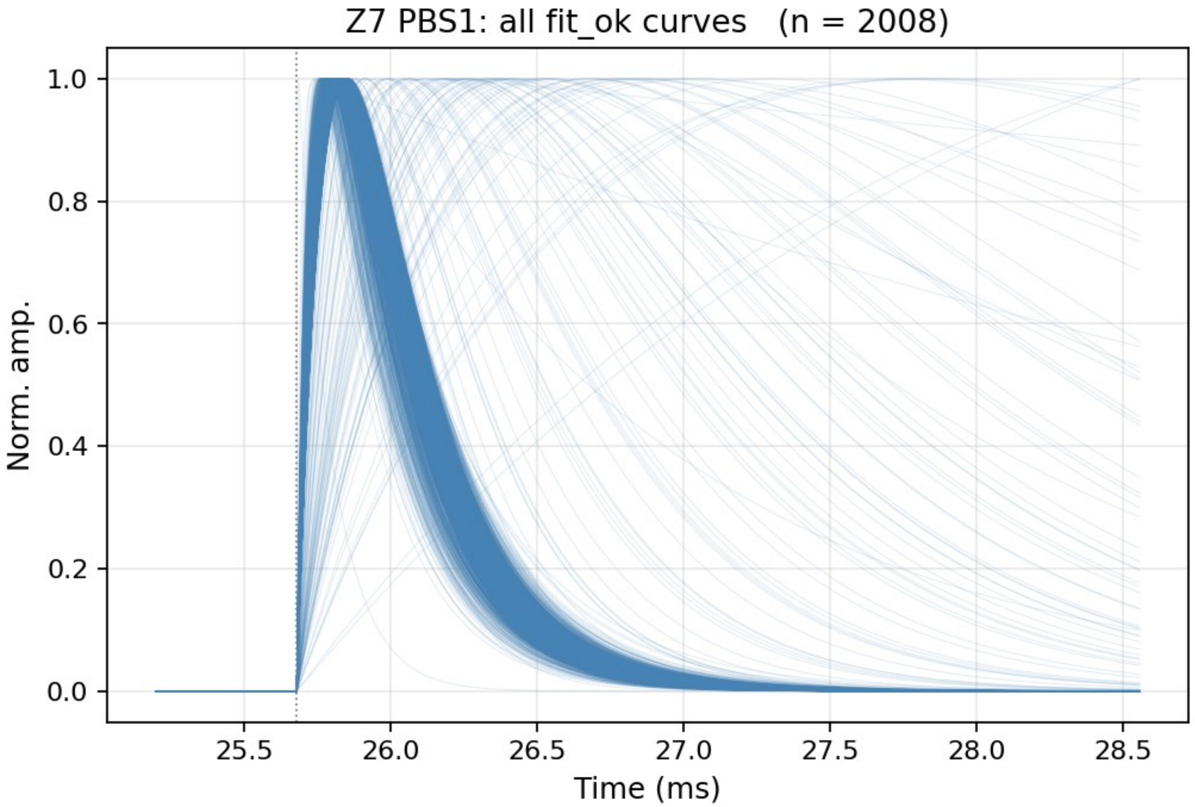
Quality cut 2: NRMSE ≤ 0.4 — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05–0.1) vs noise triggers (≈ 1–2), valley at ≈ 0.4–0.5
- The cut sits in the valley: eyeball read off the distribution
- Worse detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates

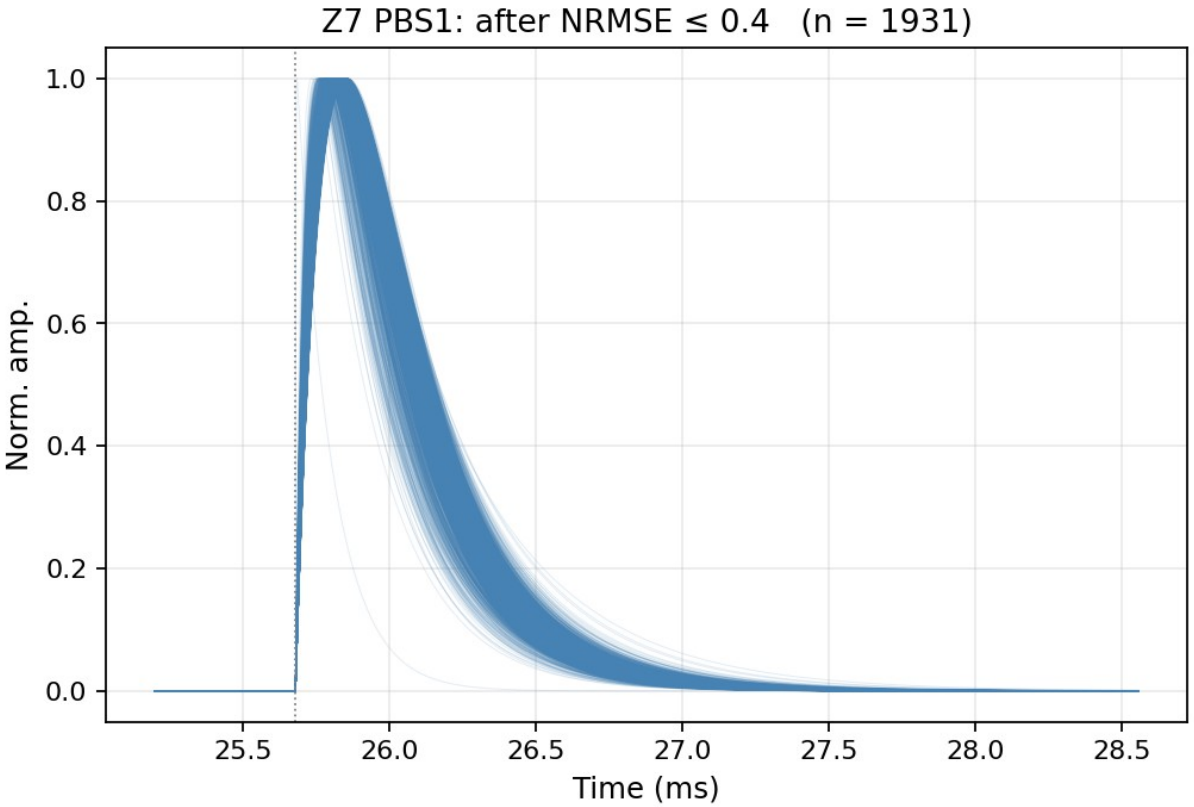


Z7 PBS1: two clean populations, log-log axes

Aligned fitted curves: before and after the NRMSE cut

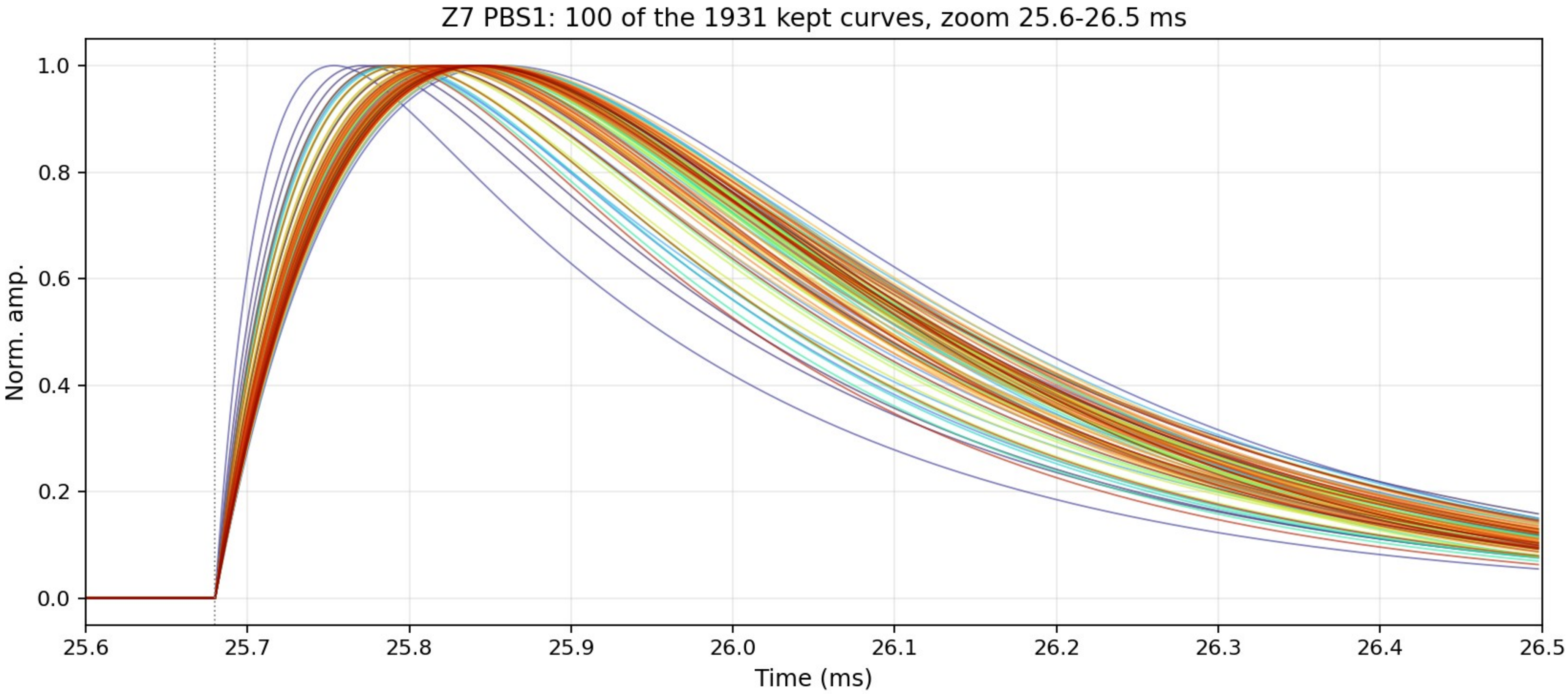


All fit_ok fitted curves (Z7 PBS1, n = 2008, every curve drawn)



After NRMSE ≤ 0.4 (n = 1931): one tight shape family

After NRMSE ≤ 0.4: a closer look

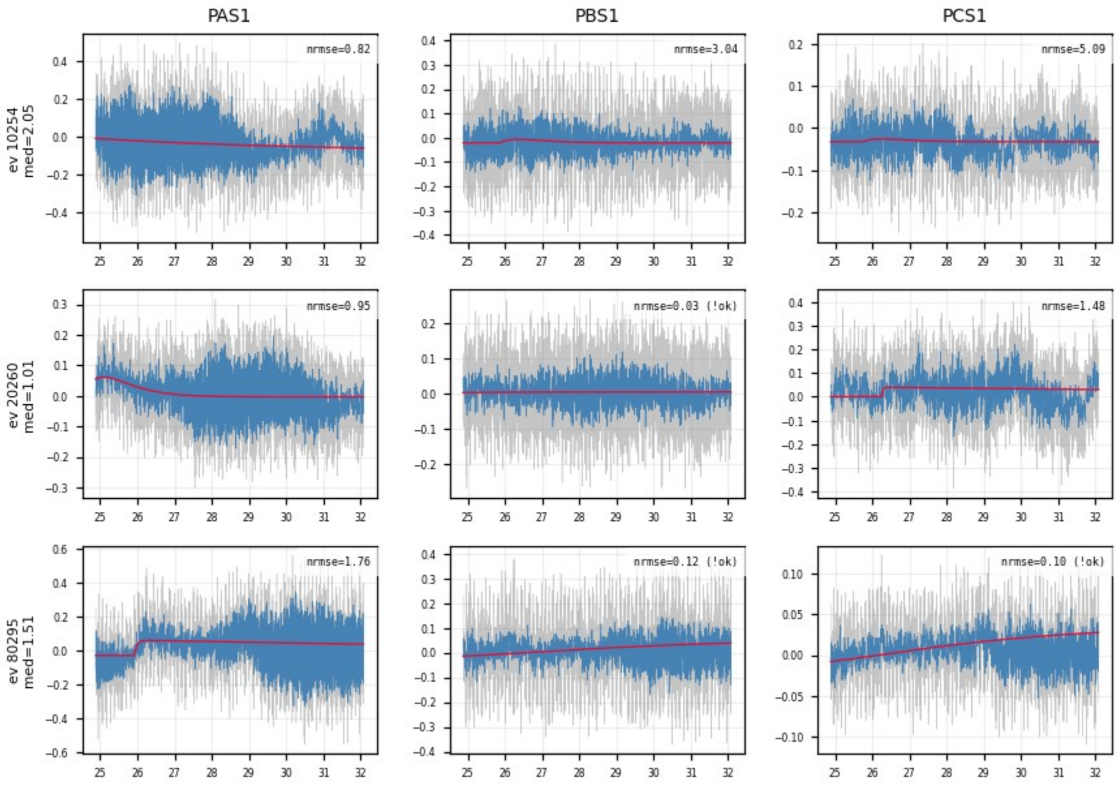


Z7 PBS1: 100 randomly drawn curves of the 1931 kept, zoomed to 25.6–26.5 ms

The rejected population is noise — verified in the raw traces

- Example of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse
- the cut removes noise triggers, not physics

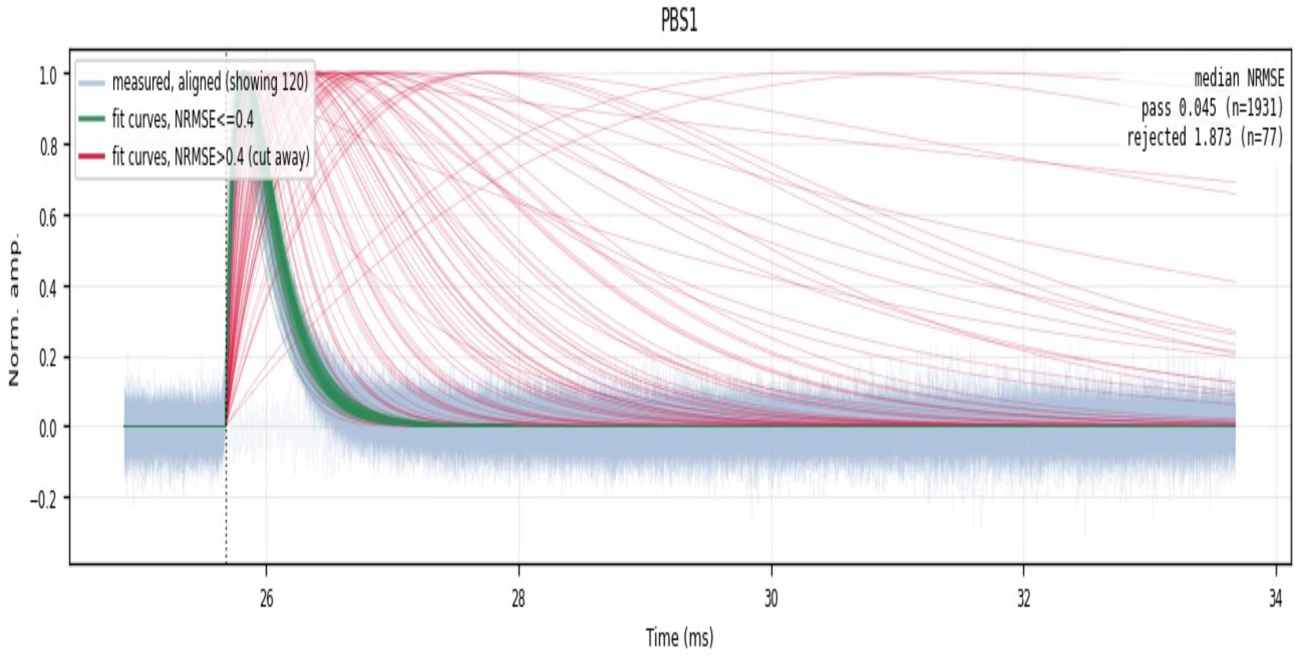
Rejected events: raw vs fit



Z7: NRMSE-rejected events — raw traces are noise

Aligned curves: kept vs rejected

median NRMSE: kept 0.045 (n = 1931) · rejected 1.873 (n = 77)

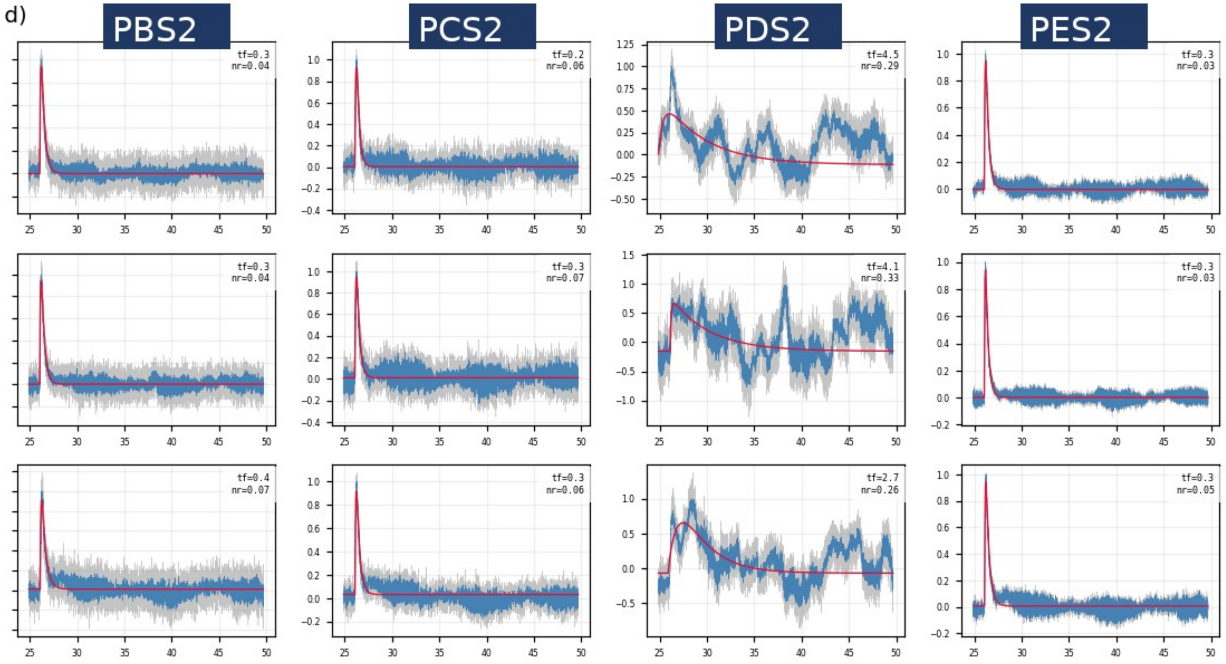


Z7 PBS1 —— passes the cut (kept) —— cut away = rejected

faint blue = measured traces 1931 kept / 77 cut (~3.8%)

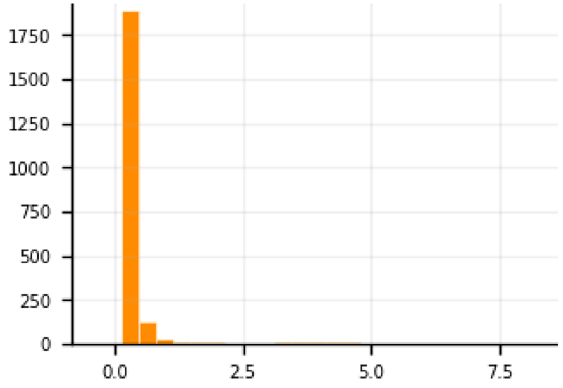
Slow-fall tail in one particular channel

- PDS2 alone: can not make useful templates; hard to fix in analysis/software, the noise itself needs to be fixed
- Temporary solution: use the PDS1 template in place of PDS2 (applied in the ROOT files)

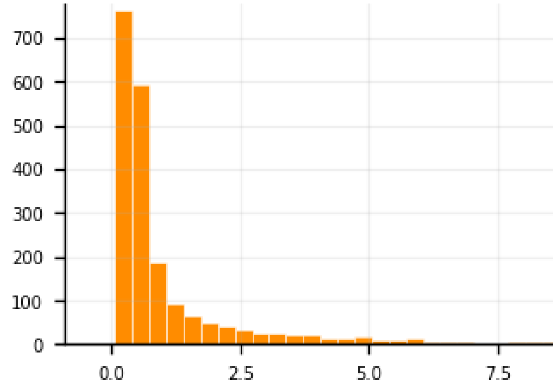


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_fall distribution, same 0–7 ms axis:



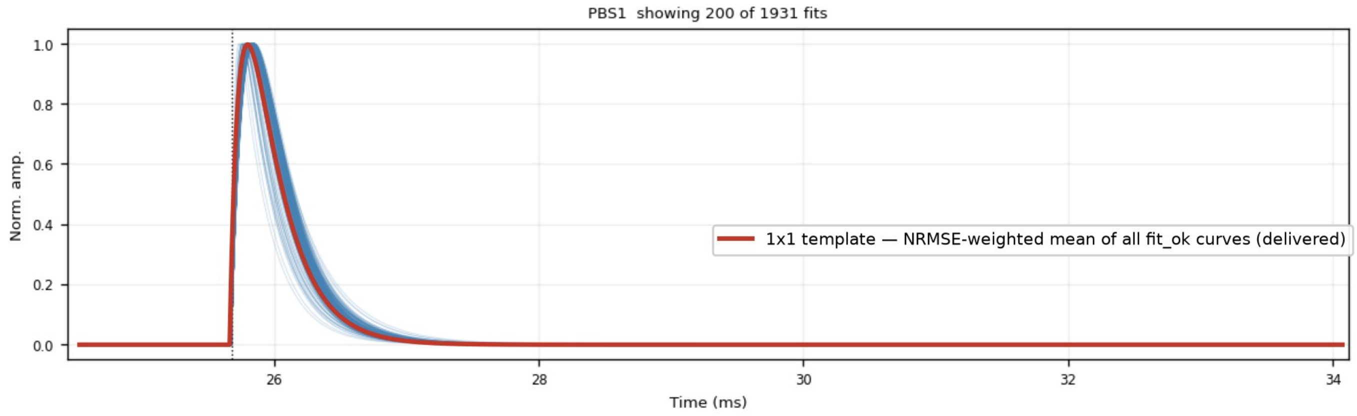
Z7 PAS1 — normal channel: narrow (median 0.25 ms)



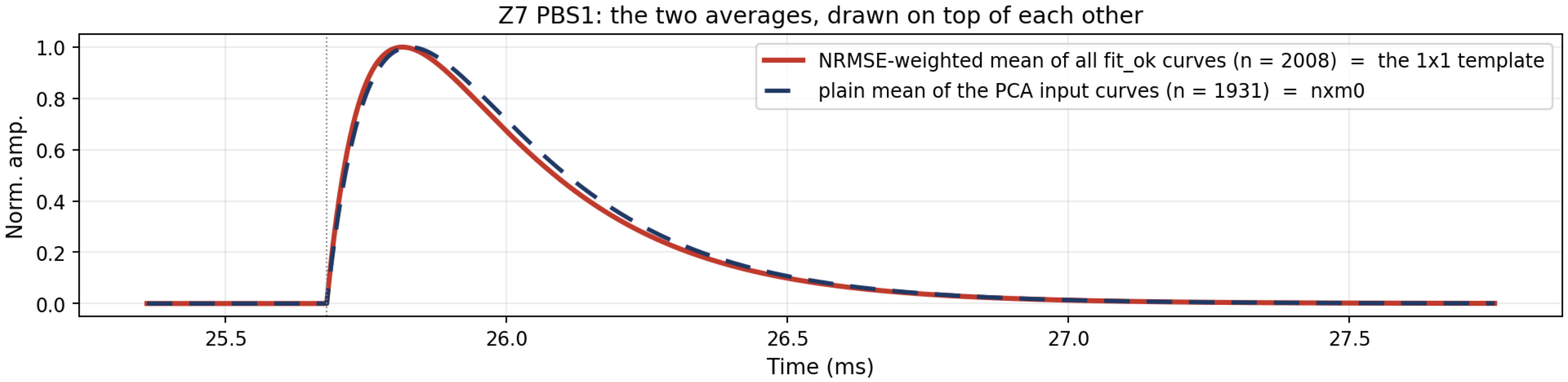
Z7 PDS2 — bad channel: broad tail (median 0.51 ms)

Results: 1x1 Templates

- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction



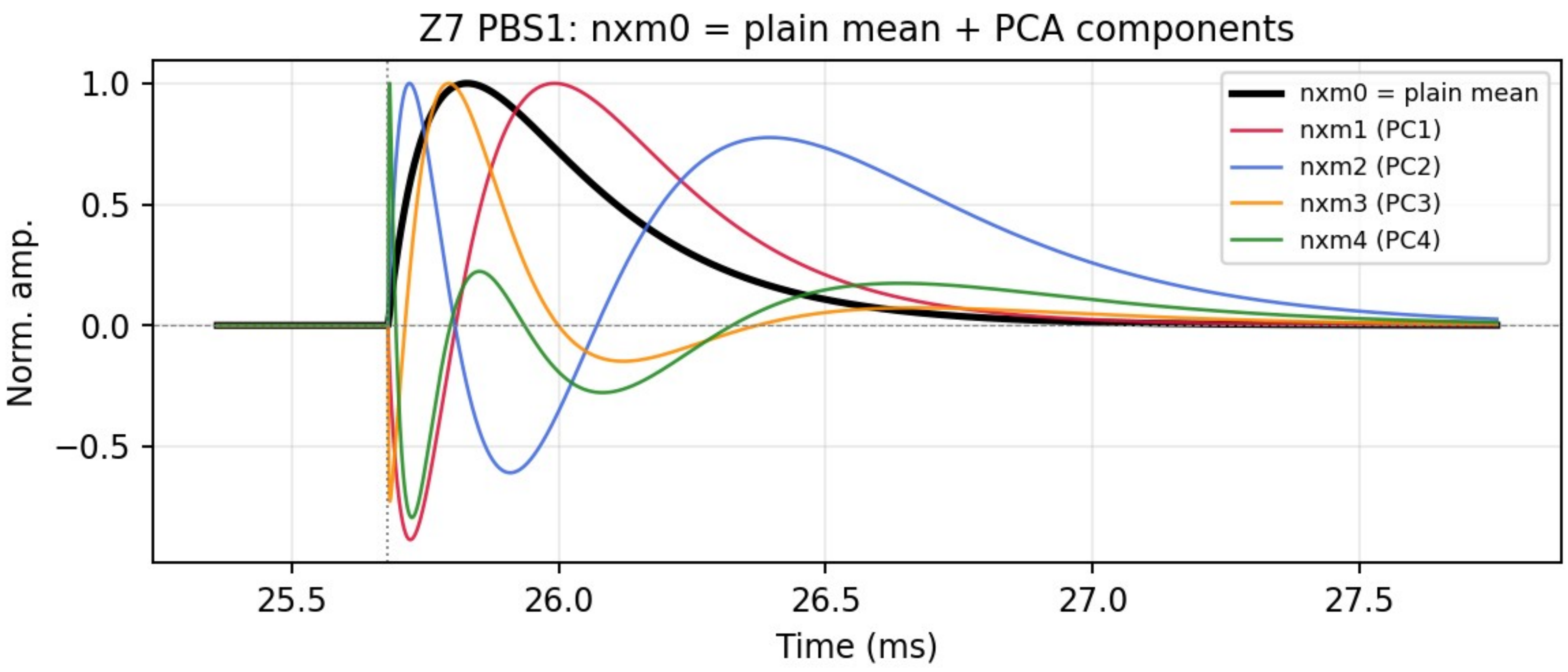
Z7 PBS1, blue: the fitted curves (200 of 1931 drawn); red: the 1x1 template



Solid red: NRMSE-weighted mean (= the 1x1 template); dashed navy: plain mean of the PCA input curves (= nxm0)

Results: NxM Templates

- channel-specific PCA over the fitted curves (fit_ok + NRMSE ≤ 0.4)
- nxm0 = mean shape; nxm1–4 = principal components; real pulse = $\Sigma_i \, amp_i \cdot nxm_i$
- PC1 + PC2 ≈ 96–98% of the shape variance



Z7 PBS1: nxm0 (plain mean over the fit_ok + NRMSE-cut curves) + nxm1–4, zoomed to the pulse

Template file

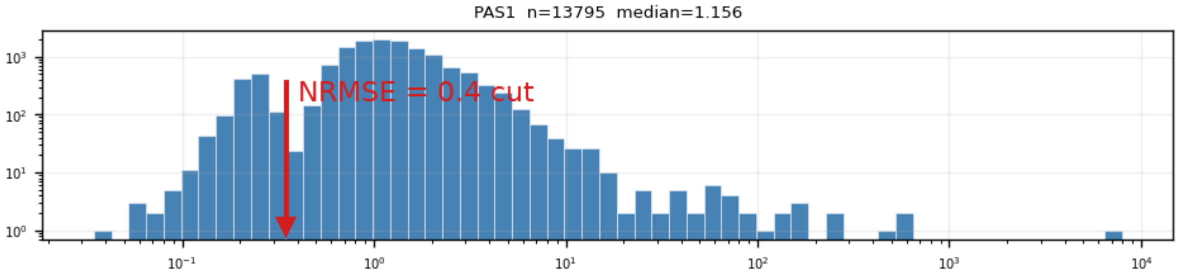
- Two template sets for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- In cdmsbats_config feature branch, ready for merge

Future Steps

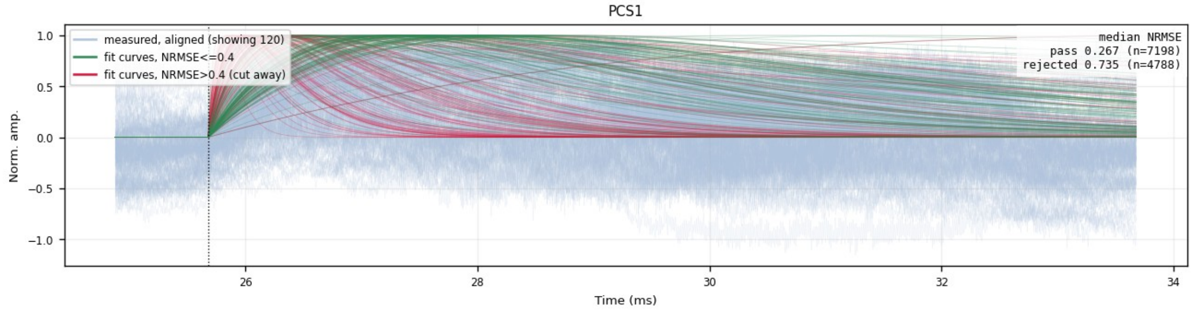
- Three-, four-exponential fits
- Exercise NxM processing
- Other detectors: worse noise, a few more cuts added, in the backup slides

Backup · weak detectors (example: Z22)

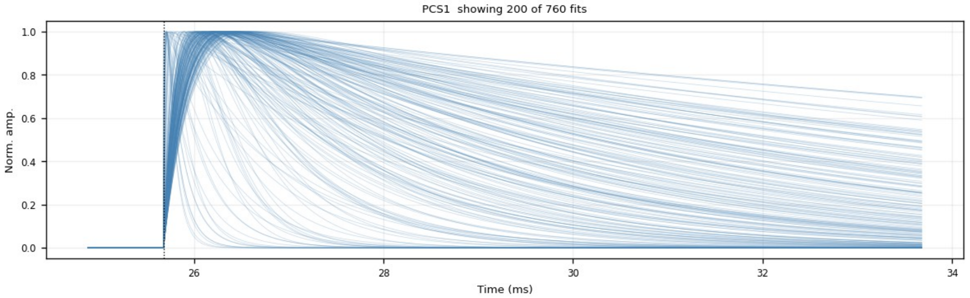
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line inside the noise population, the window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominant; the 0.4 cut digs the real pulses out (Z22 PCS1: 7198 kept / 4788 cut, ~40%)
- Kept bundle broader; steep red curves = fits latching onto noise spikes, no real pulses lost
- $\tau_{rise} \le 0.3$ ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7; removes residual slow drift



Z22 PAS1 NRMSE histogram: noise dominates



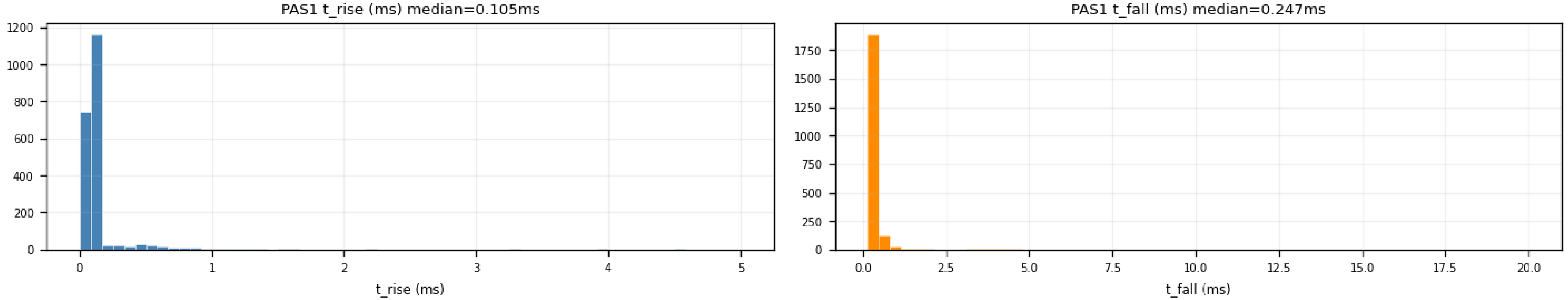
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after NRMSE ≤ 0.4 + $\tau_{rise} \le 0.3$ ms: final shape family (760 fits)

Backup · τ_rise ≤ 0.3 ms ceiling (PCA input)

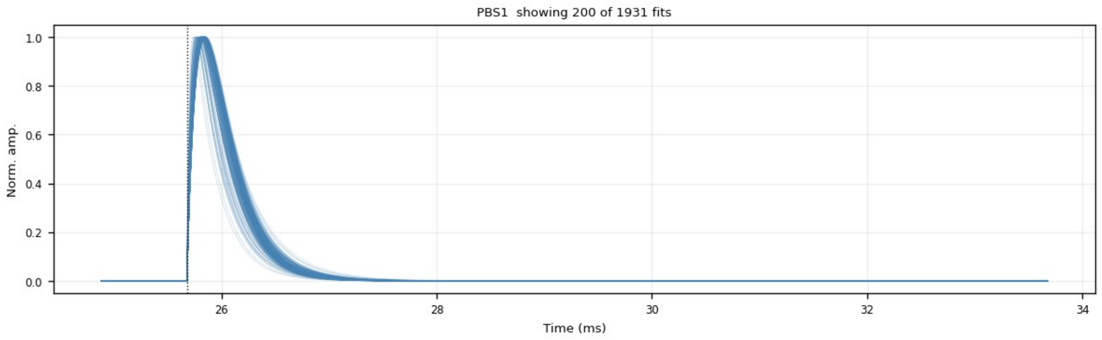
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at τ_rise ≈ 0.1 ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%), weak 55–74% (Z22 71%), all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



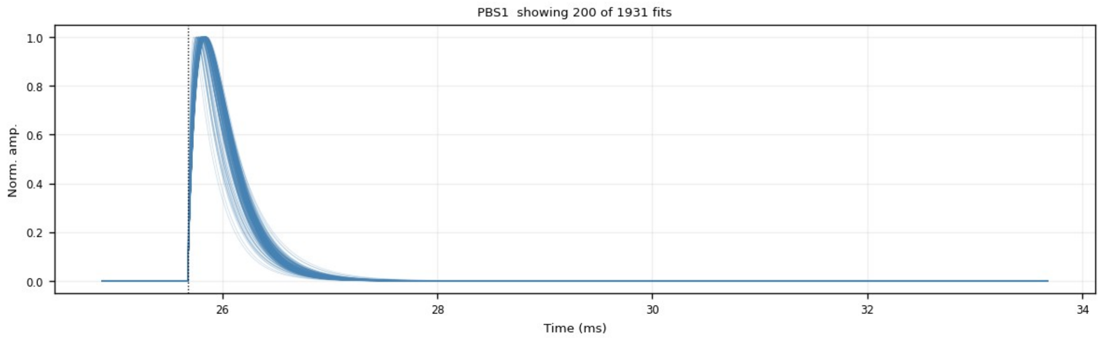
Z7 PAS1: fitted τ_rise (median 0.105 ms) and τ_fall distributions

Backup · τ_rise cut: good vs bad channel (Z7)

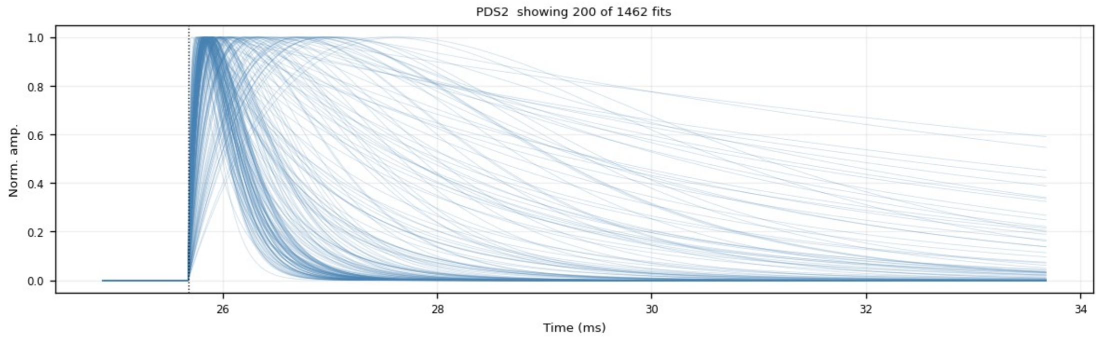
- Good channel PBS1: tight clean bundle; the cut removes ~0% (nearly free on quiet channels)
- Bad channel PDS2: broad messy fan, the channel itself is bad; the cut trims slow risers, 1462 → 1154 (21%)



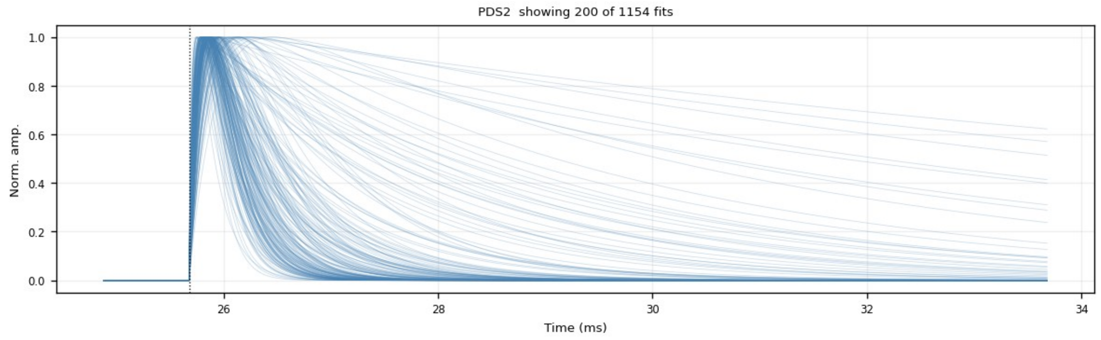
PBS1 (good): NRMSE ≤ 0.4



PBS1 (good): + τ_rise ≤ 0.3 ms (unchanged)



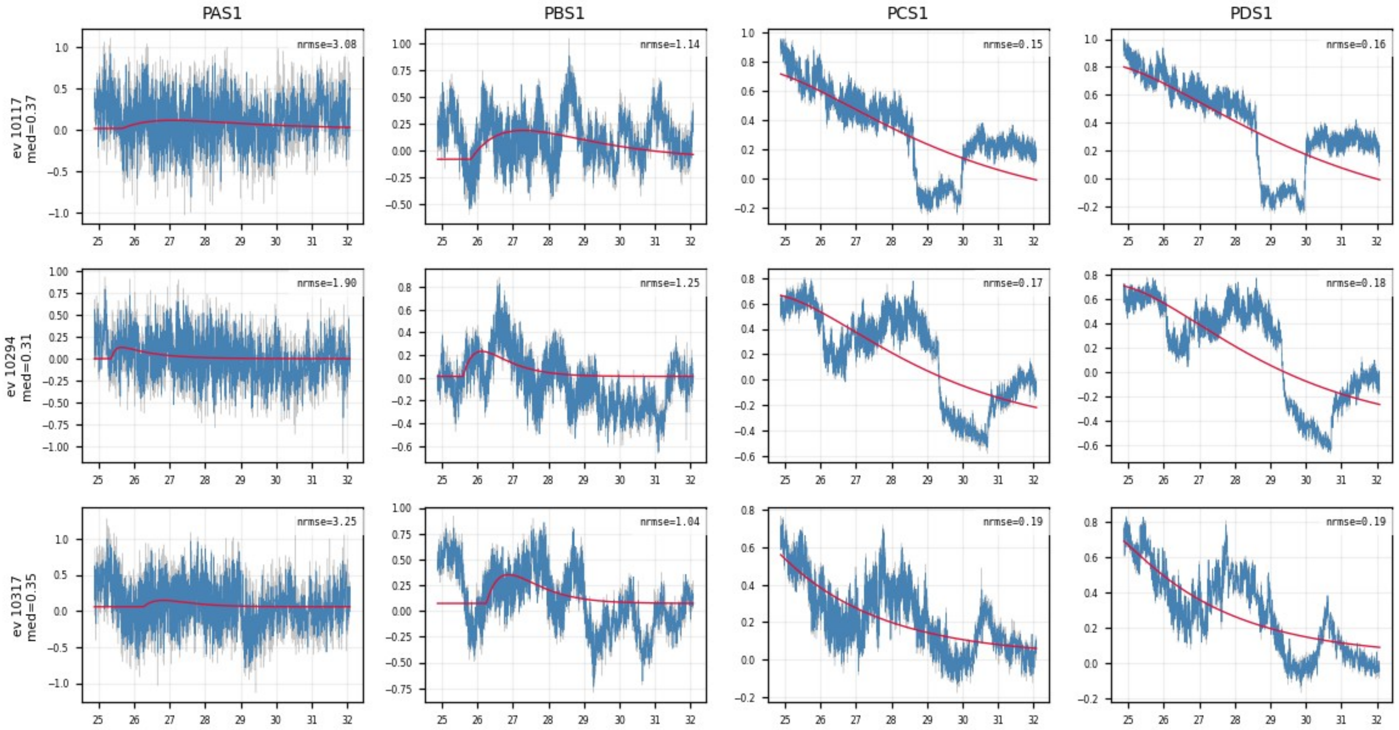
PDS2 (bad): NRMSE ≤ 0.4



PDS2 (bad): + τ_rise ≤ 0.3 ms (slow risers trimmed)

Backup · what the τ_rise cut removes (Z22)

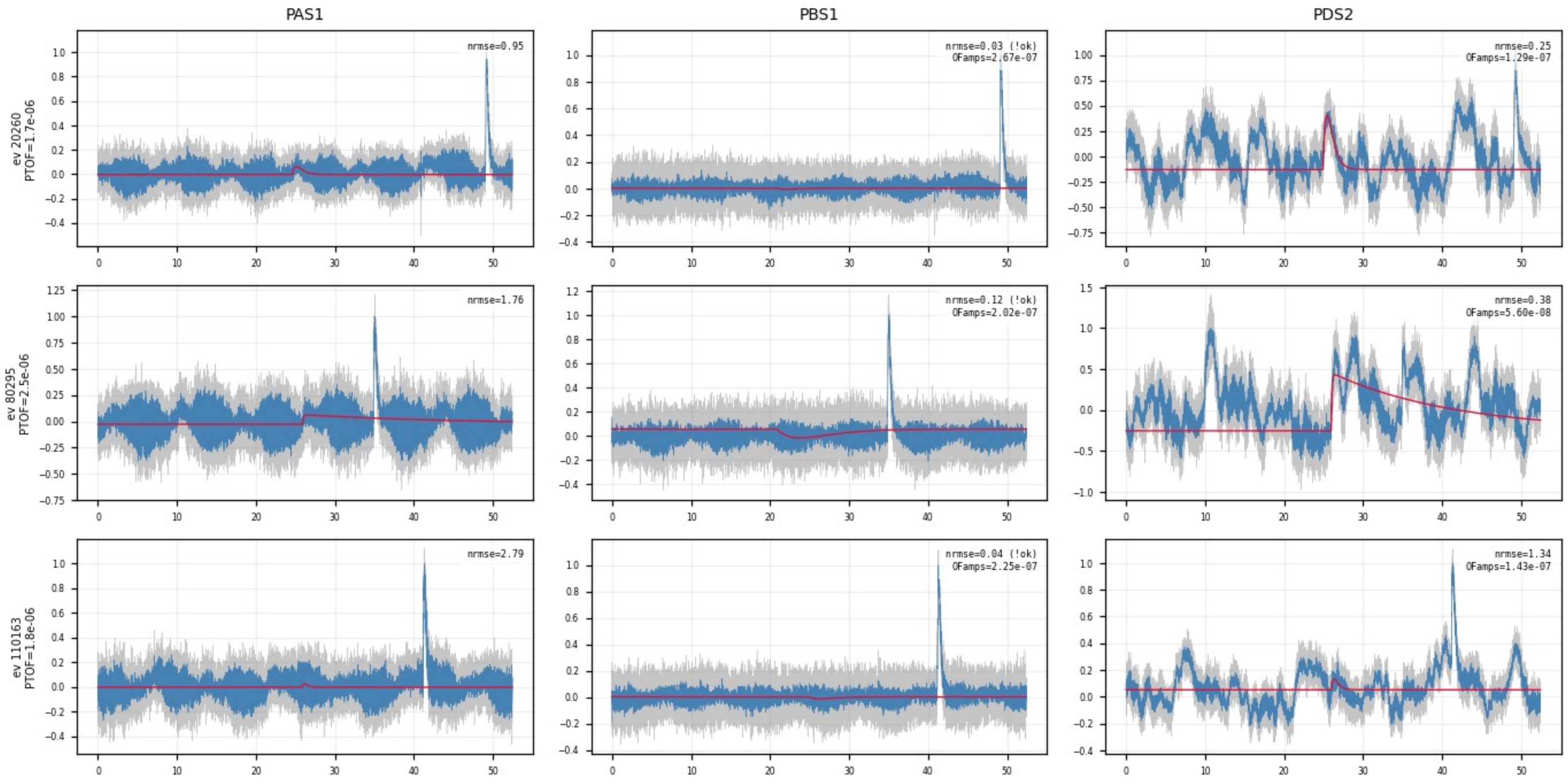
- Events that PASS NRMSE ≤ 0.4 but are REMOVED by τ_rise ≤ 0.3 ms: the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at NRMSE ≈ 0.15 (a slow 2-exp hugs the drift); the raw trace has no clean fast pulse
- Trade-off: also trims genuine slow-rise pulses; decision pending



Z22, 3 events × PAS1/PBS1/PCS1/PDS1: pass NRMSE ≤ 0.4, removed by τ_rise ≤ 0.3 ms; PCS1/PDS1 = pure slow drift

Backup · fit_ok-rejected events: normal amplitude, late pulse

- These events have normal K-line amplitude (PBS1OFamps ≈ 2×10^{-7}, same as kept events); the pulse simply arrives at 35–50 ms
- The fit searches the pretrigger only within 16050 ± 3000 samples (≈ 21–30 ms): the window holds noise only, the fit falls into the swapped-τ negative solution
- fit_ok removes them correctly: a pulse outside the fit window can not be used for templates



Z7, 3 fit_ok-rejected events × PAS1/PBS1/PDS2, full 0–52 ms trace: the real pulse sits far right of the fit window

Backup · Z7 full results gallery

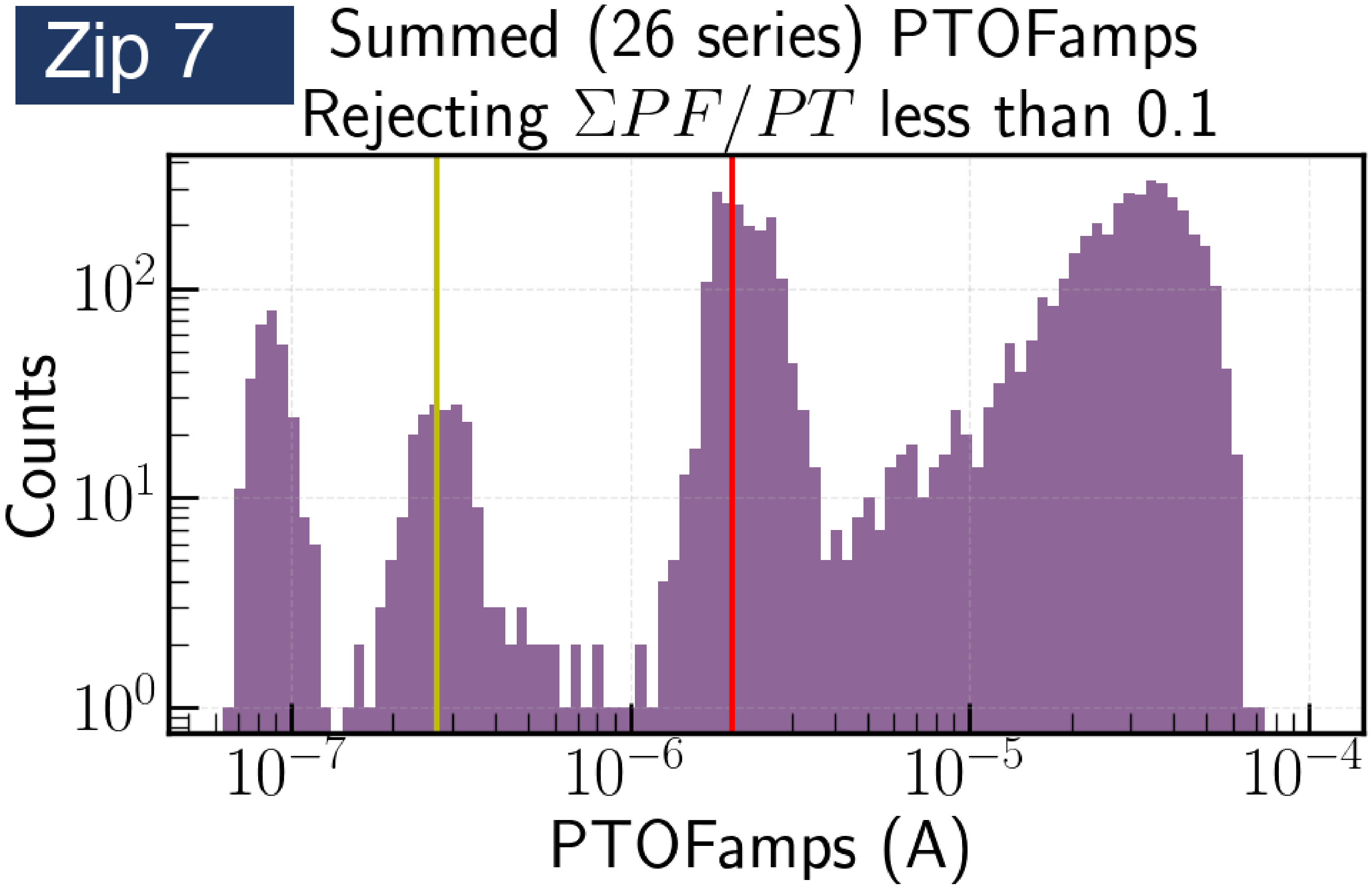
every diagnostic and result figure of the Z7 pipeline, in processing order

- Dataset: the K-line selection window
- Example grids: LP + fit and raw vs fit, event × channel
- Fit diagnostics: NRMSE, fitted pretrigger, τ_{rise} / τ_{fall} distributions
- Aligned fans: no cut → NRMSE ≤ 0.4 → + τ_{rise} ≤ 0.3 ms; kept vs rejected; measured traces
- Special populations: rejected events, well-fit slow-rise events, PDS2 slow-fall study
- Templates: PCA nxm0–4 per channel, summed PT / PS1 / PS2
- Tall figures are cut at white space into full-width panels, two per page; grids into 3-event × half-channel blocks

Backup · Z7 · K-line selection window

figure: kline_zip7.png

Summed PTOFamps spectrum of Z7; red line = Prof. Saab's 10.37 keV K-line position, shaded band = the ×/÷ 1.35 selection window; everything inside is cached raw.

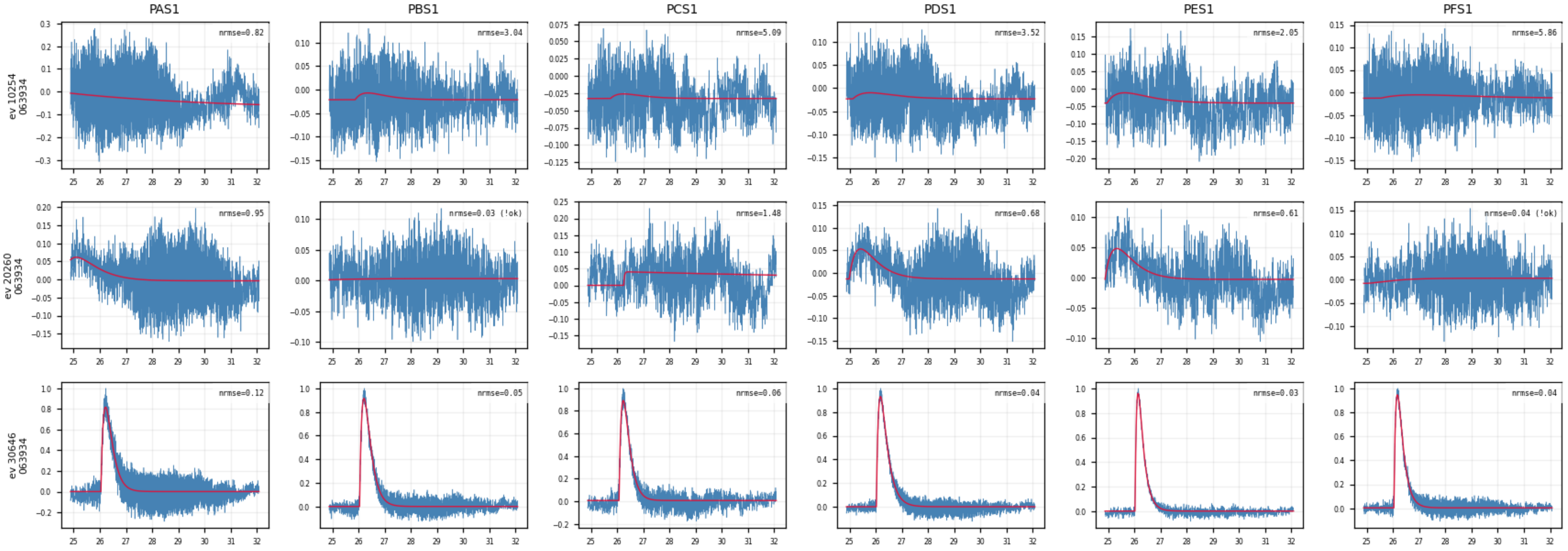


Z7 PTOFamps spectrum with the K-line selection window

Backup · Z7 · fit_examples: LP trace vs 2-exp fit (1/10)

figure: zip7_fit_examples.png, event rows 1–3 of 15, left half of the channels

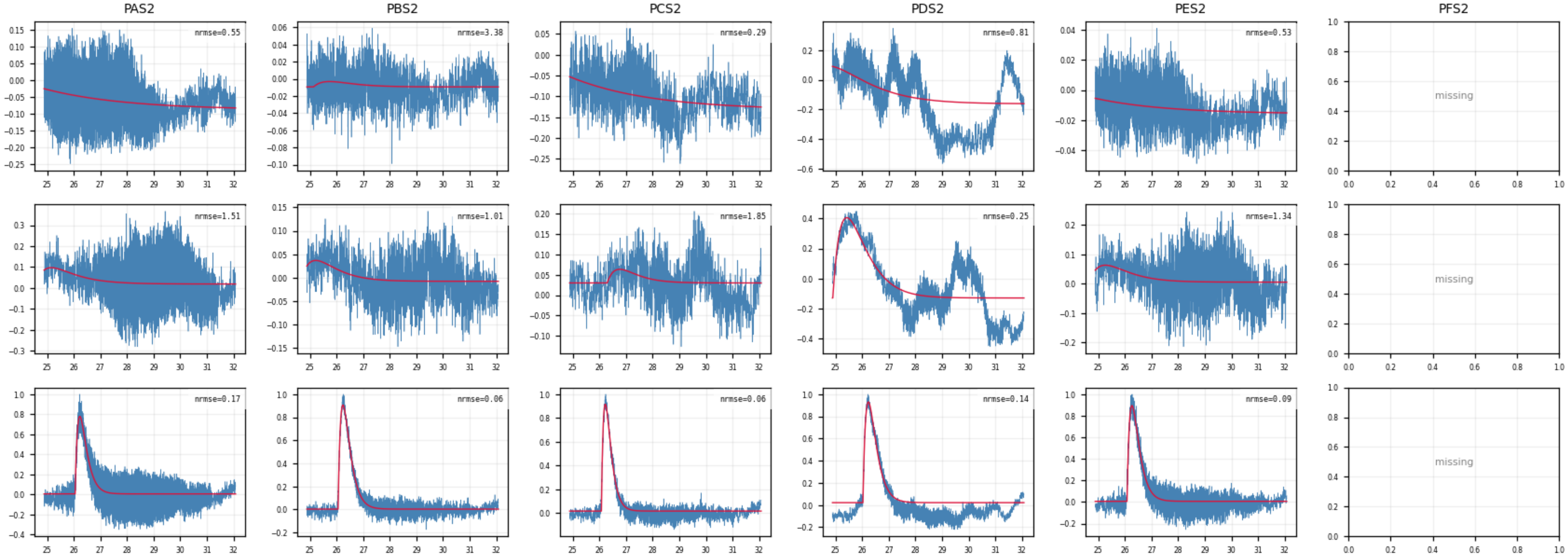
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (2/10)

figure: zip7_fit_examples.png, event rows 1–3 of 15, right half of the channels

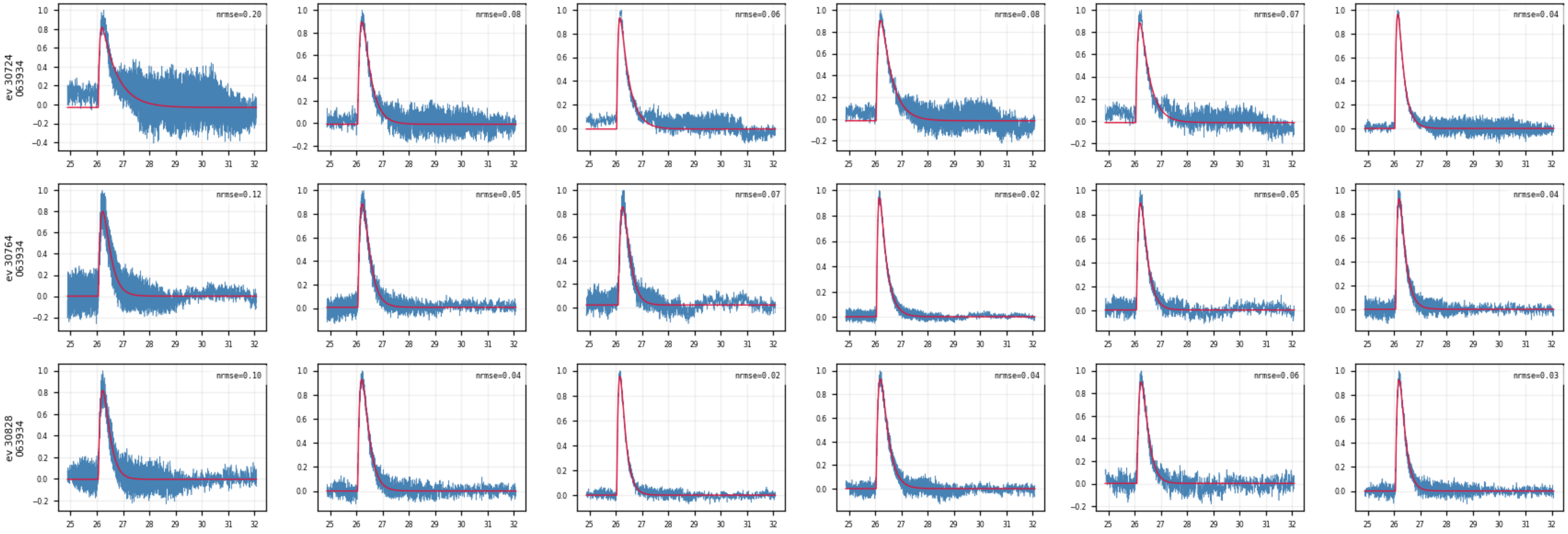
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (3/10)

figure: zip7_fit_examples.png, event rows 4–6 of 15, left half of the channels

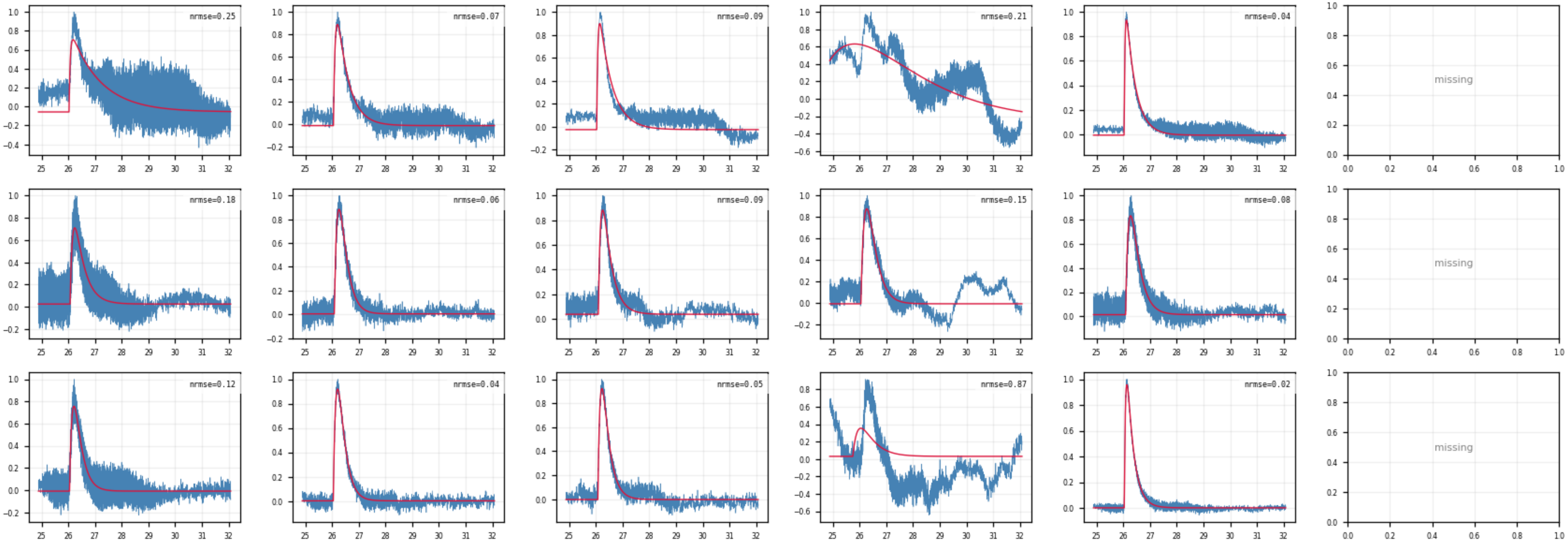
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (4/10)

figure: zip7_fit_examples.png, event rows 4–6 of 15, right half of the channels

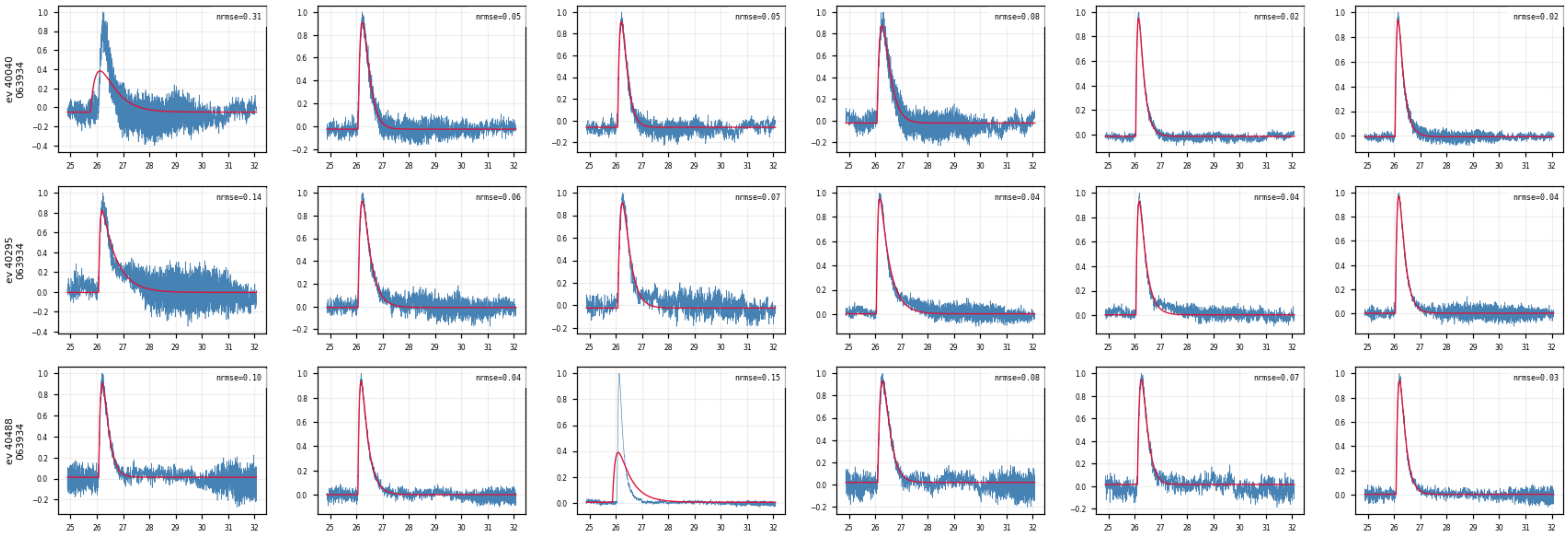
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (5/10)

figure: zip7_fit_examples.png, event rows 7–9 of 15, left half of the channels

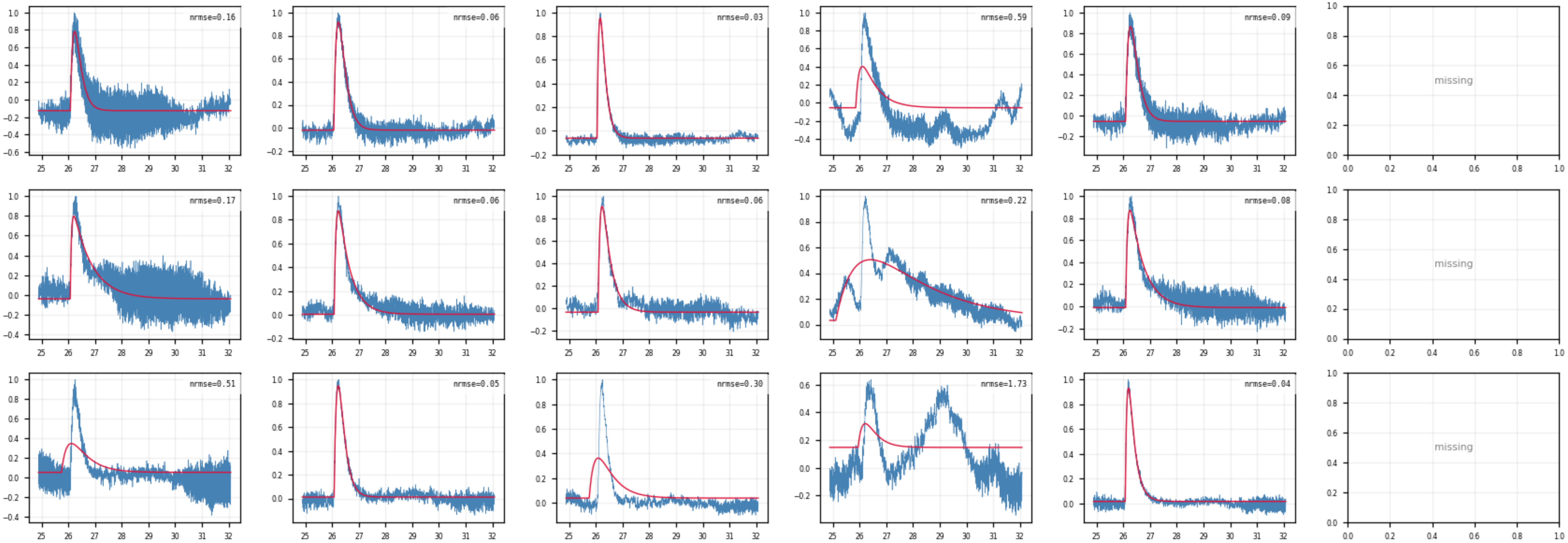
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (6/10)

figure: zip7_fit_examples.png, event rows 7–9 of 15, right half of the channels

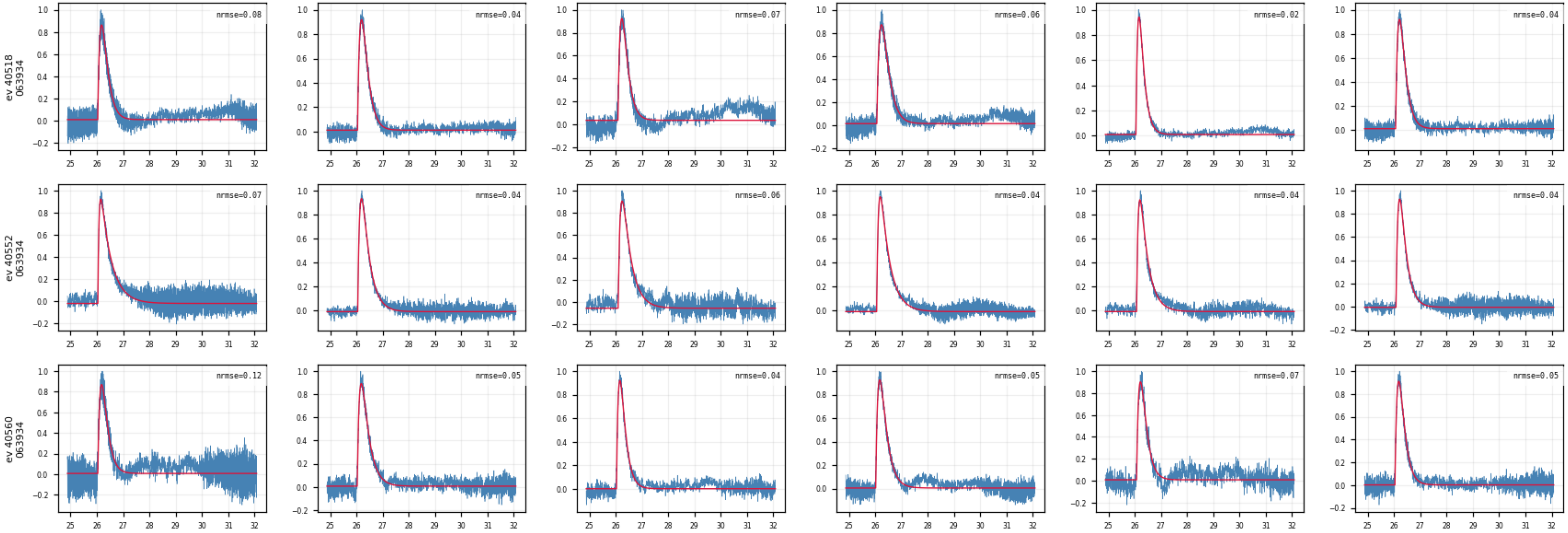
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (7/10)

figure: zip7_fit_examples.png, event rows 10–12 of 15, left half of the channels

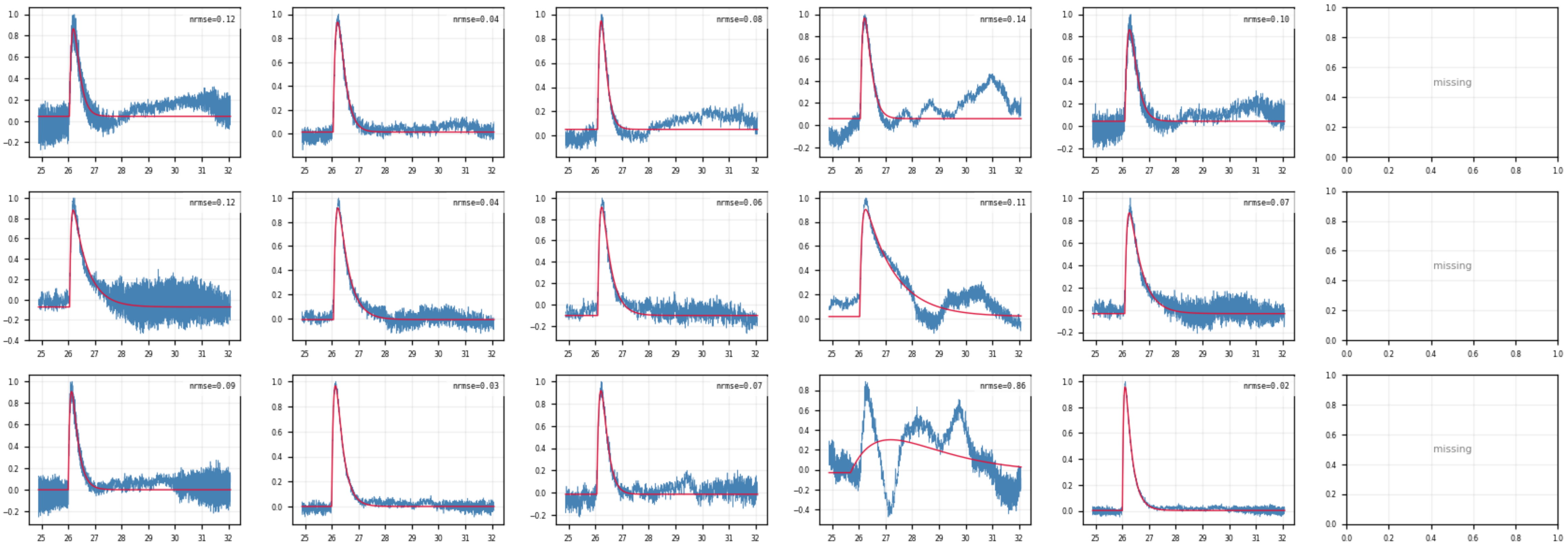
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (8/10)

figure: zip7_fit_examples.png, event rows 10–12 of 15, right half of the channels

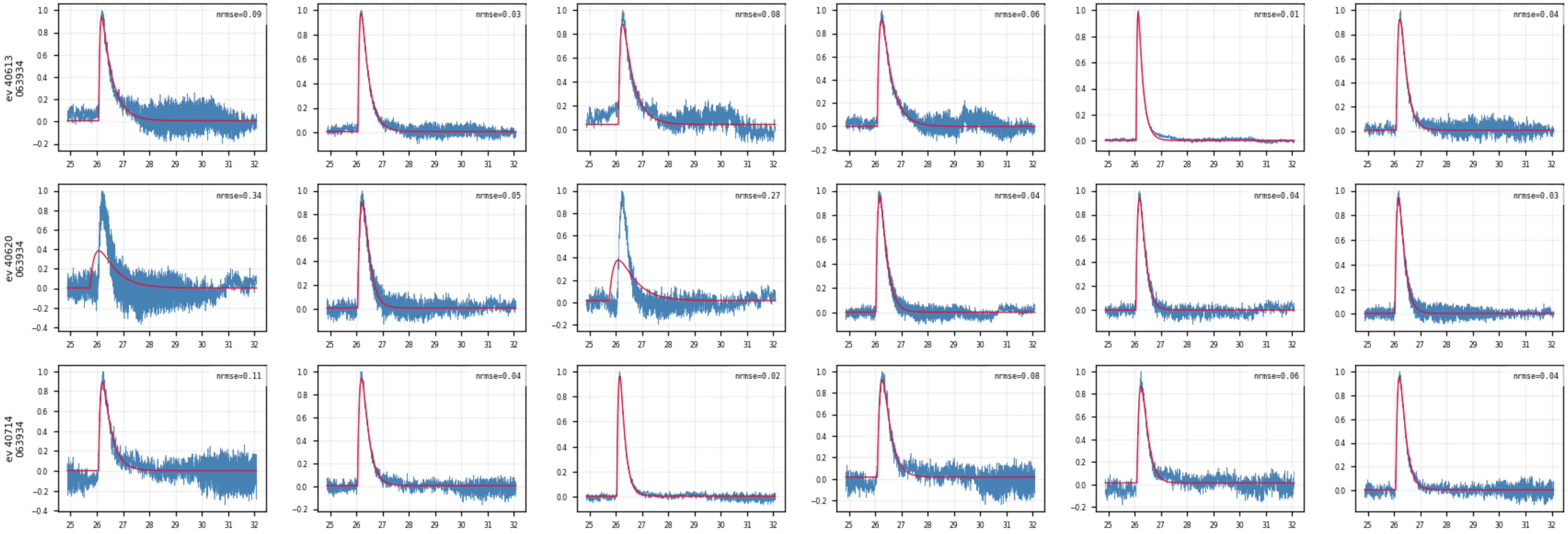
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (9/10)

figure: zip7_fit_examples.png, event rows 13–15 of 15, left half of the channels

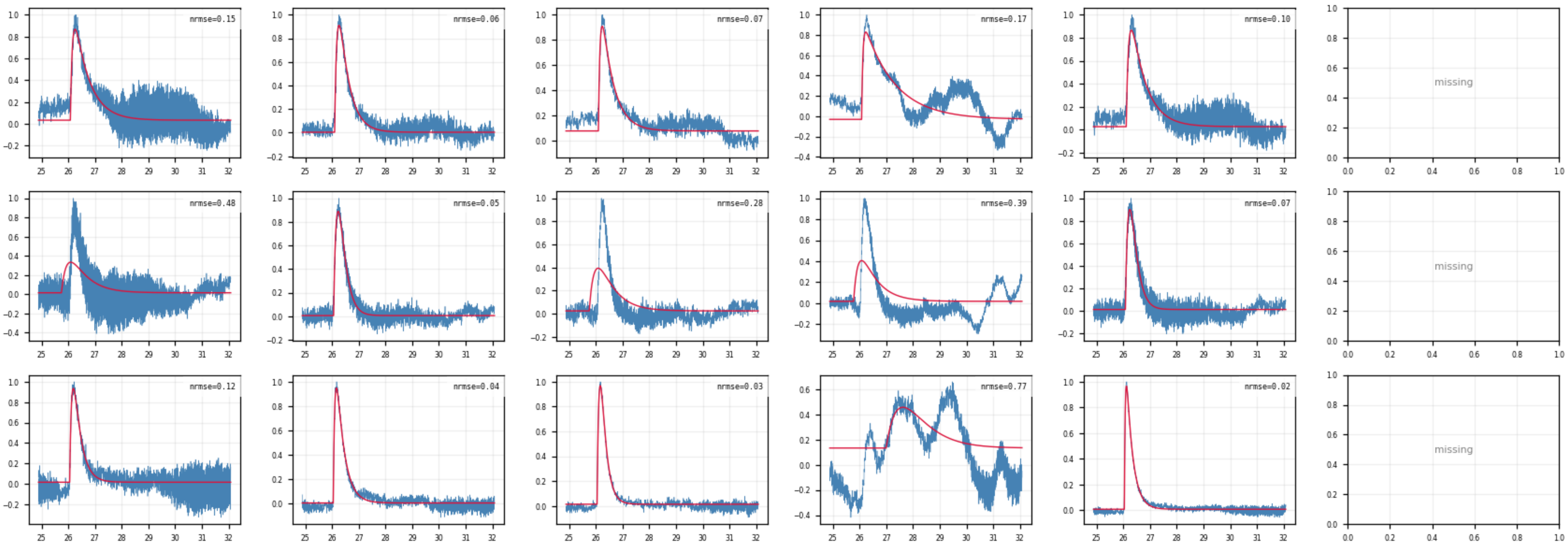
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (10/10)

figure: zip7_fit_examples.png, event rows 13–15 of 15, right half of the channels

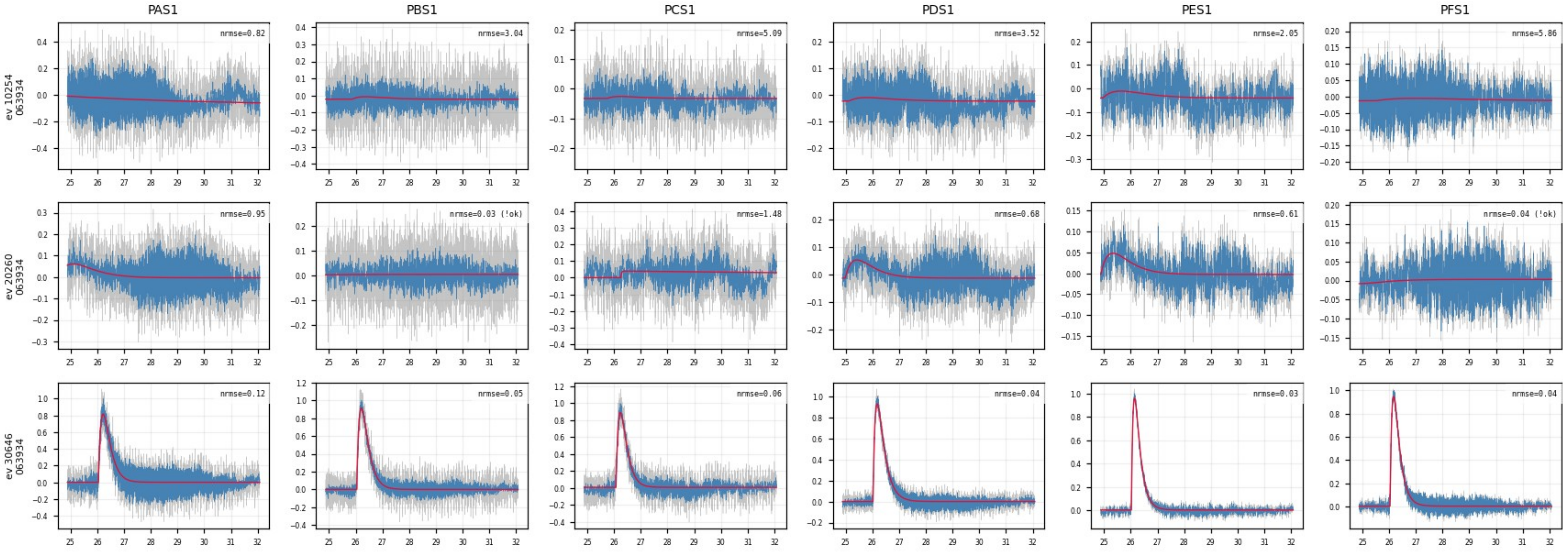
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (1/10)

figure: zip7_raw_vs_fit_examples.png, event rows 1–3 of 15, left half of the channels

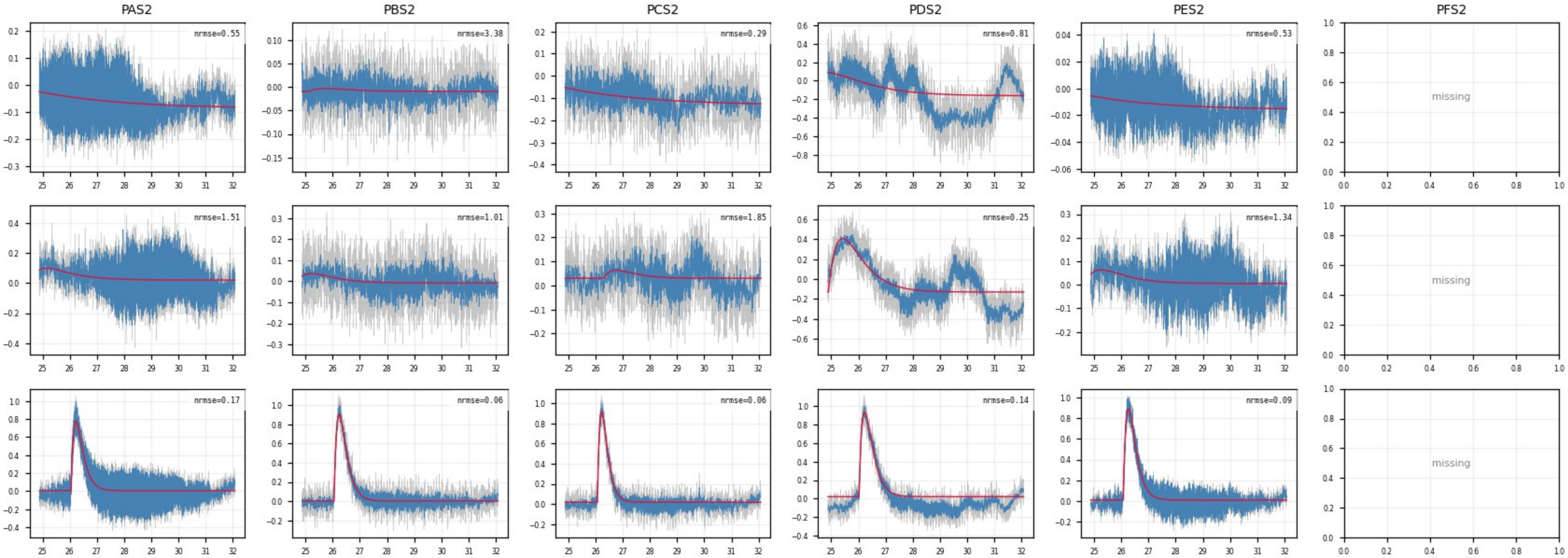
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (2/10)

figure: zip7_raw_vs_fit_examples.png, event rows 1–3 of 15, right half of the channels

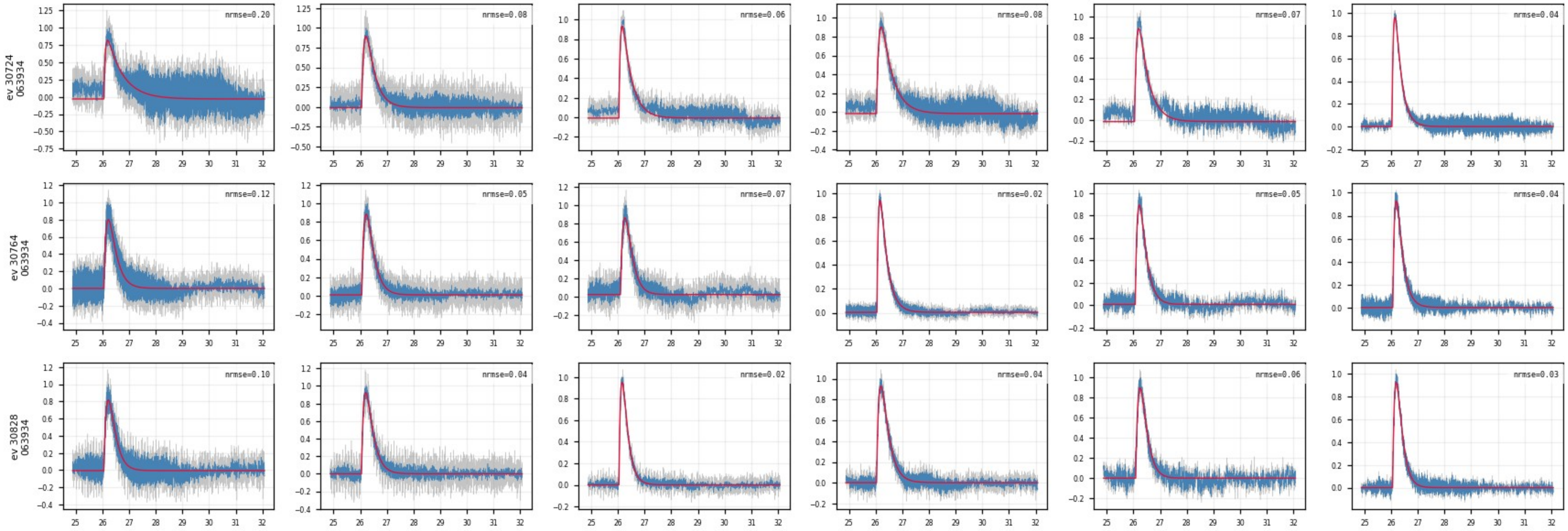
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (3/10)

figure: zip7_raw_vs_fit_examples.png, event rows 4–6 of 15, left half of the channels

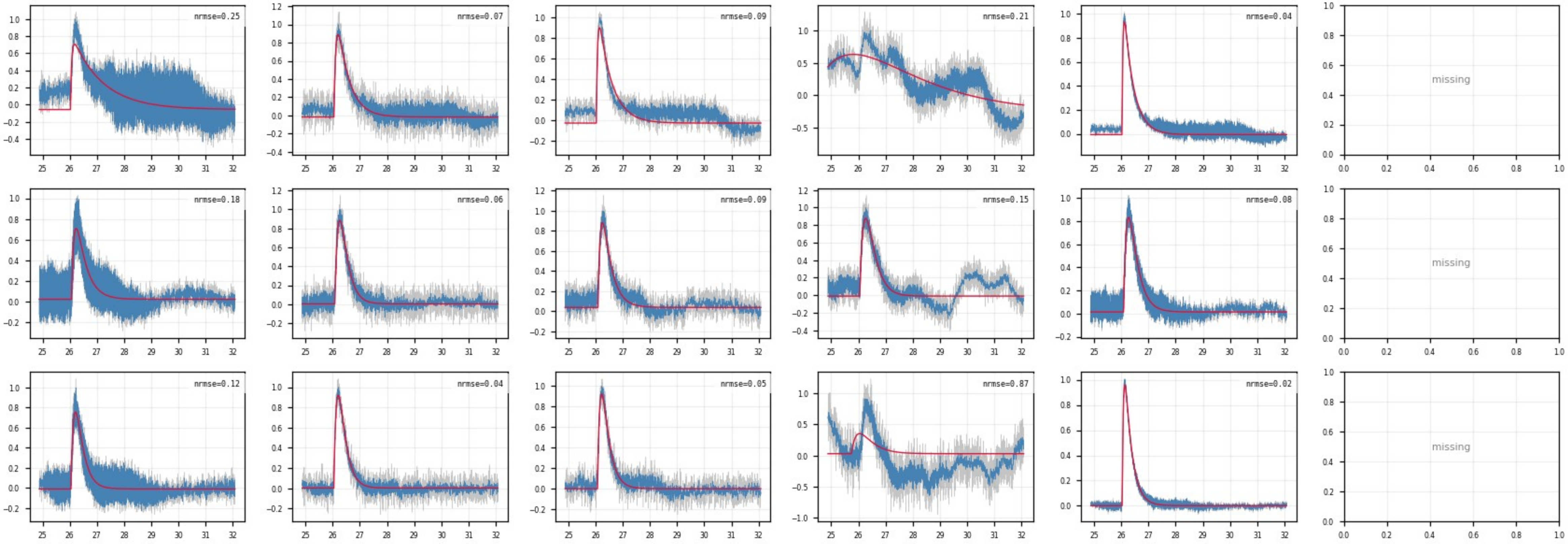
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (4/10)

figure: zip7_raw_vs_fit_examples.png, event rows 4–6 of 15, right half of the channels

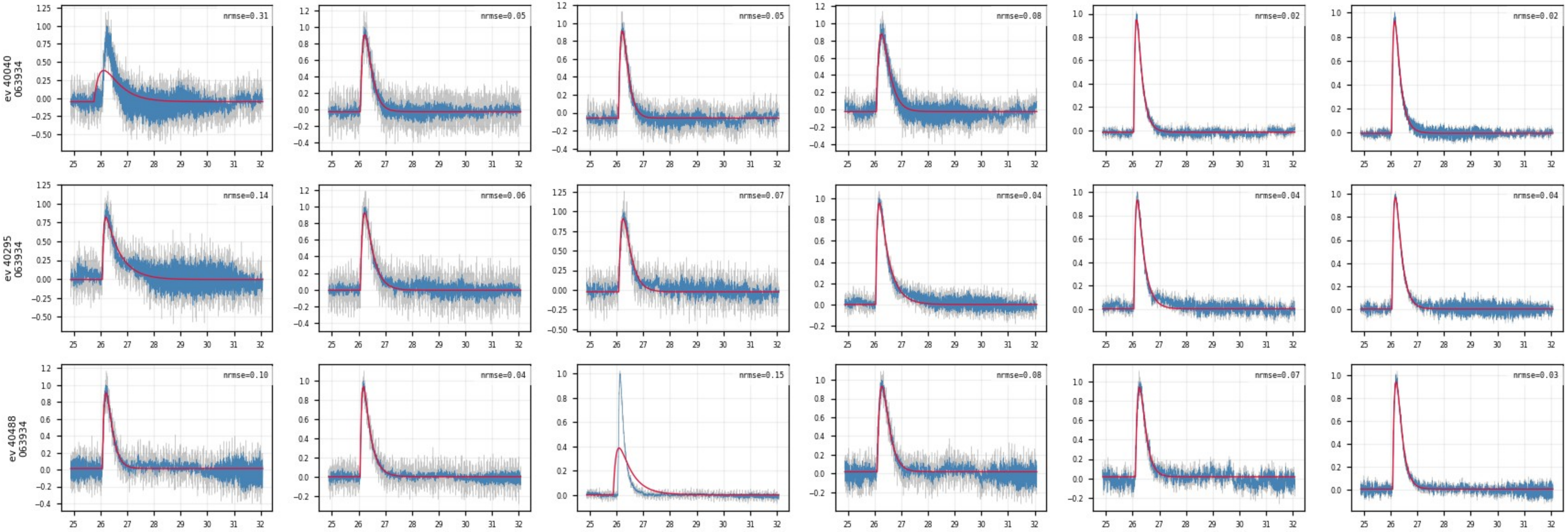
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (5/10)

figure: zip7_raw_vs_fit_examples.png, event rows 7–9 of 15, left half of the channels

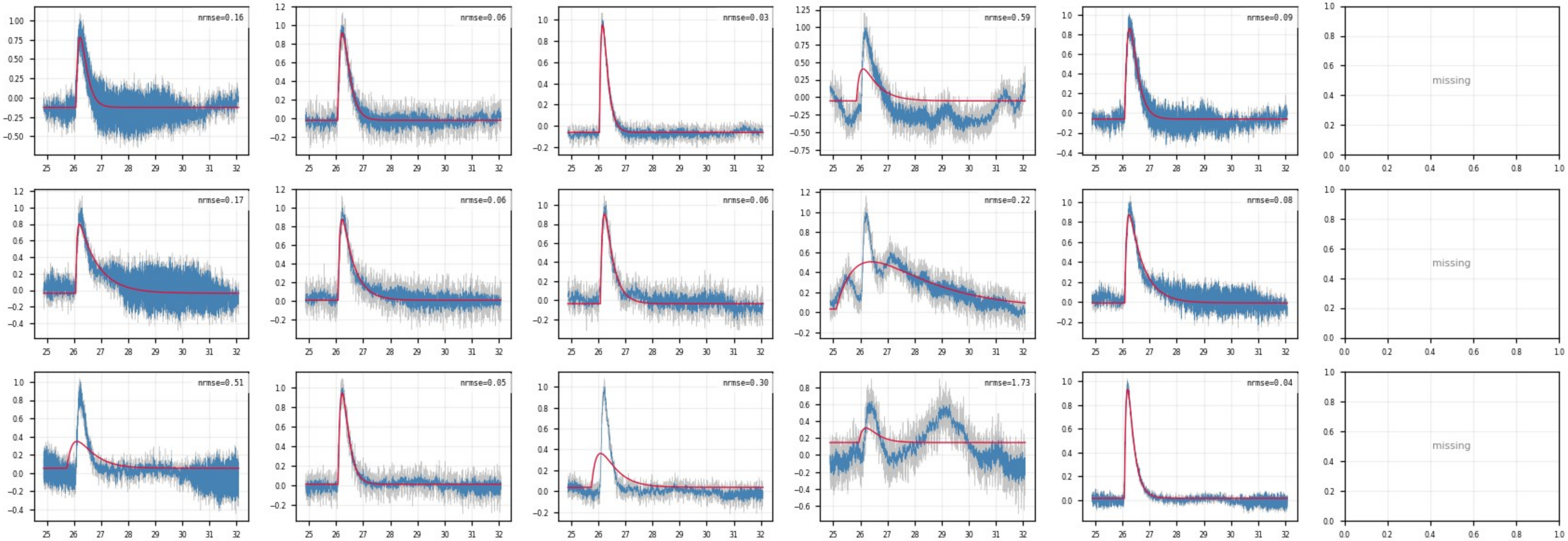
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (6/10)

figure: zip7_raw_vs_fit_examples.png, event rows 7–9 of 15, right half of the channels

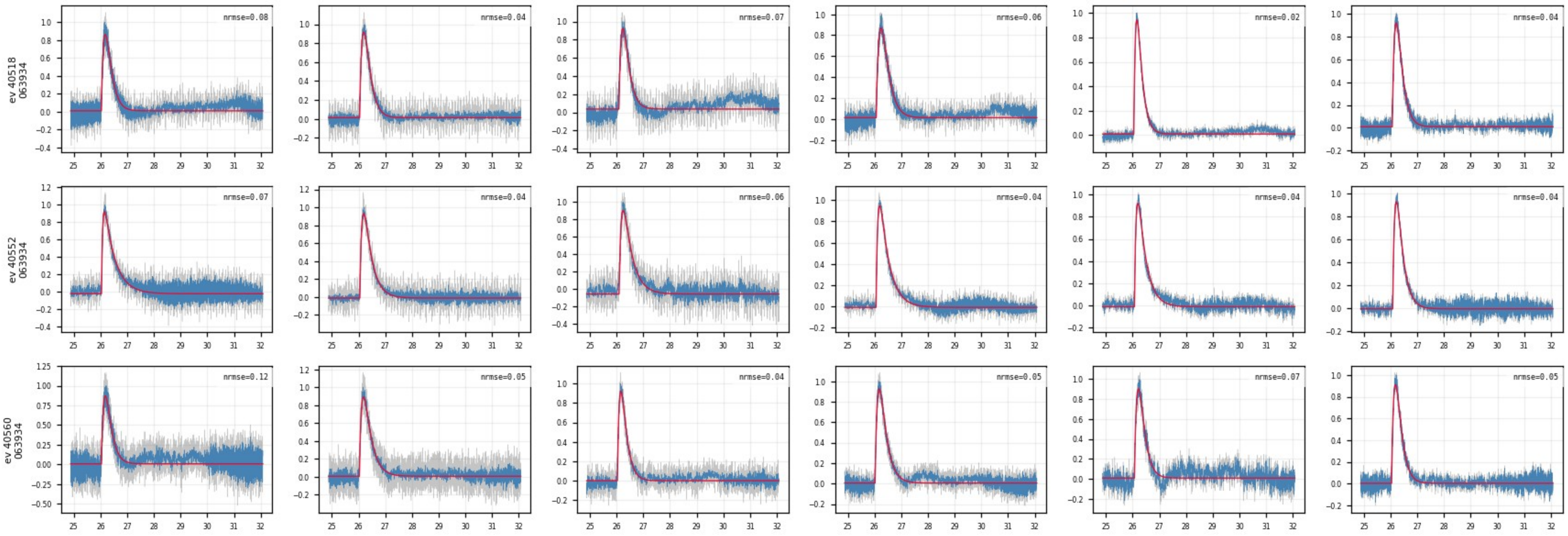
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (7/10)

figure: zip7_raw_vs_fit_examples.png, event rows 10–12 of 15, left half of the channels

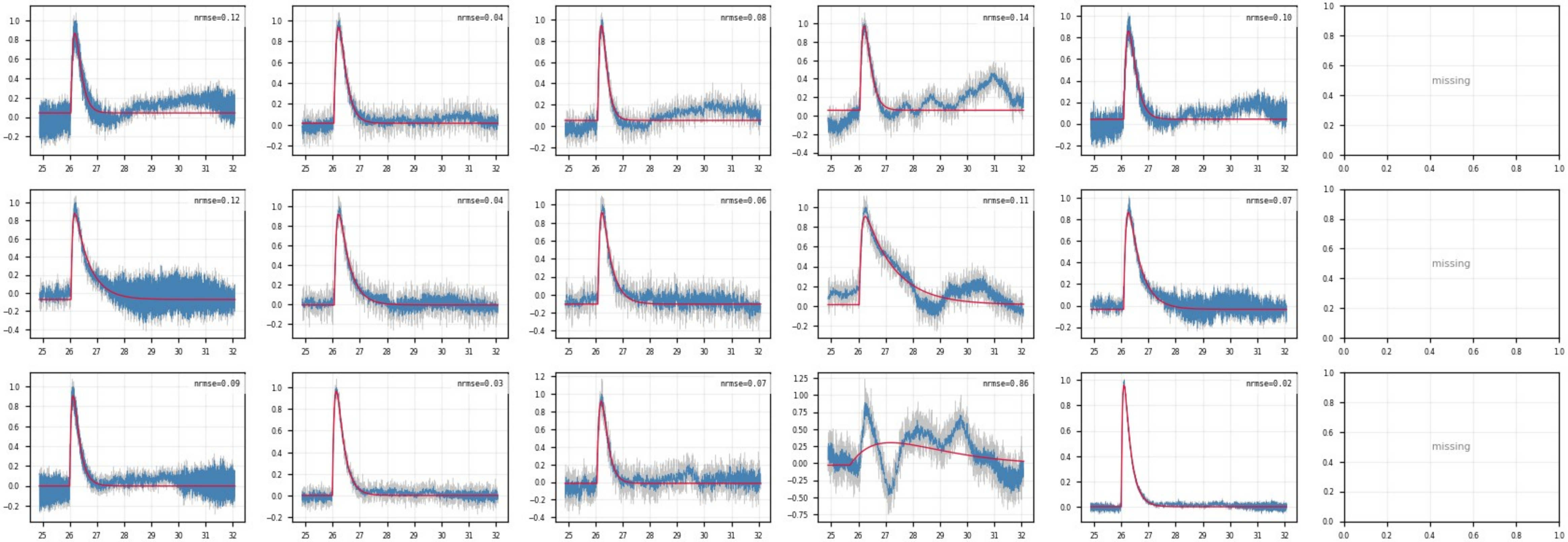
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (8/10)

figure: zip7_raw_vs_fit_examples.png, event rows 10–12 of 15, right half of the channels

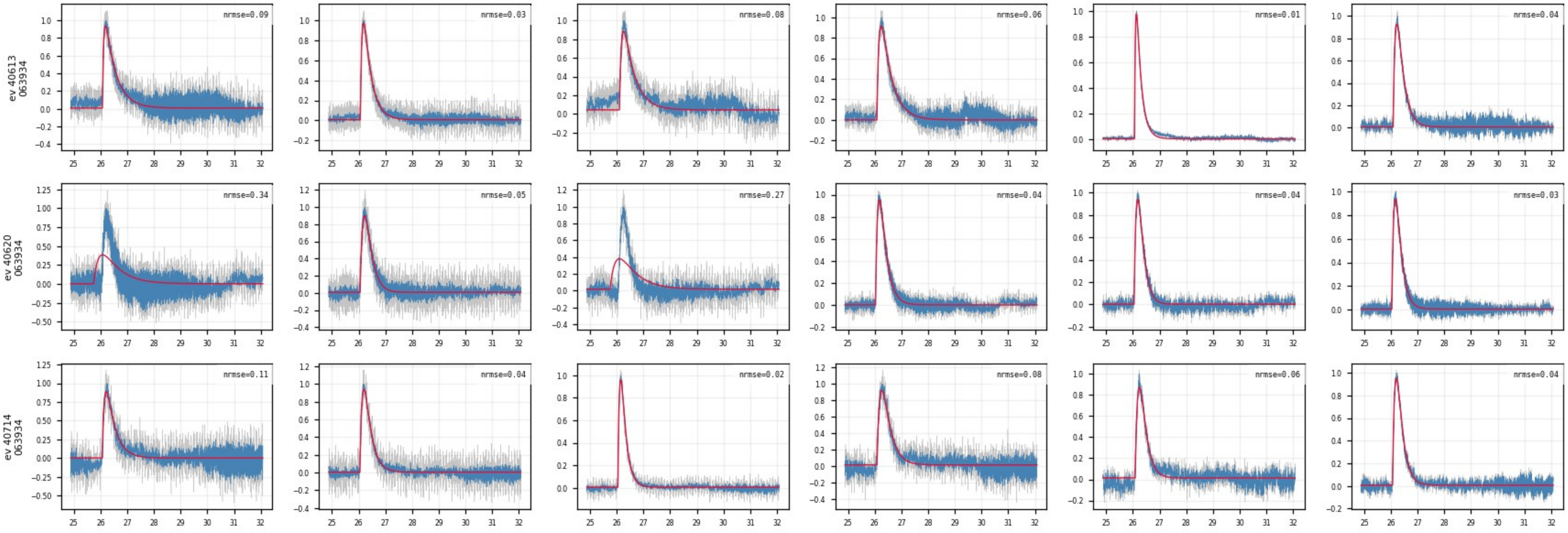
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (9/10)

figure: zip7_raw_vs_fit_examples.png, event rows 13–15 of 15, left half of the channels

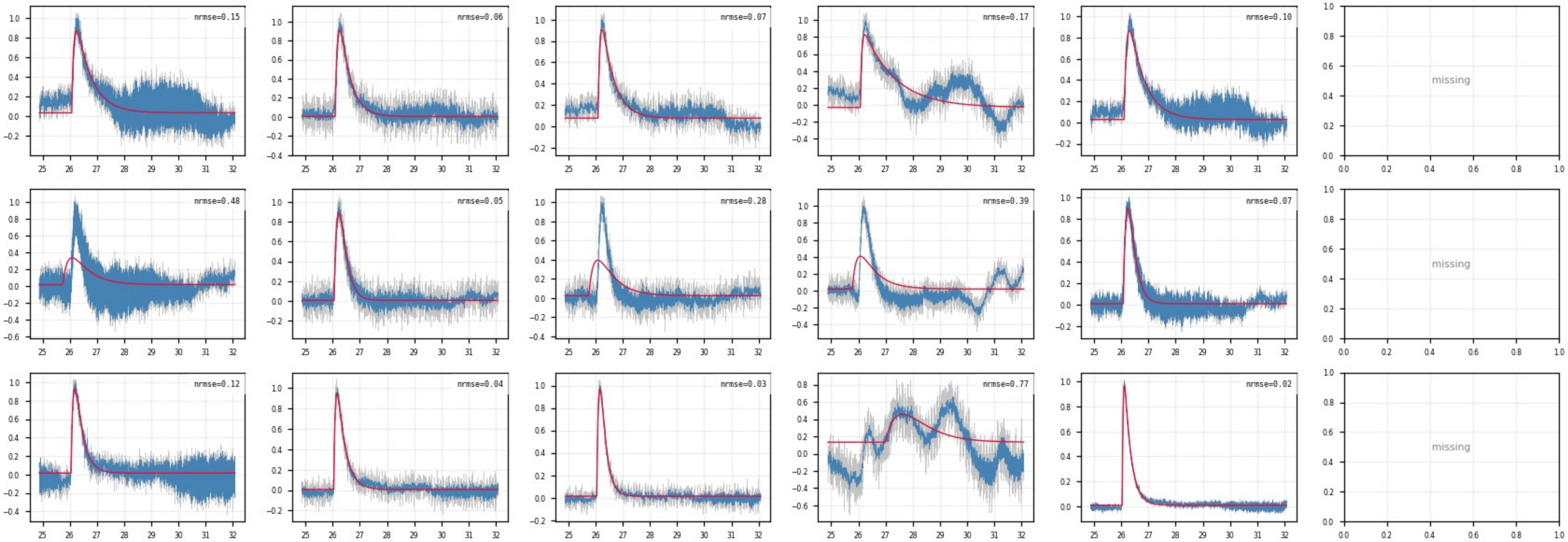
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (10/10)

figure: zip7_raw_vs_fit_examples.png, event rows 13–15 of 15, right half of the channels

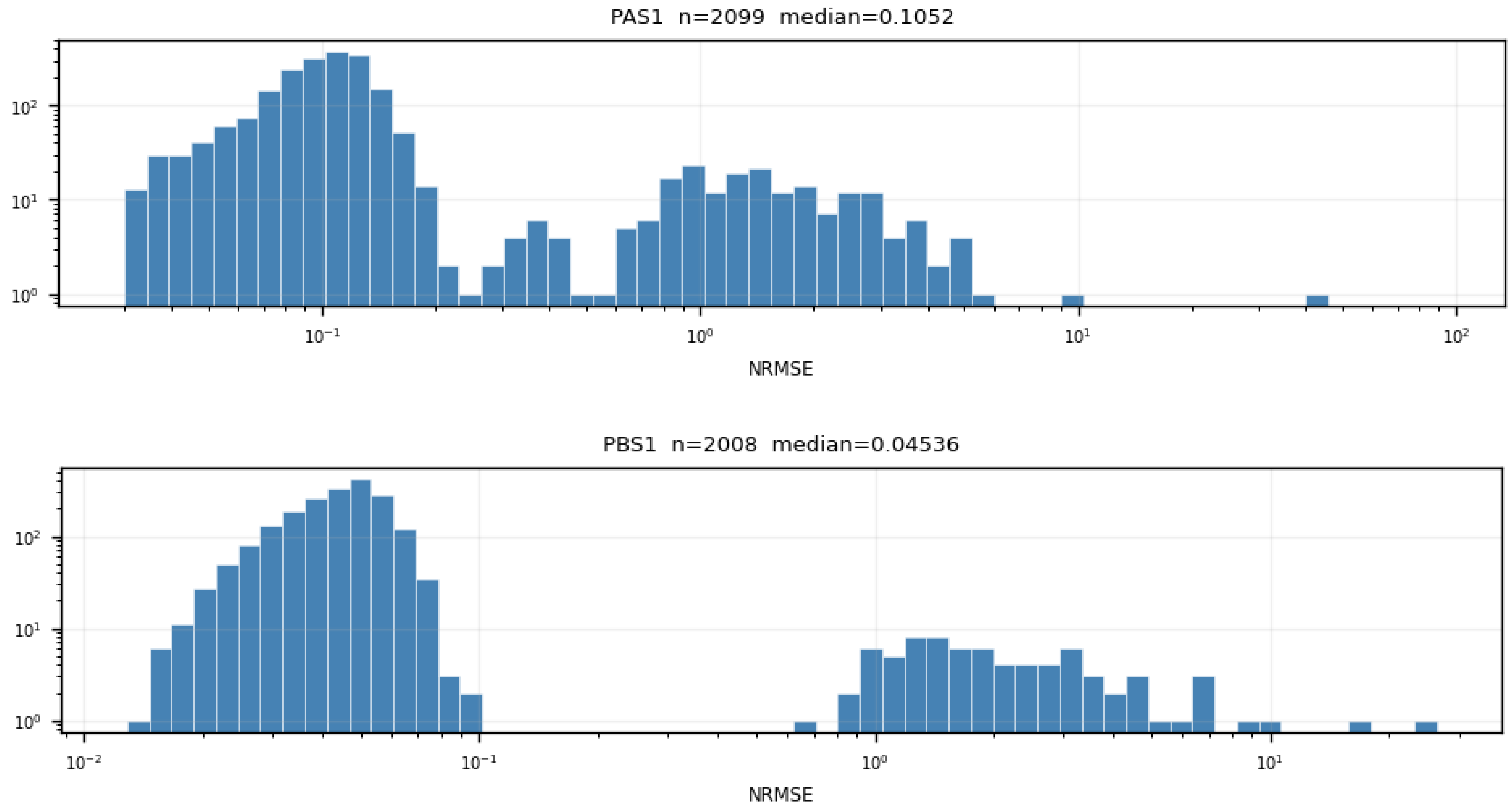
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · nrmse: NRMSE distribution per channel (1/6)

figure: zip7_nrmse.png, channel panels 1–2 of 11, top to bottom

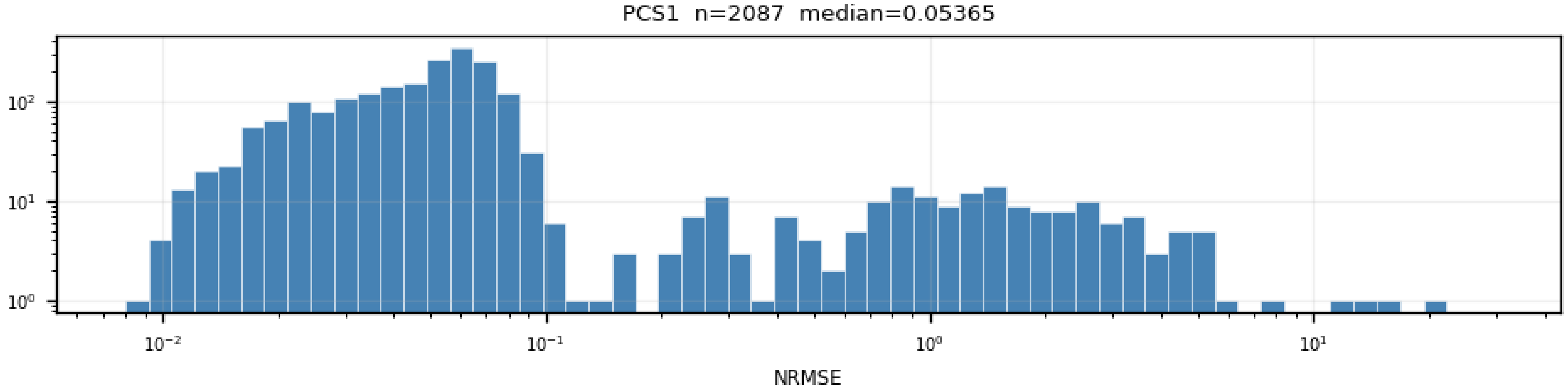
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.

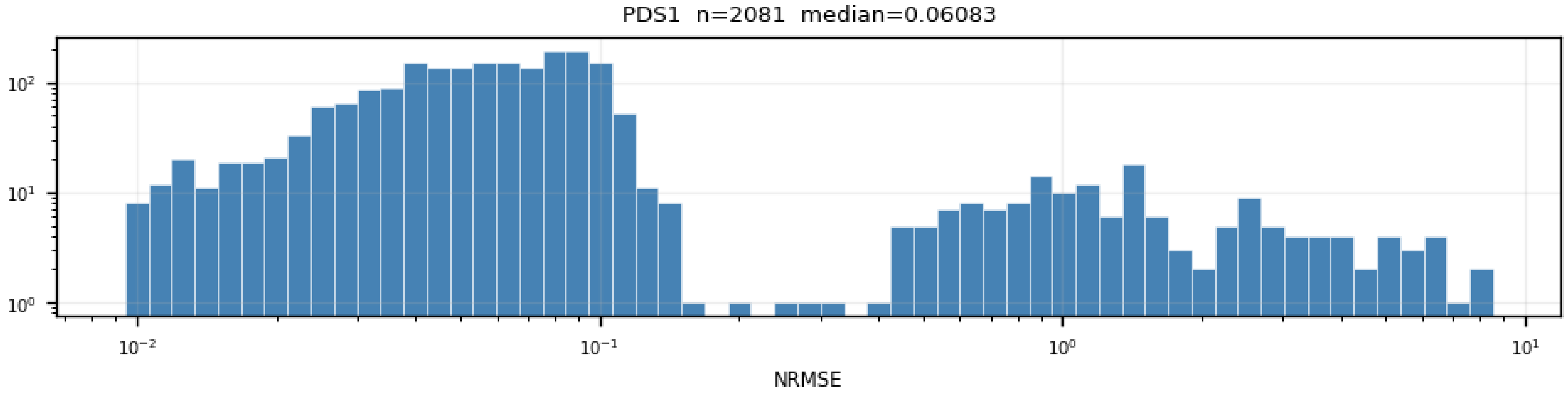


Backup · Z7 · nrmse: NRMSE distribution per channel (2/6)

figure: zip7_nrmse.png, channel panels 3–4 of 11, top to bottom

NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.

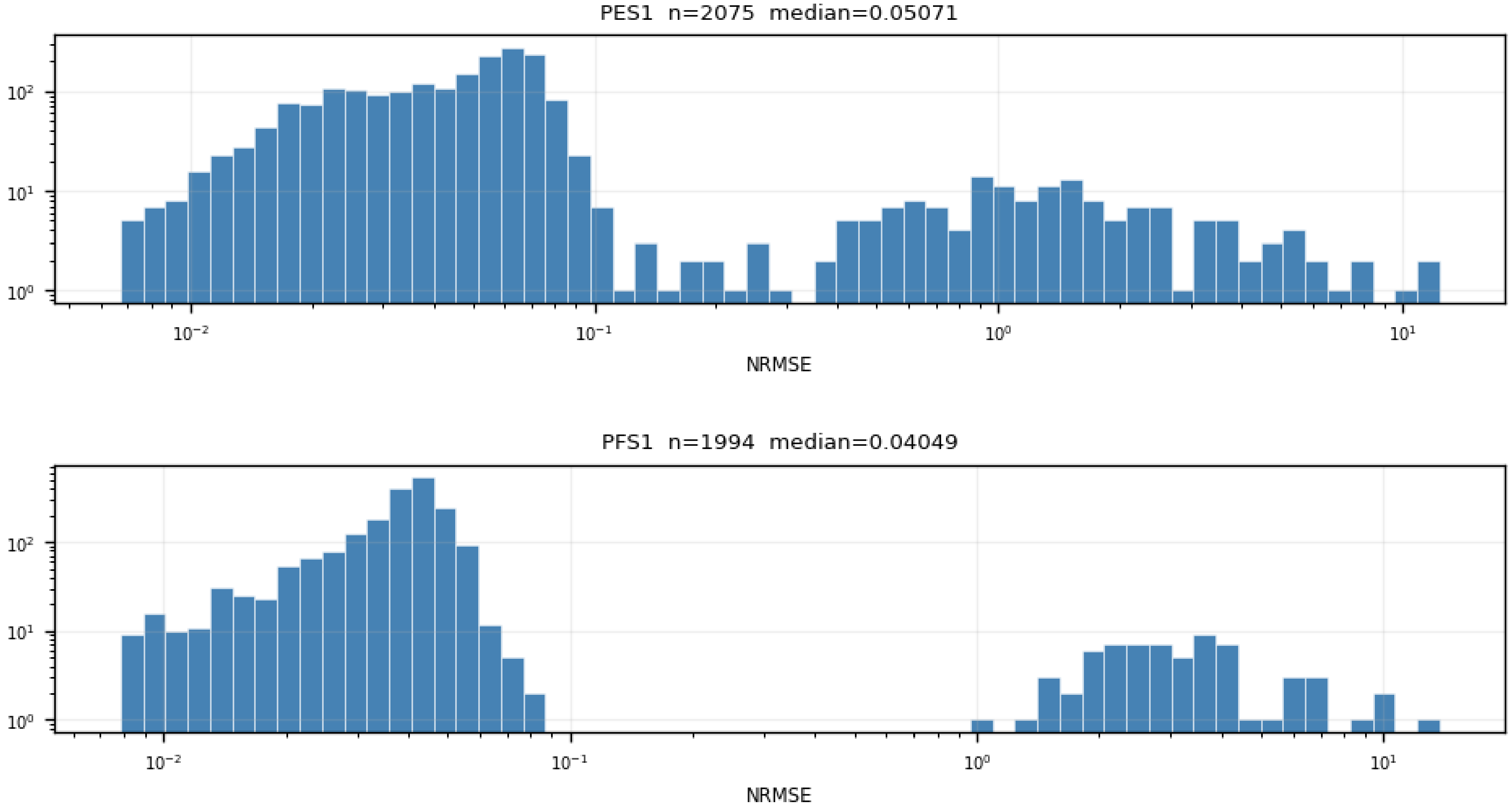




Backup · Z7 · nrmse: NRMSE distribution per channel (3/6)

figure: zip7_nrmse.png, channel panels 5–6 of 11, top to bottom

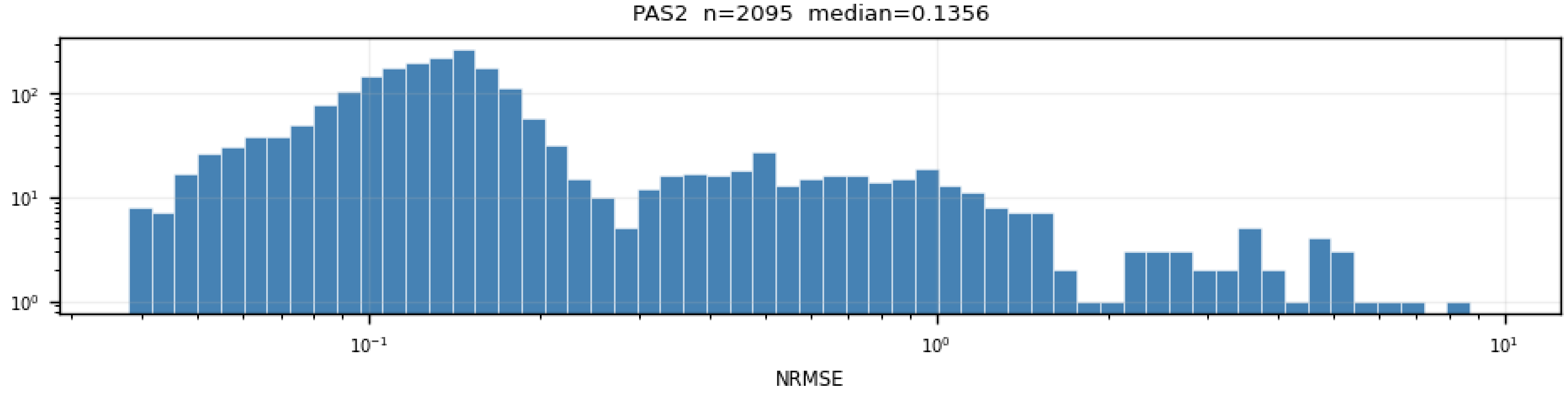
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.

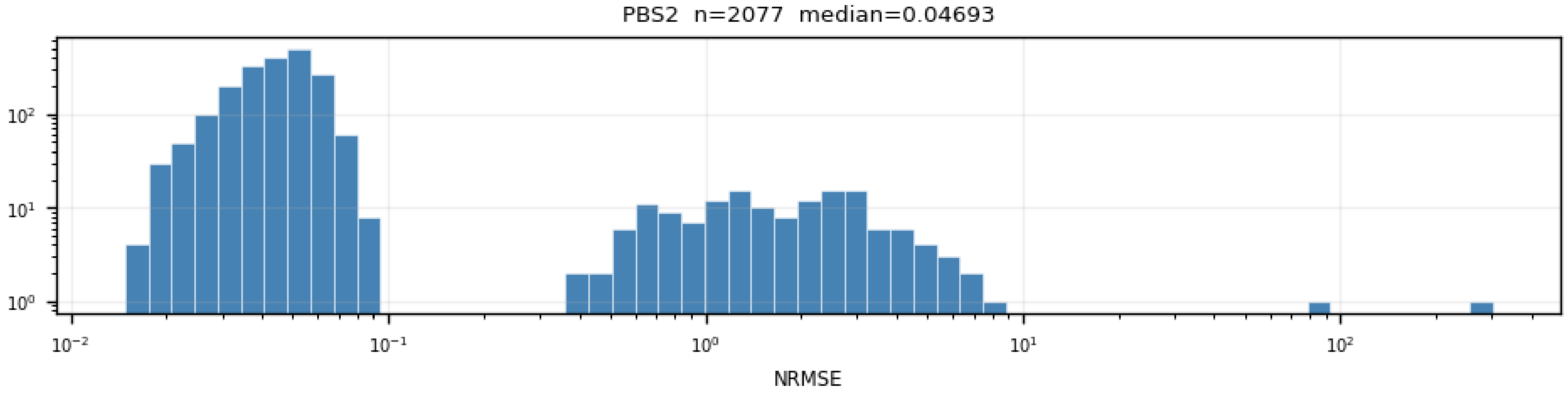


Backup · Z7 · nrmse: NRMSE distribution per channel (4/6)

figure: zip7_nrmse.png, channel panels 7–8 of 11, top to bottom

NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.

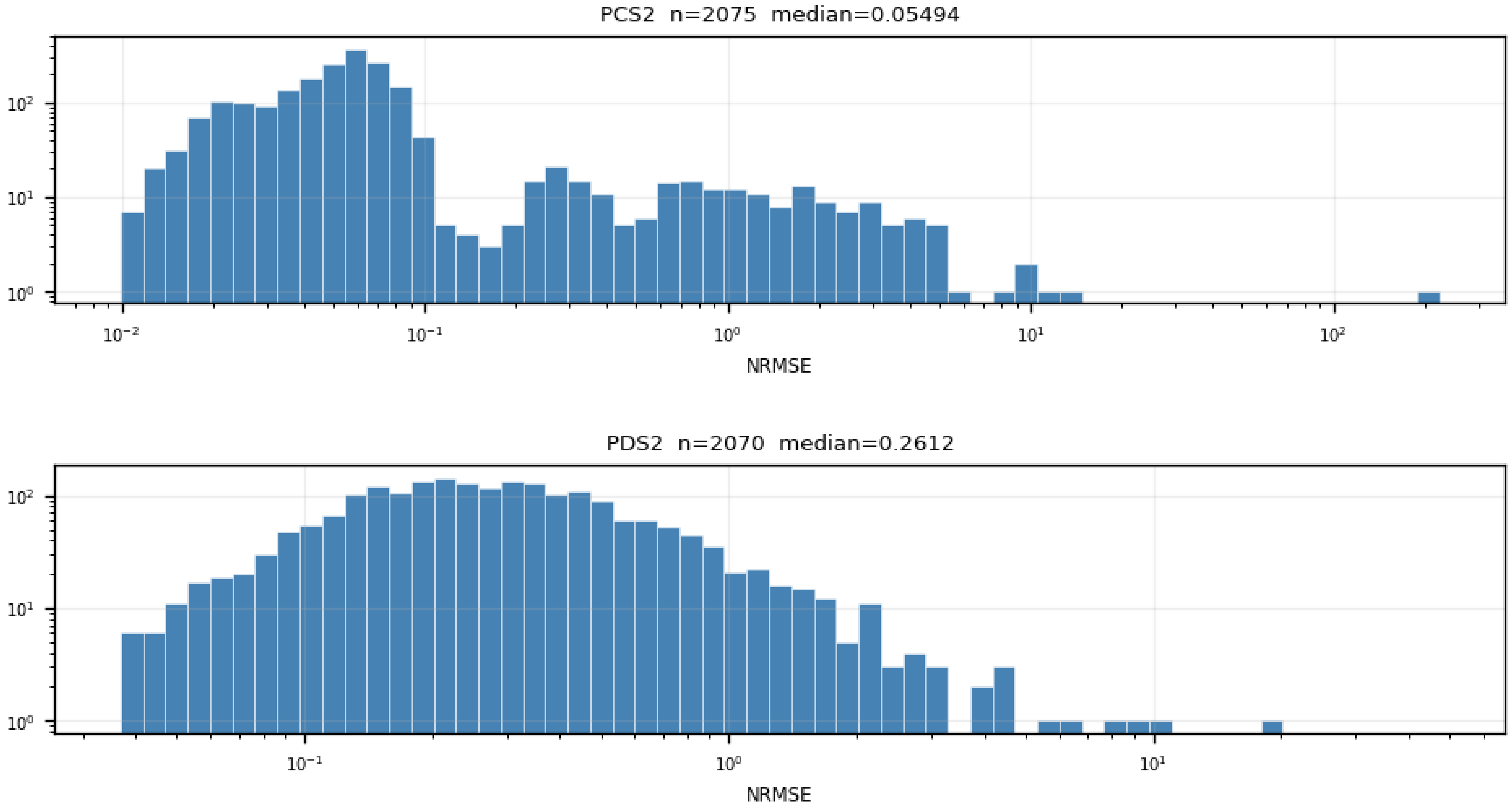




Backup · Z7 · nrmse: NRMSE distribution per channel (5/6)

figure: zip7_nrmse.png, channel panels 9–10 of 11, top to bottom

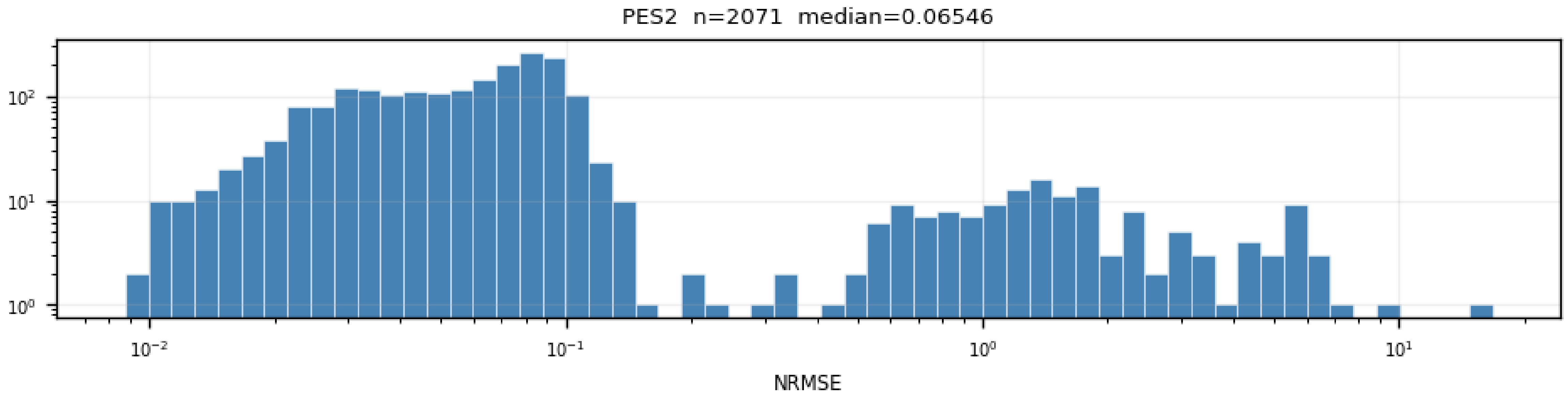
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.



Backup · Z7 · nrmse: NRMSE distribution per channel (6/6)

figure: zip7_nrmse.png, channel panels 11–11 of 11, top to bottom

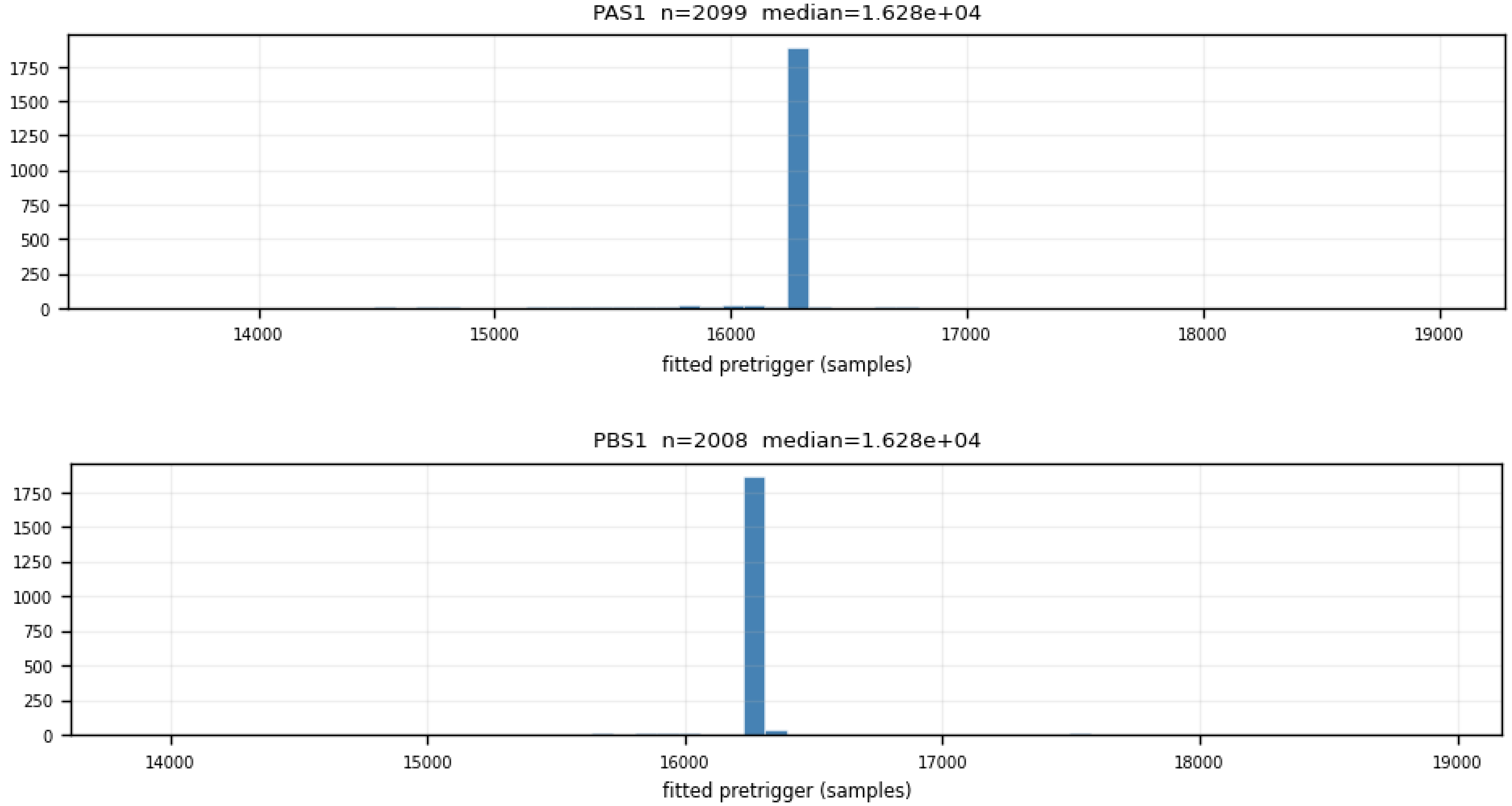
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.



Backup · Z7 · pretrigger: fitted onset per channel (1/6)

figure: zip7_pretrigger.png, channel panels 1–2 of 11, top to bottom

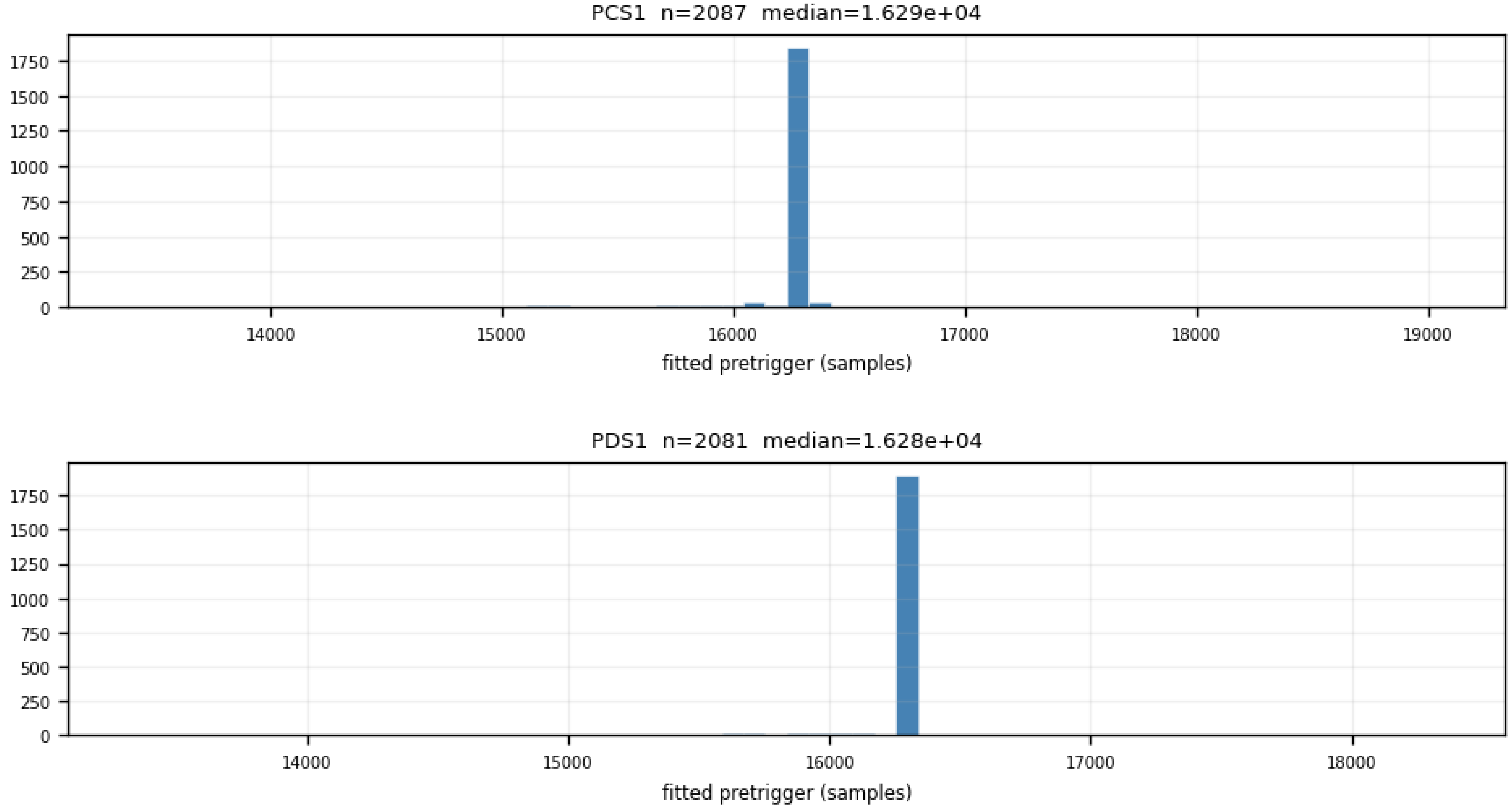
The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (2/6)

figure: zip7_pretrigger.png, channel panels 3–4 of 11, top to bottom

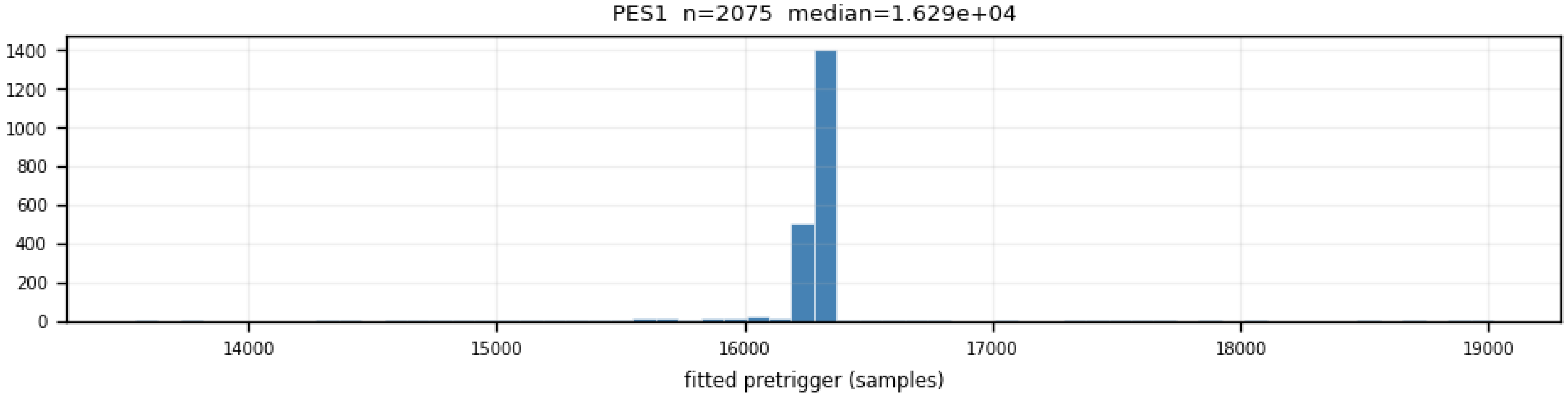
The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.

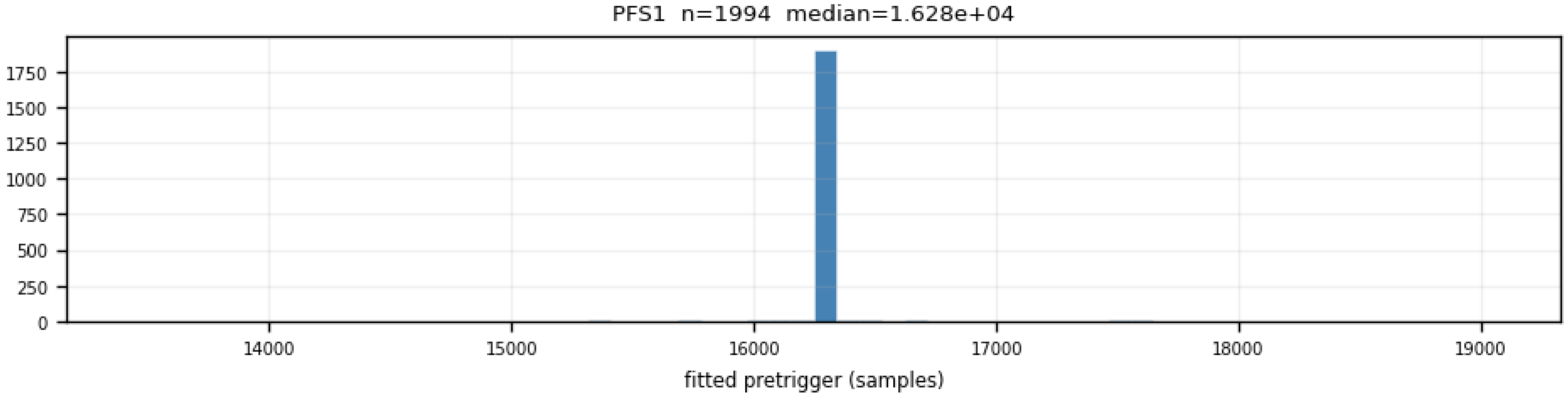


Backup · Z7 · pretrigger: fitted onset per channel (3/6)

figure: zip7_pretrigger.png, channel panels 5–6 of 11, top to bottom

The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.

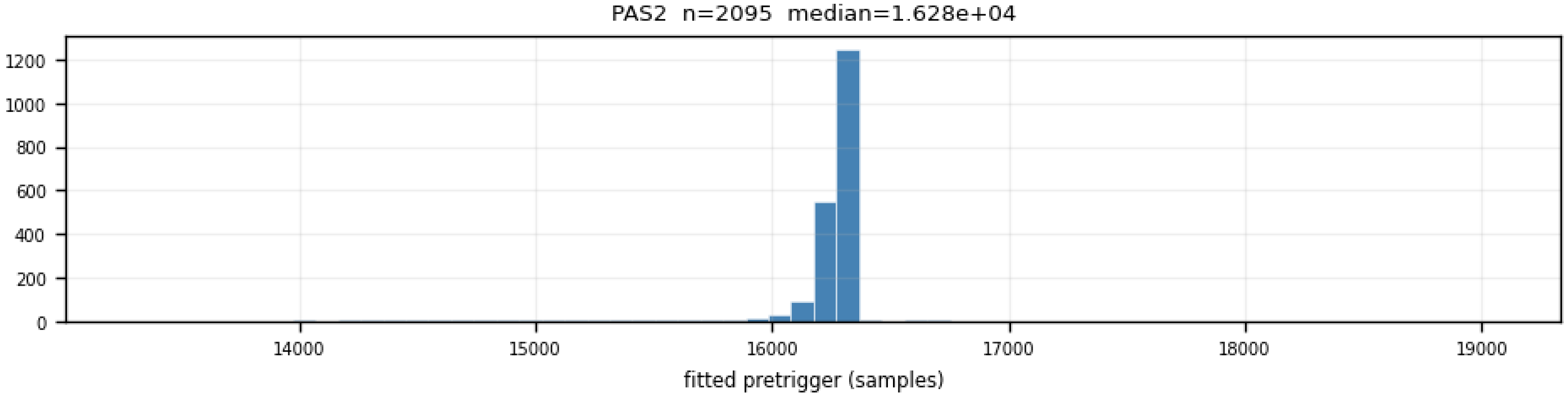


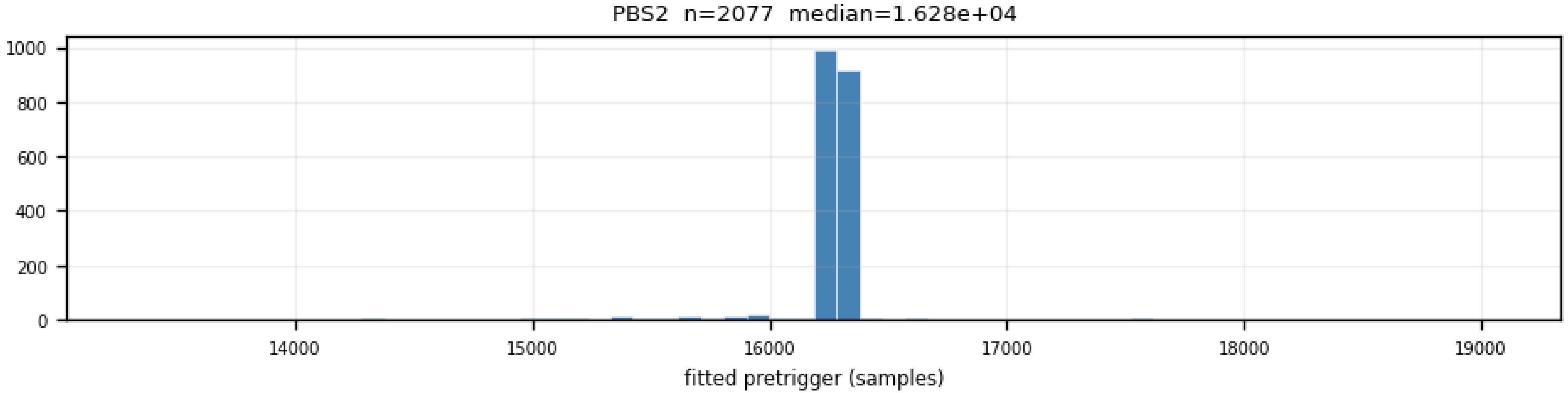


Backup · Z7 · pretrigger: fitted onset per channel (4/6)

figure: zip7_pretrigger.png, channel panels 7–8 of 11, top to bottom

The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.

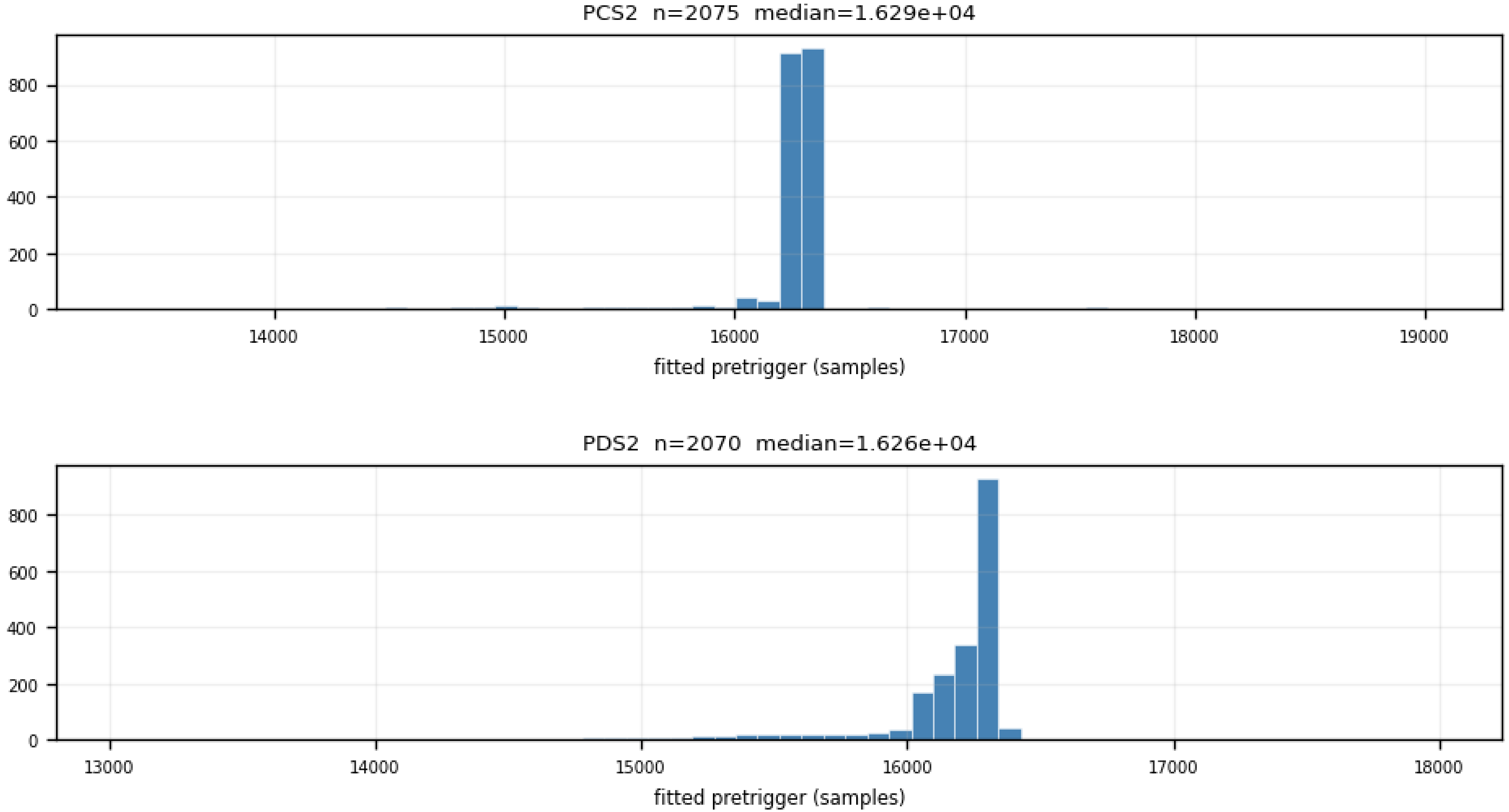




Backup · Z7 · pretrigger: fitted onset per channel (5/6)

figure: zip7_pretrigger.png, channel panels 9–10 of 11, top to bottom

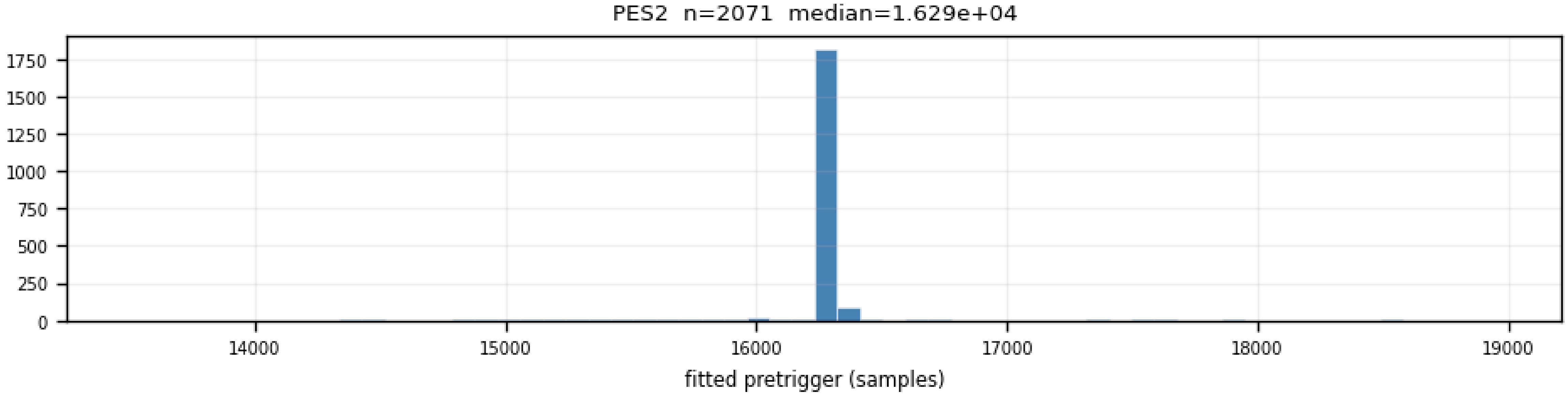
The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (6/6)

figure: zip7_pretrigger.png, channel panels 11–11 of 11, top to bottom

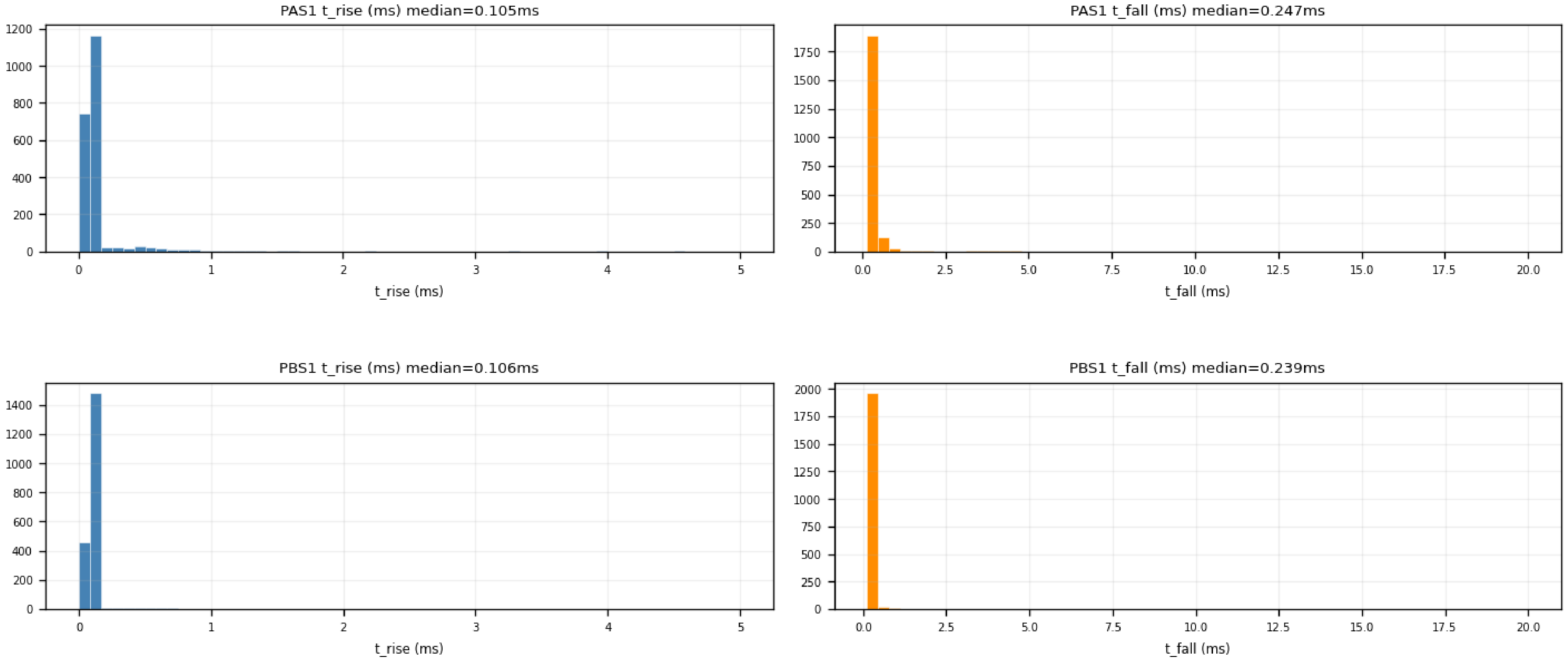
The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · time_constants: τ_rise / τ_fall per channel (1/6)

figure: zip7_time_constants.png, channel panels 1–2 of 11, top to bottom

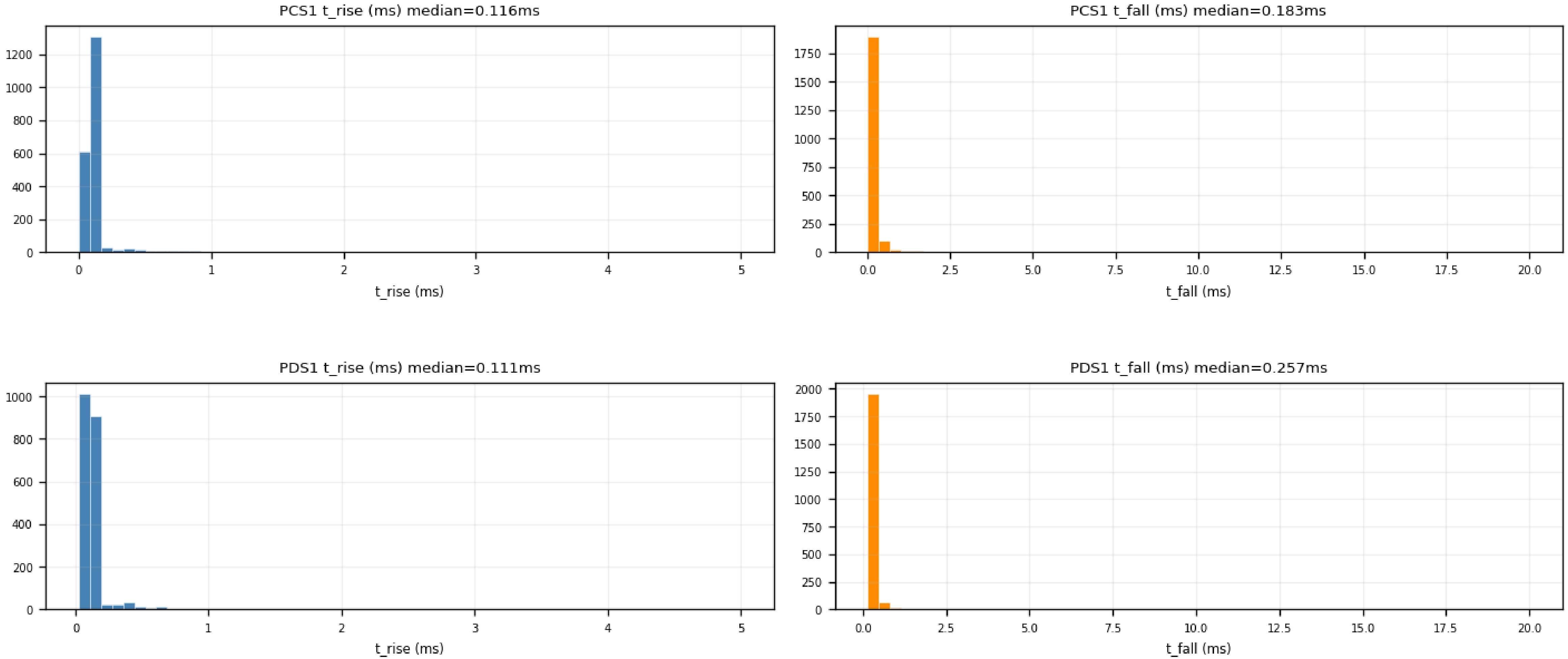
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_rise / τ_fall per channel (2/6)

figure: zip7_time_constants.png, channel panels 3–4 of 11, top to bottom

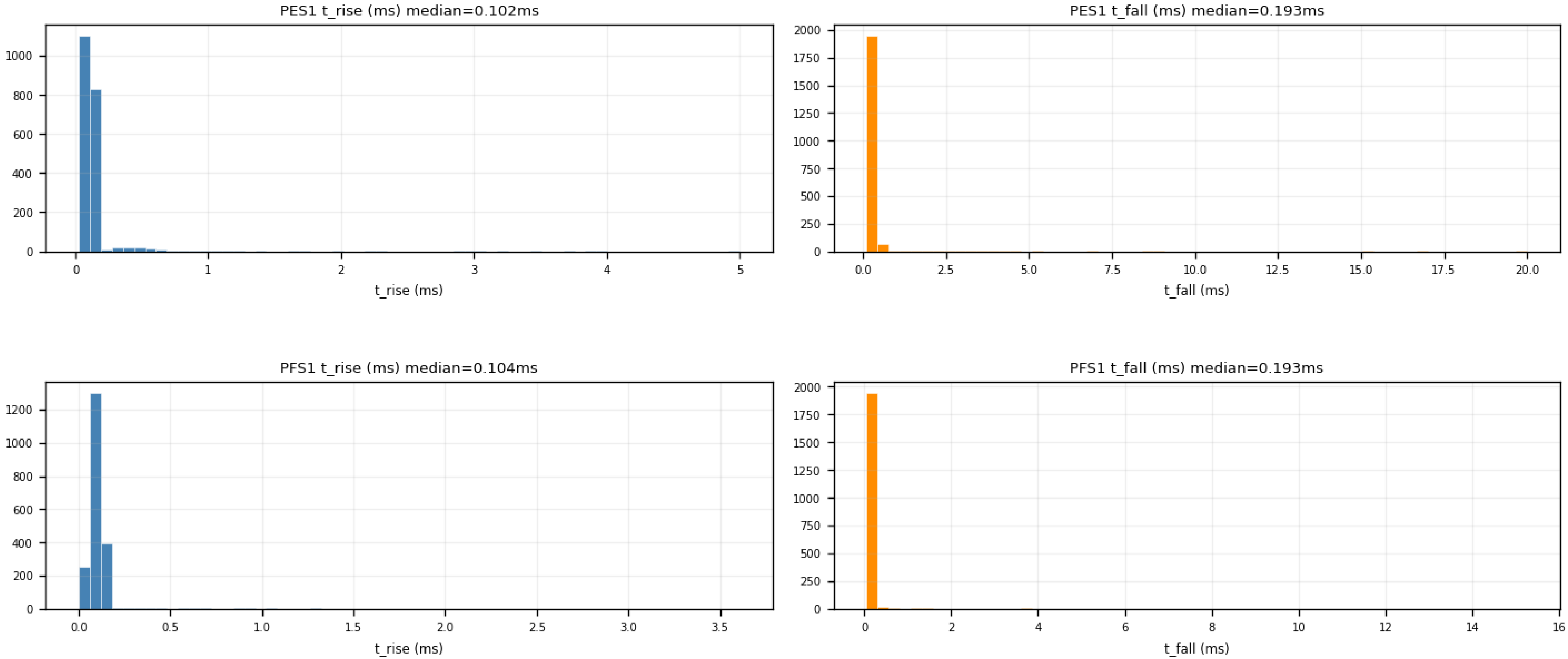
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_rise / τ_fall per channel (3/6)

figure: zip7_time_constants.png, channel panels 5–6 of 11, top to bottom

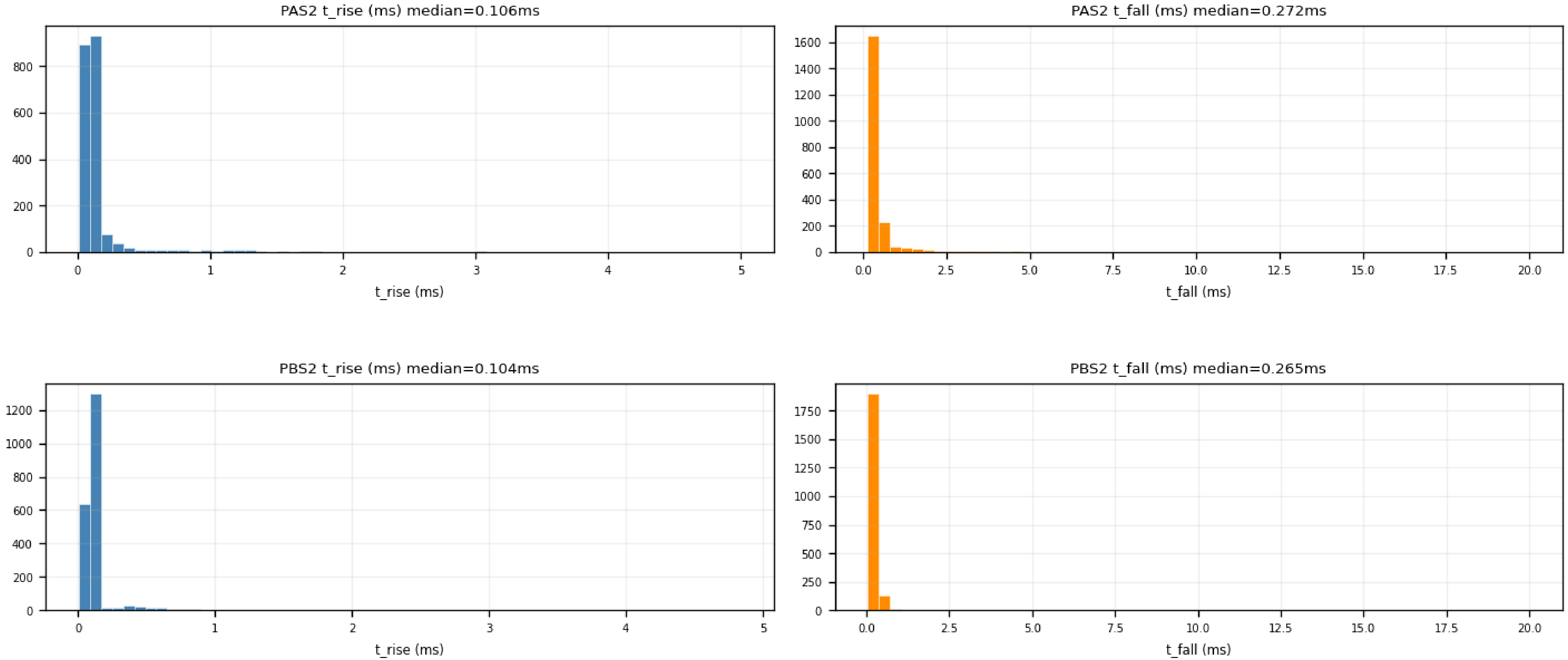
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_rise / τ_fall per channel (4/6)

figure: zip7_time_constants.png, channel panels 7–8 of 11, top to bottom

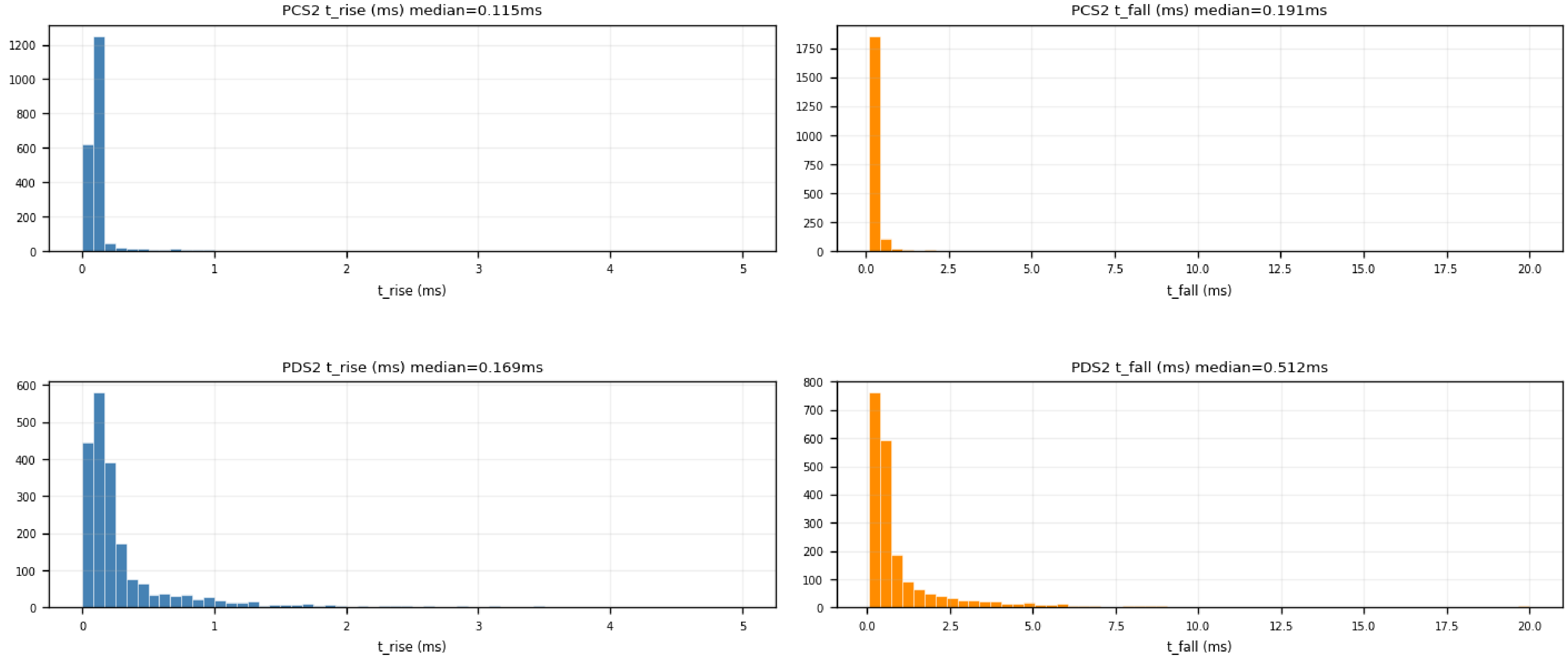
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_rise / τ_fall per channel (5/6)

figure: zip7_time_constants.png, channel panels 9–10 of 11, top to bottom

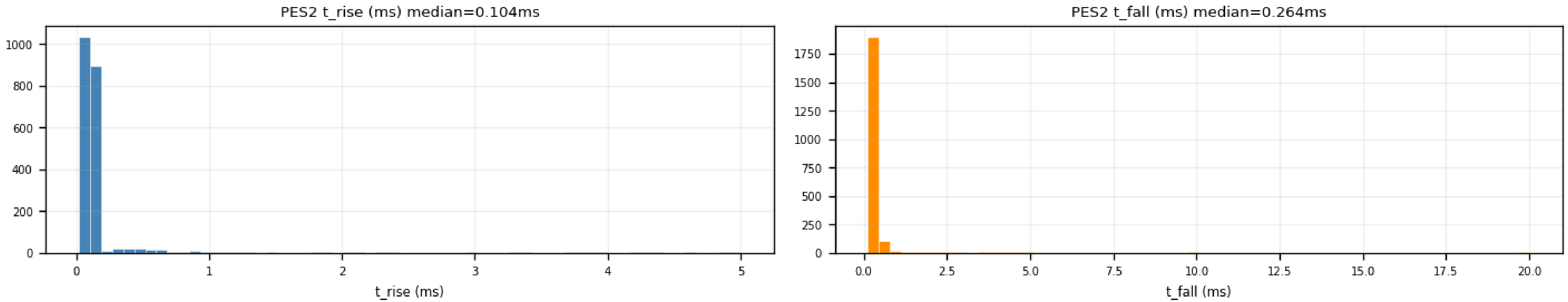
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_rise / τ_fall per channel (6/6)

figure: zip7_time_constants.png, channel panels 11–11 of 11, top to bottom

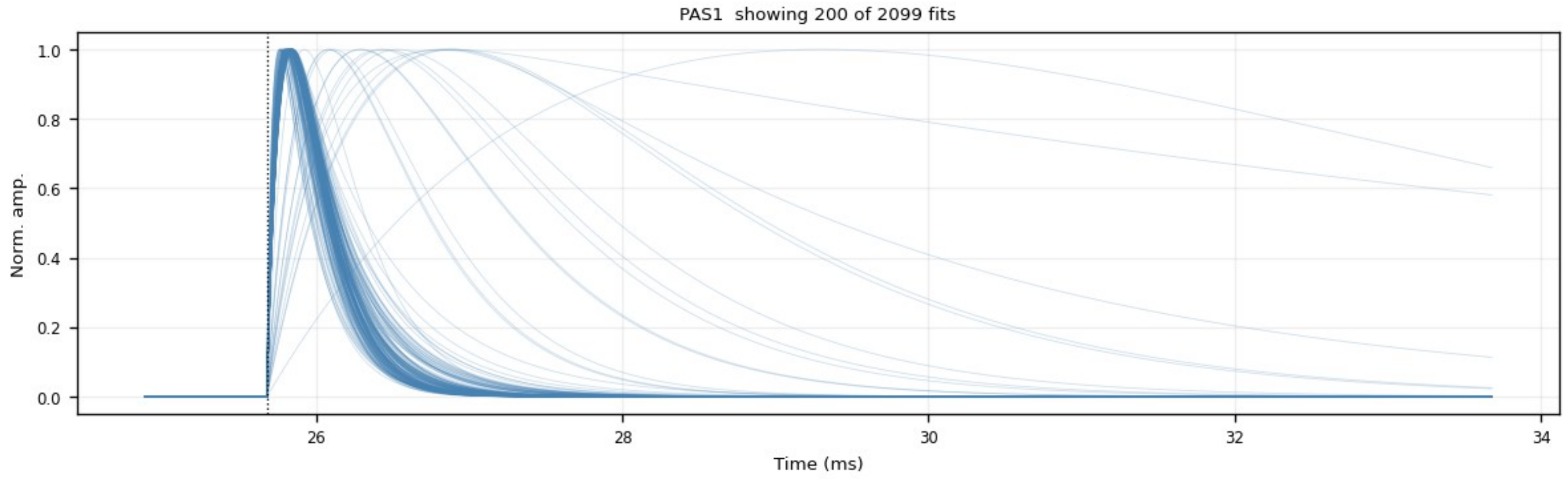
Fast-pulse population at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.

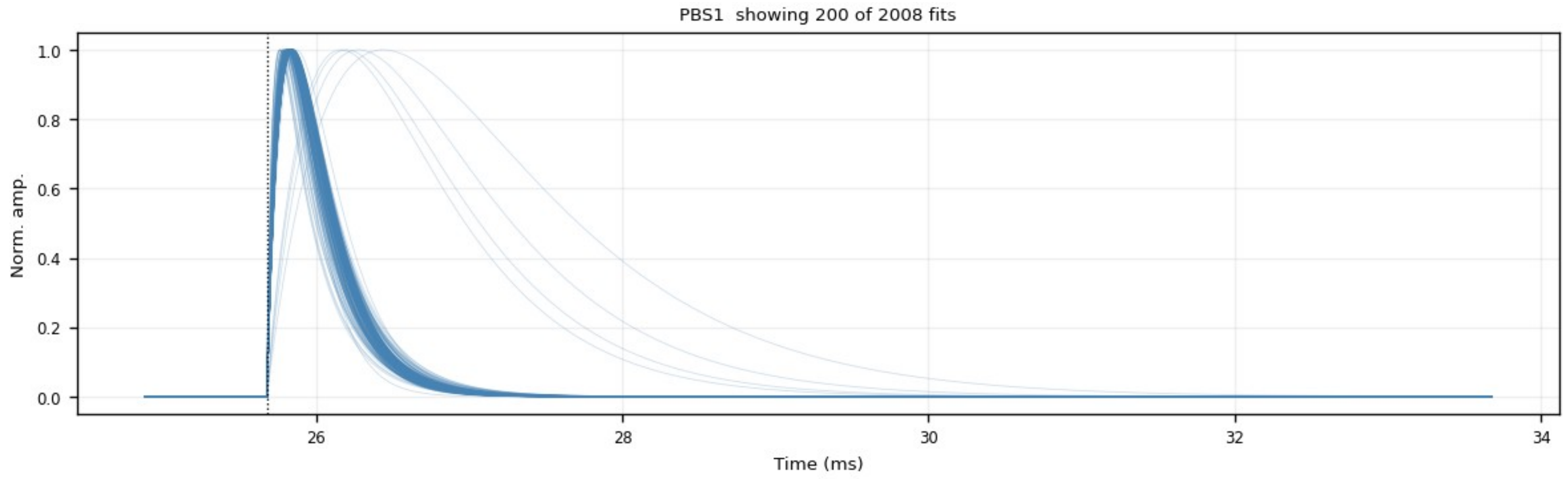


Backup · Z7 · fitted_curves_overlay: fan, no cut (1/6)

figure: zip7_fitted_curves_overlay.png, channel panels 1–2 of 11, top to bottom

Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

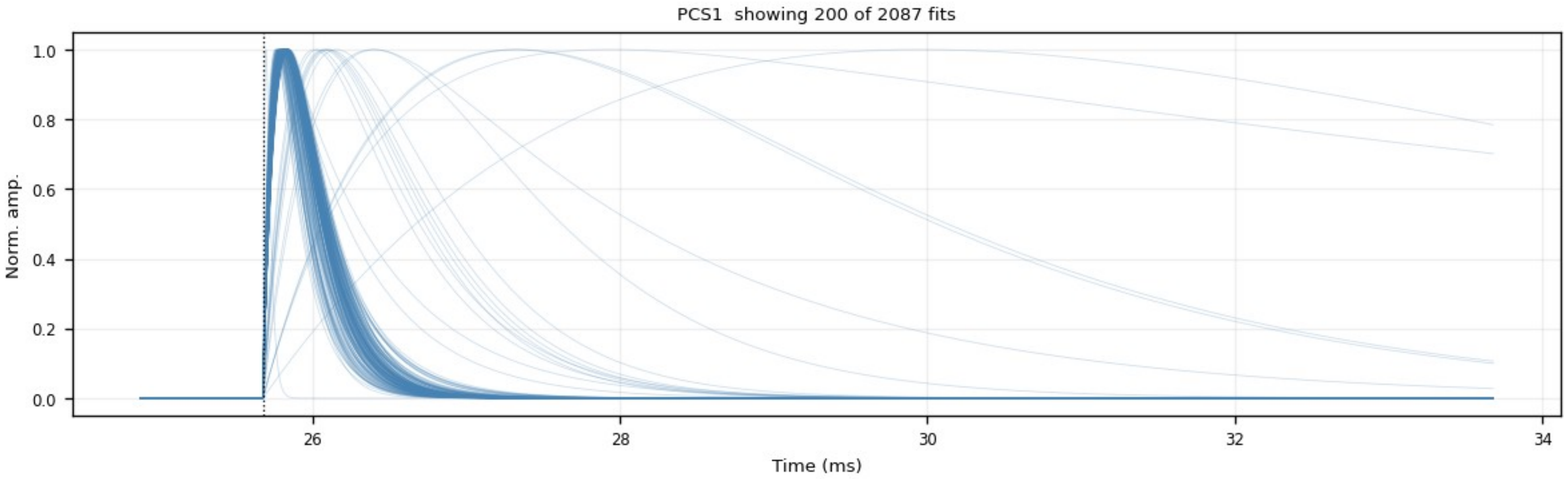


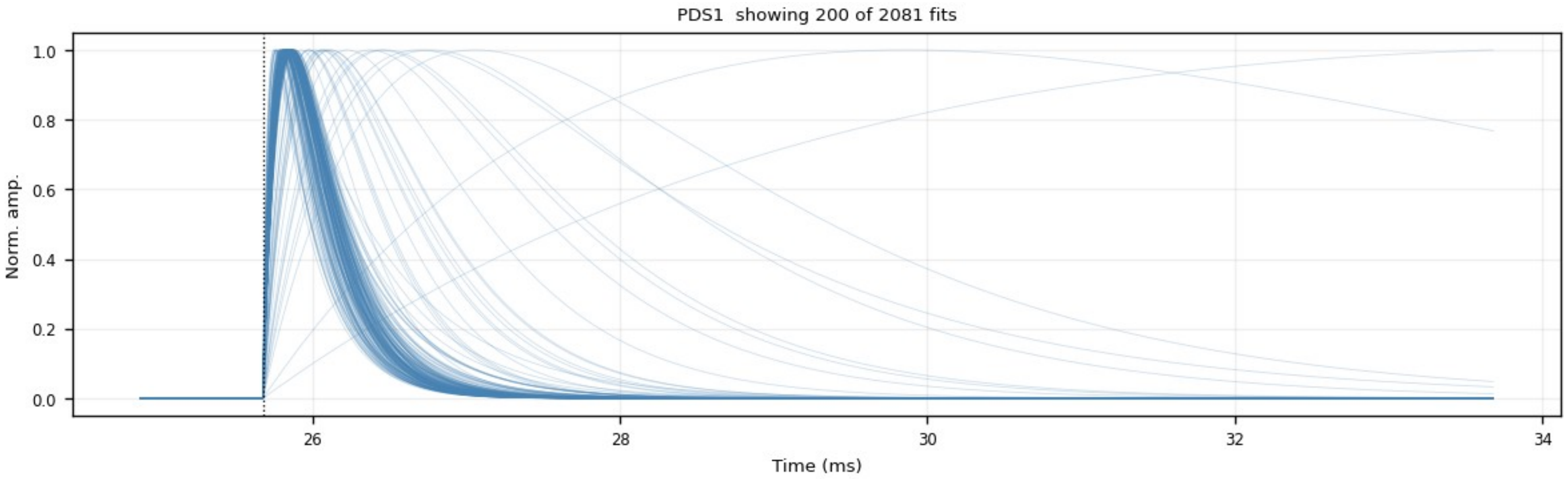


Backup · Z7 · fitted_curves_overlay: fan, no cut (2/6)

figure: zip7_fitted_curves_overlay.png, channel panels 3–4 of 11, top to bottom

Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

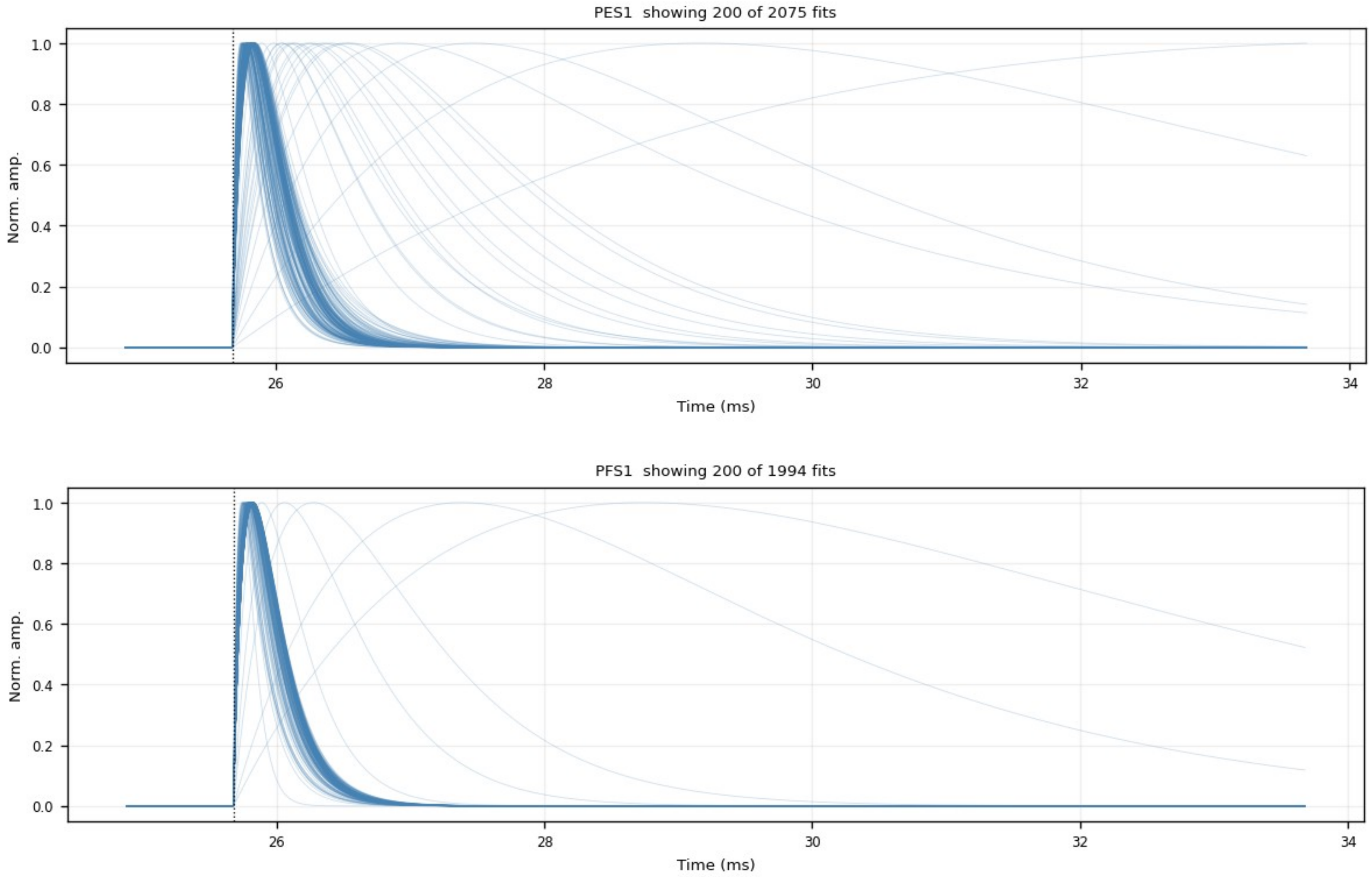




Backup · Z7 · fitted_curves_overlay: fan, no cut (3/6)

figure: zip7_fitted_curves_overlay.png, channel panels 5–6 of 11, top to bottom

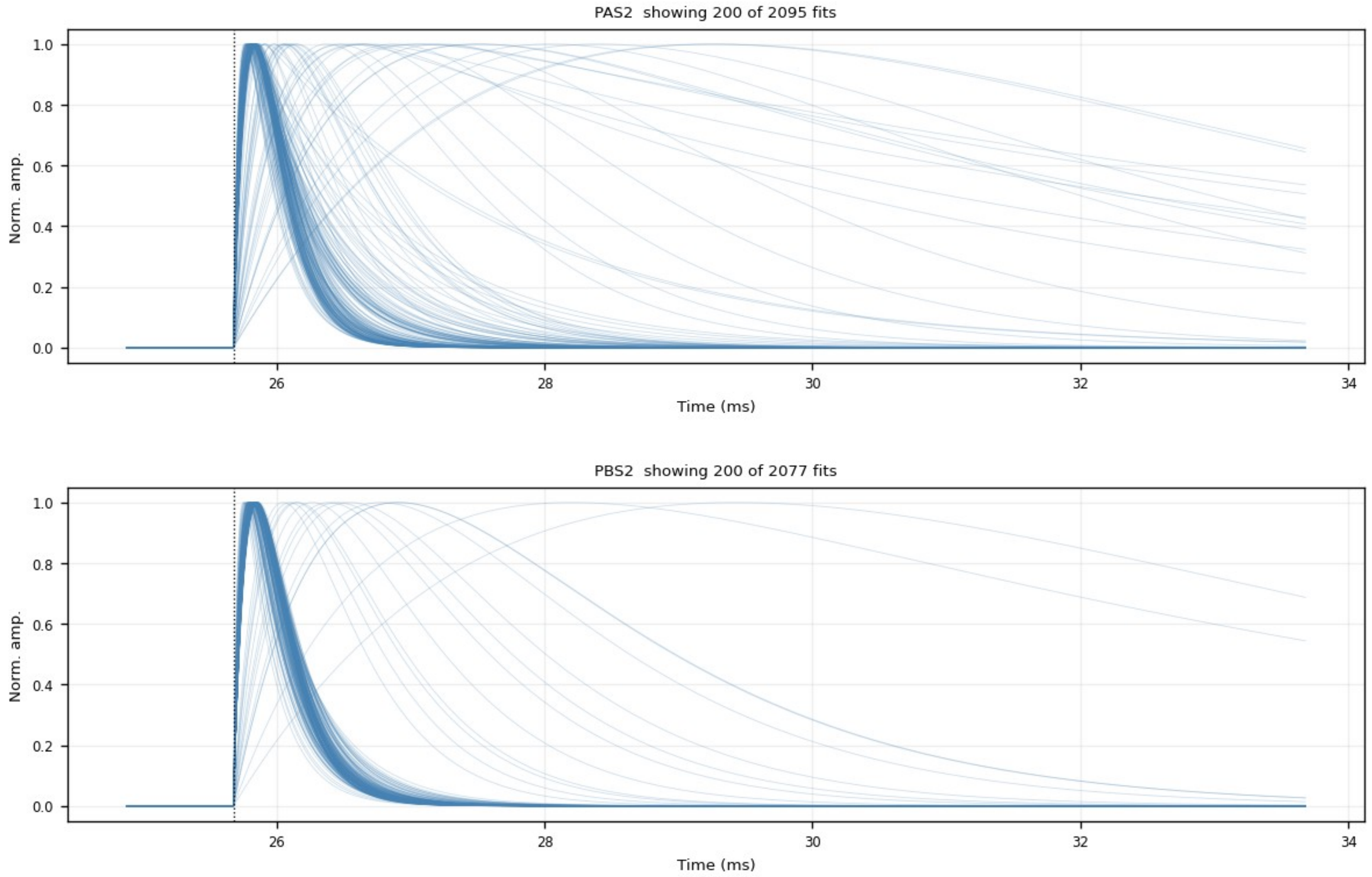
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: fan, no cut (4/6)

figure: zip7_fitted_curves_overlay.png, channel panels 7–8 of 11, top to bottom

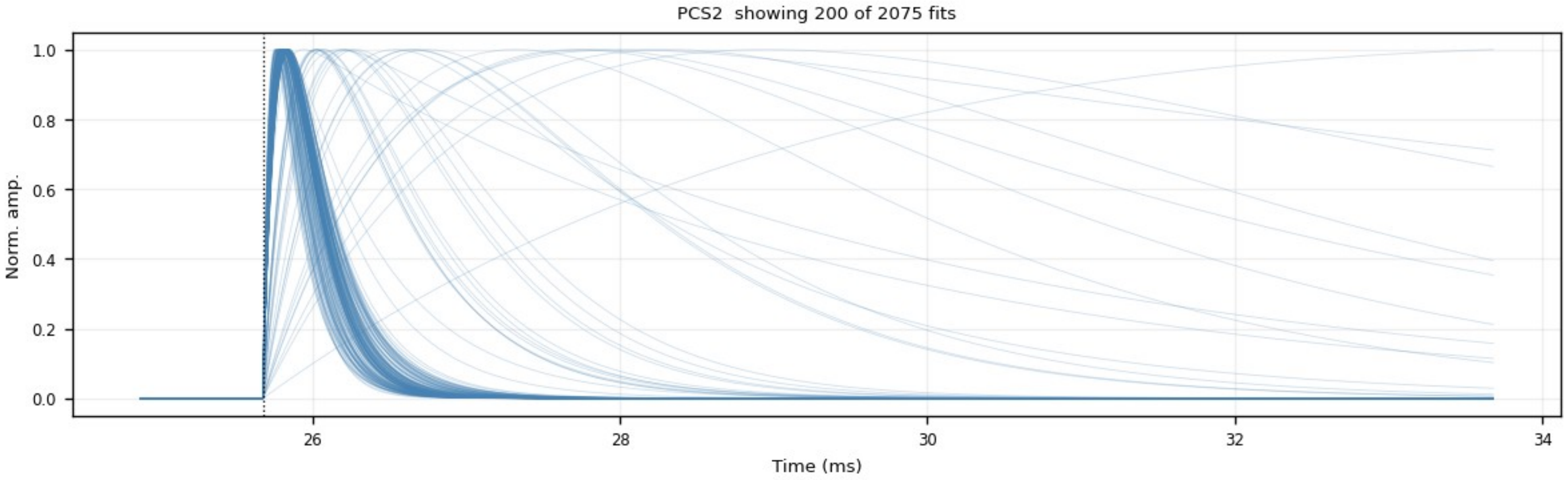
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

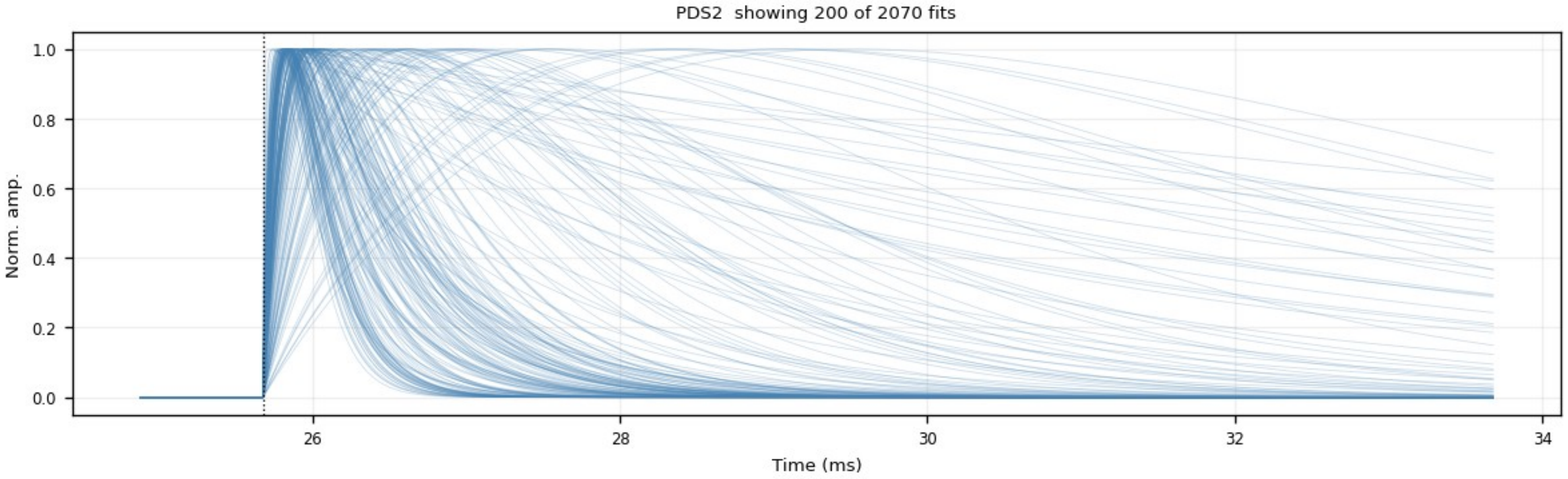


Backup · Z7 · fitted_curves_overlay: fan, no cut (5/6)

figure: zip7_fitted_curves_overlay.png, channel panels 9–10 of 11, top to bottom

Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

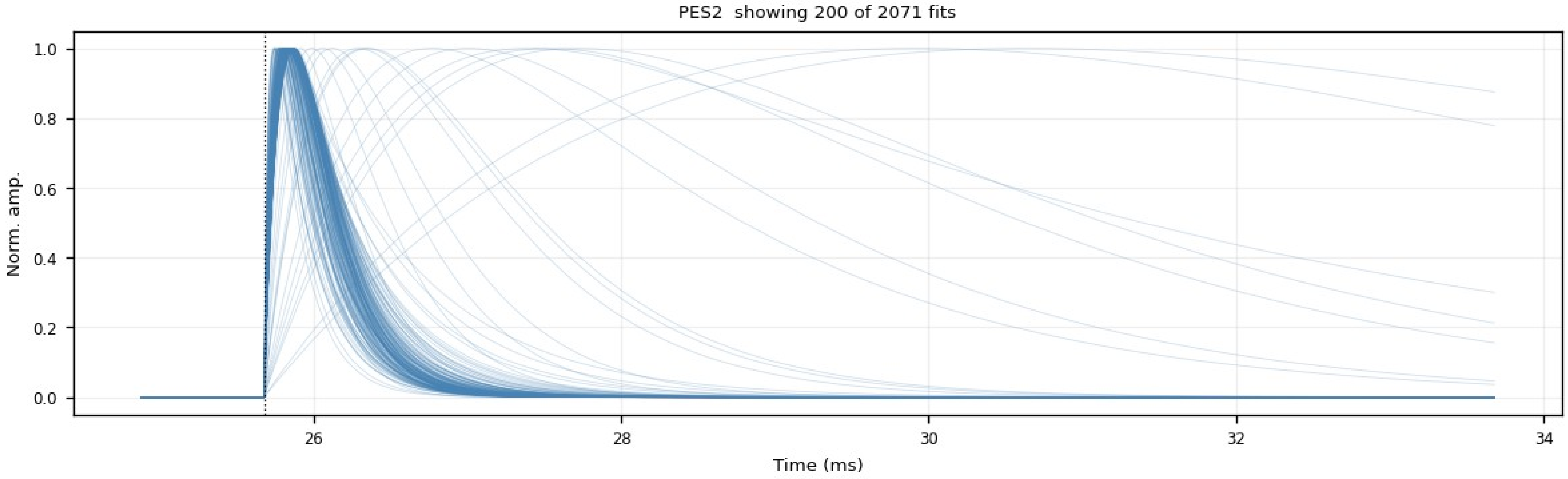




Backup · Z7 · fitted_curves_overlay: fan, no cut (6/6)

figure: zip7_fitted_curves_overlay.png, channel panels 11–11 of 11, top to bottom

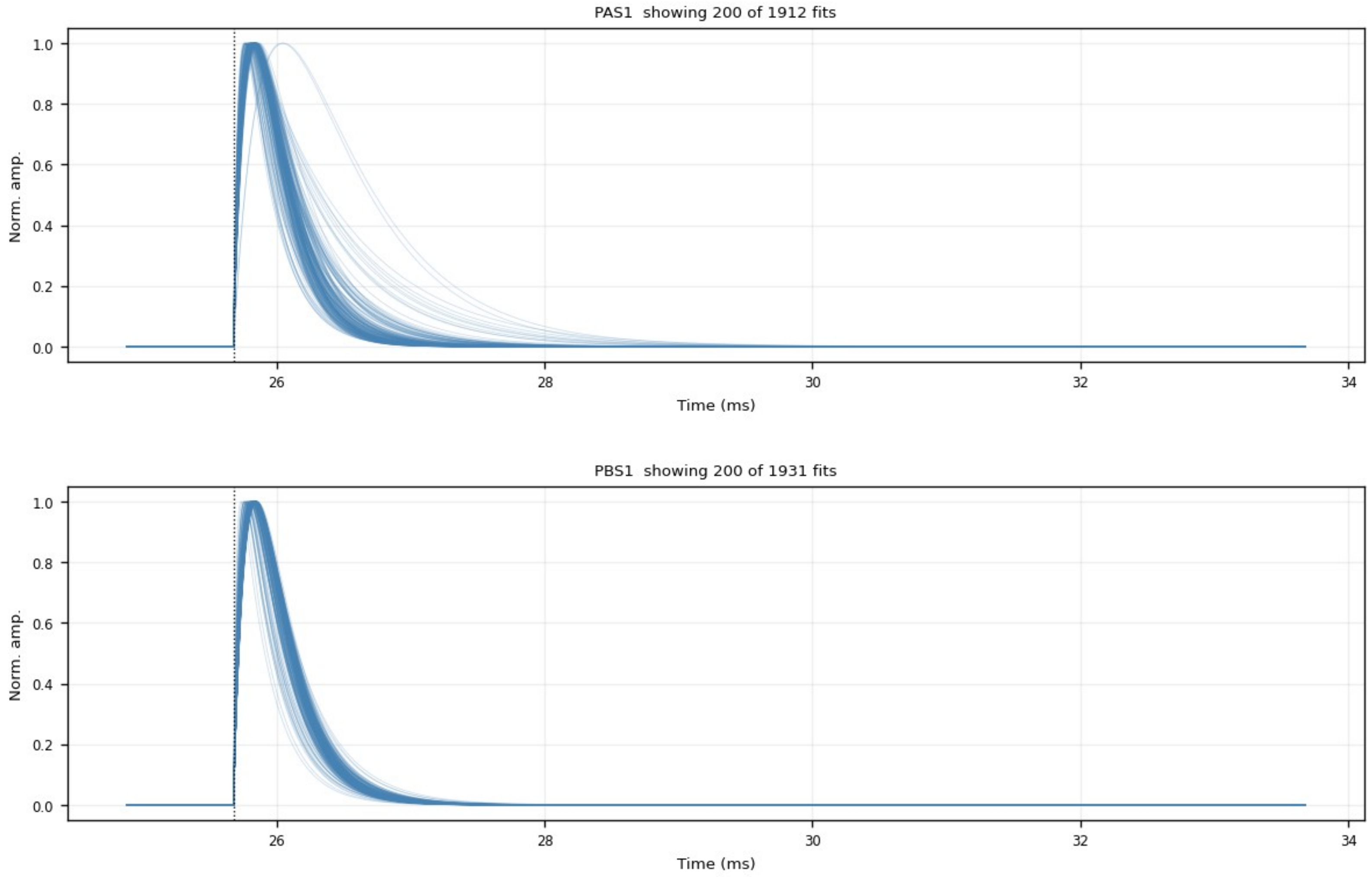
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: after NRMSE ≤ 0.4 (1/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

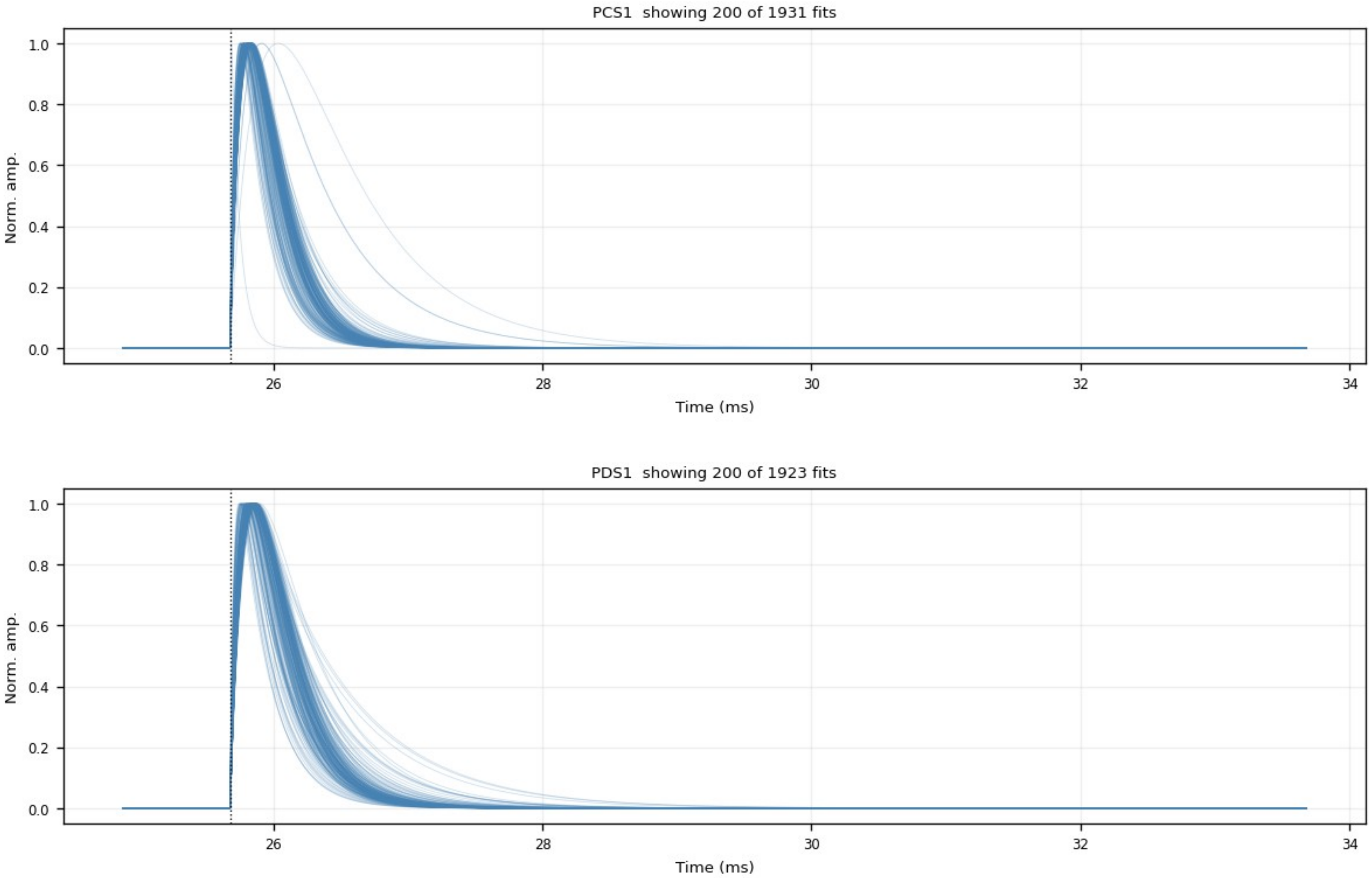
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after NRMSE ≤ 0.4 (2/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

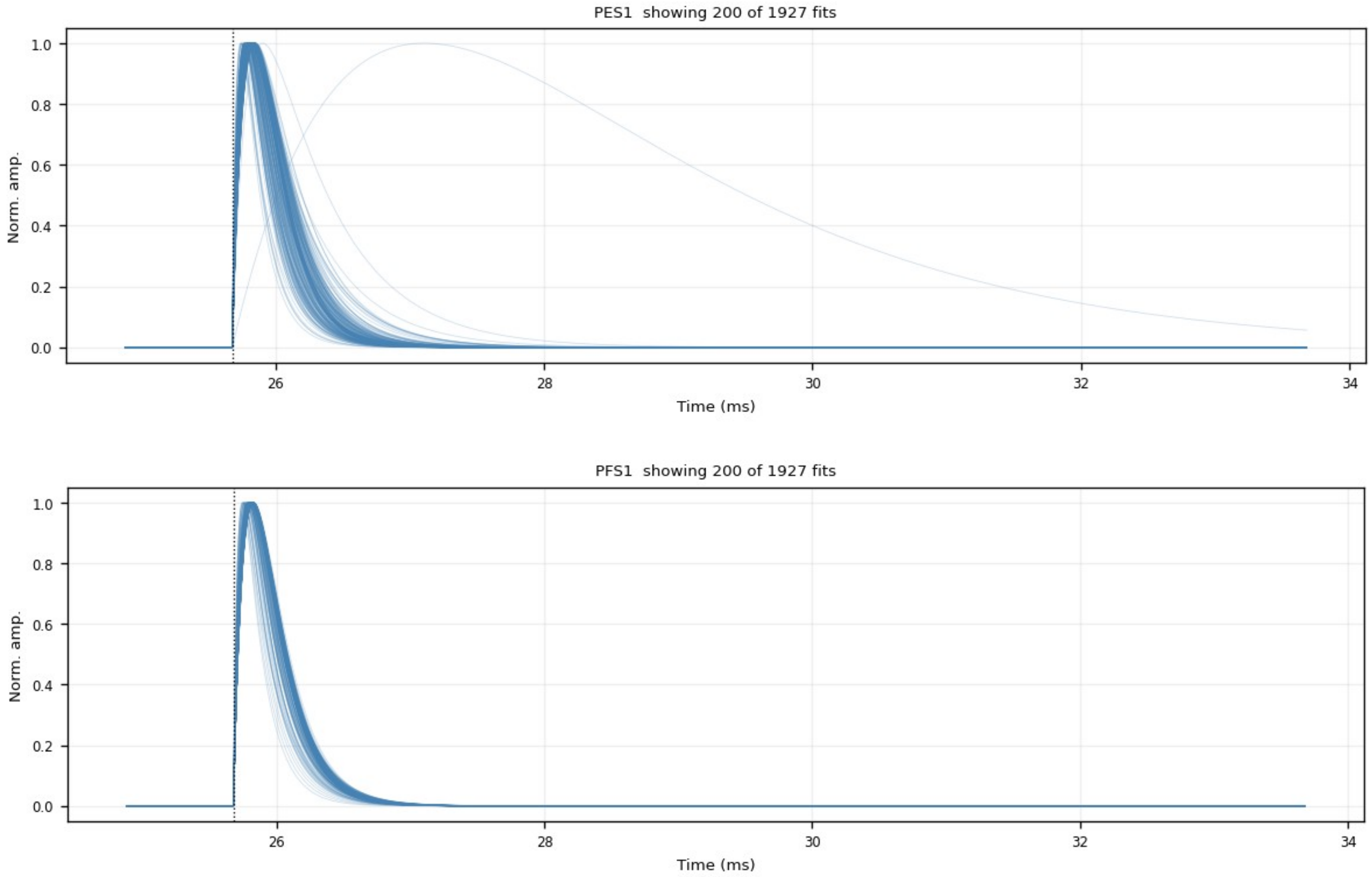
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after NRMSE ≤ 0.4 (3/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

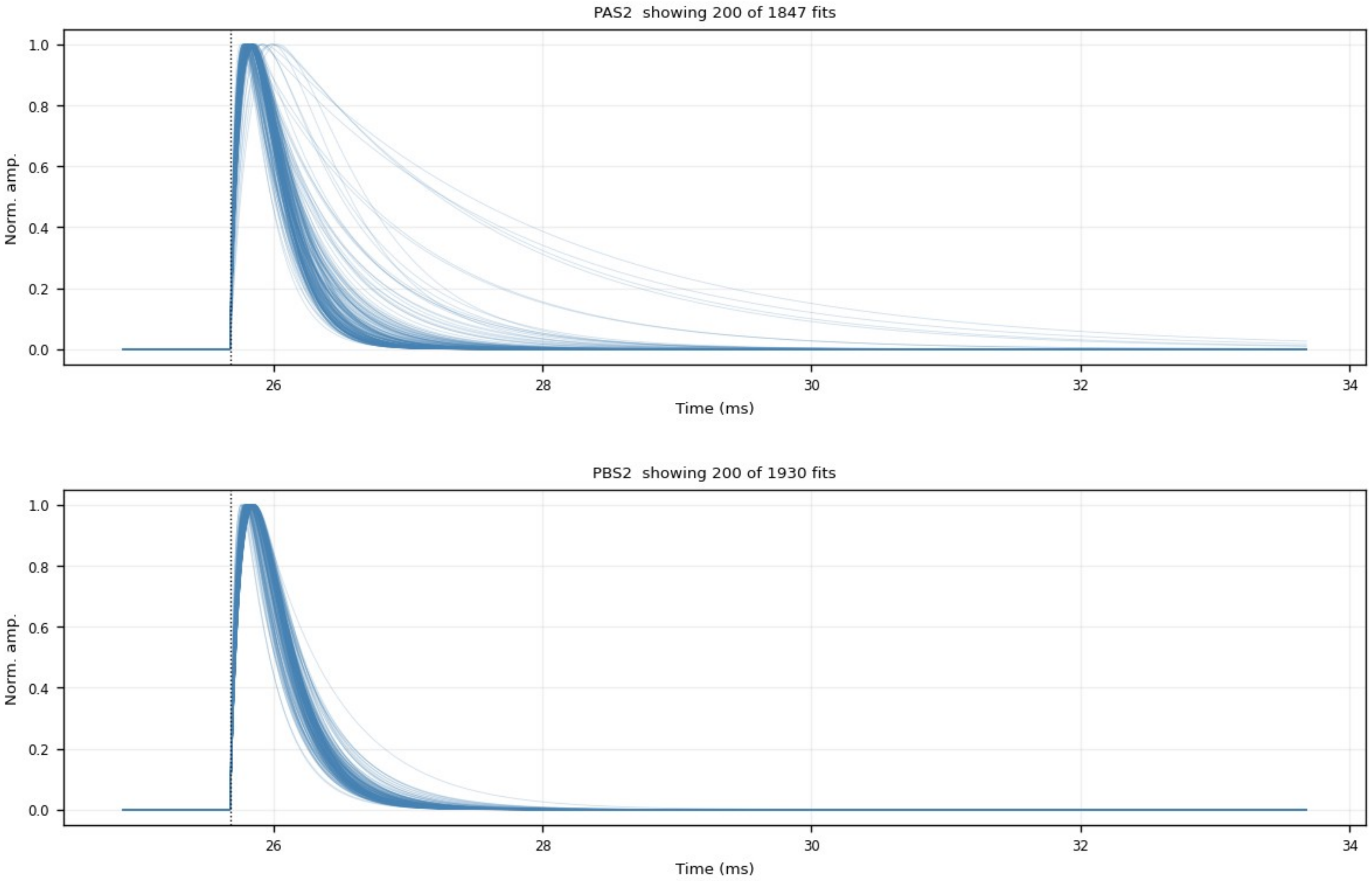
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after NRMSE ≤ 0.4 (4/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

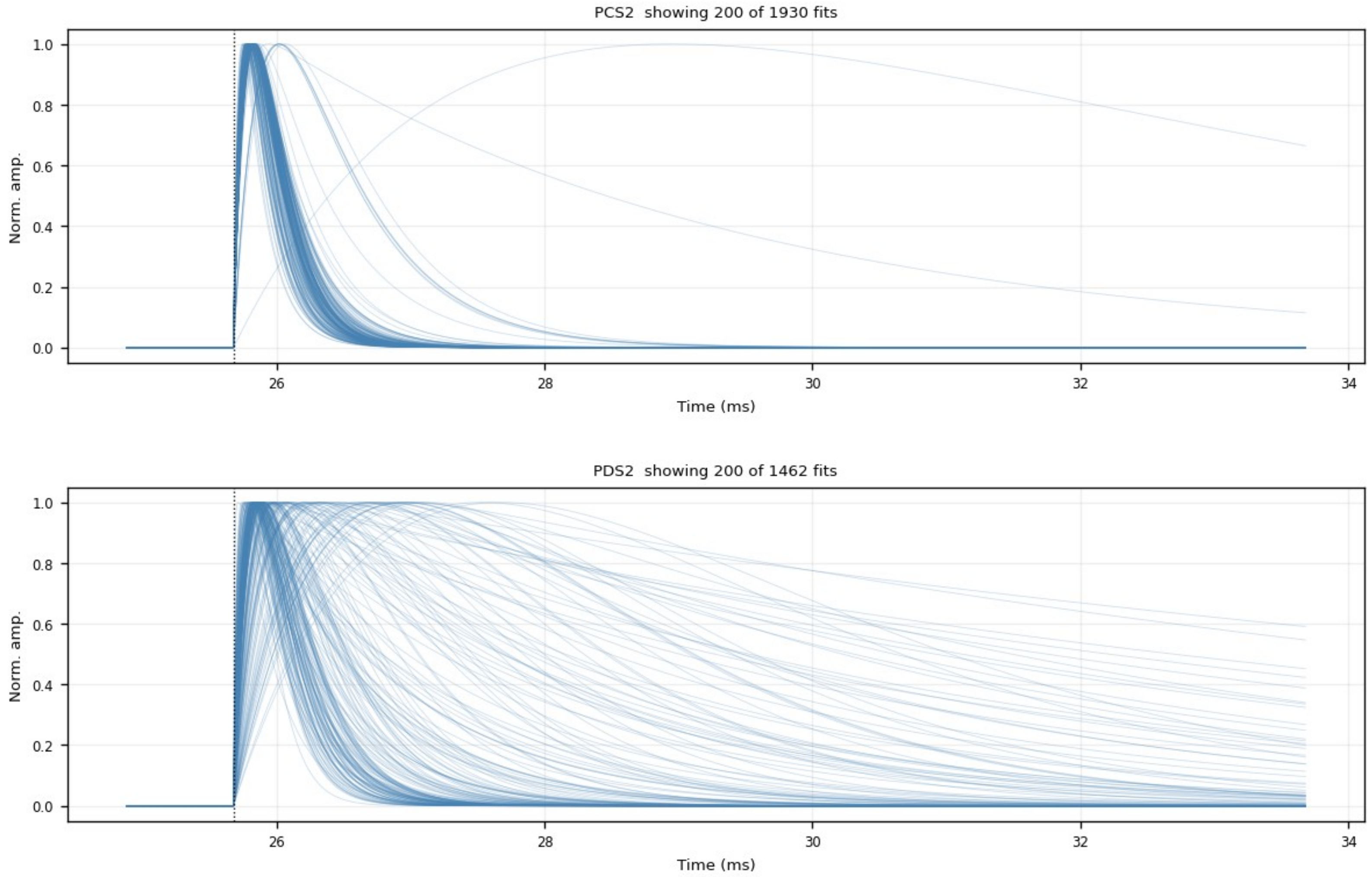
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after NRMSE ≤ 0.4 (5/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

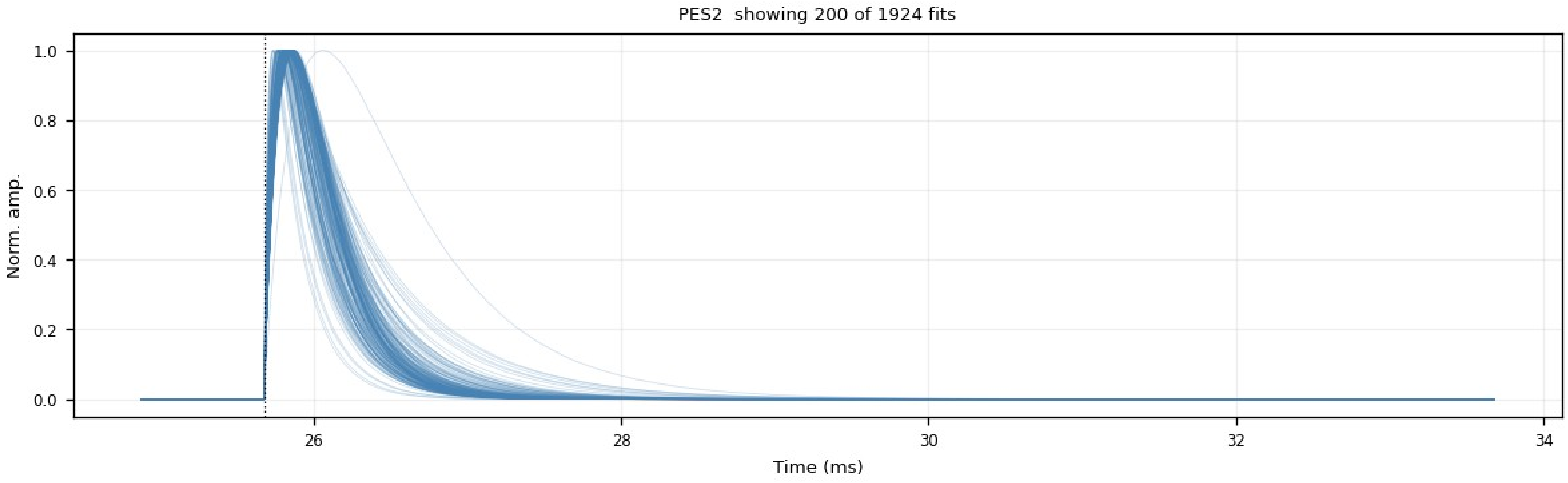
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after NRMSE ≤ 0.4 (6/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

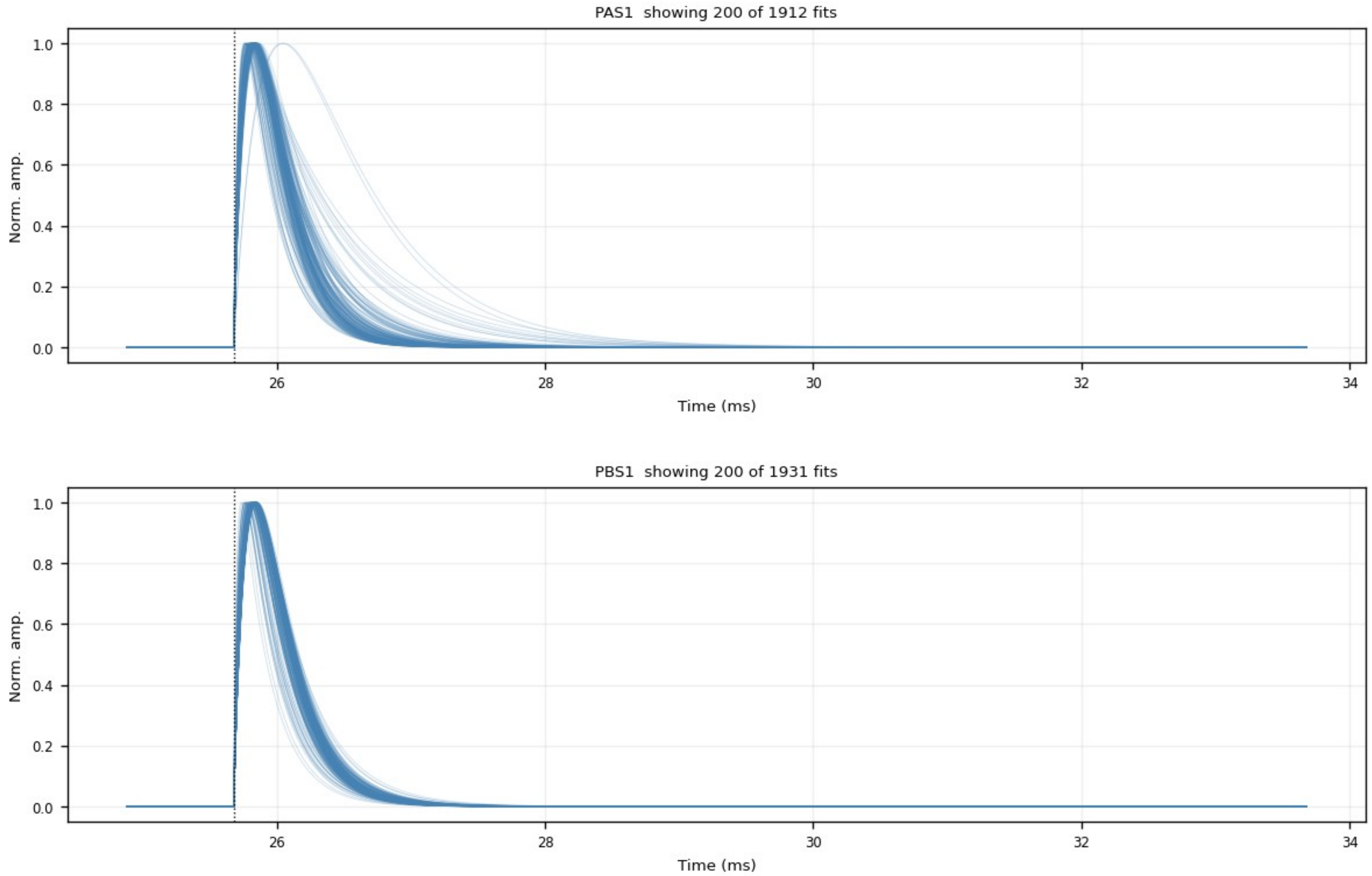
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after both cuts (1/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 1–2 of 11, top to bottom

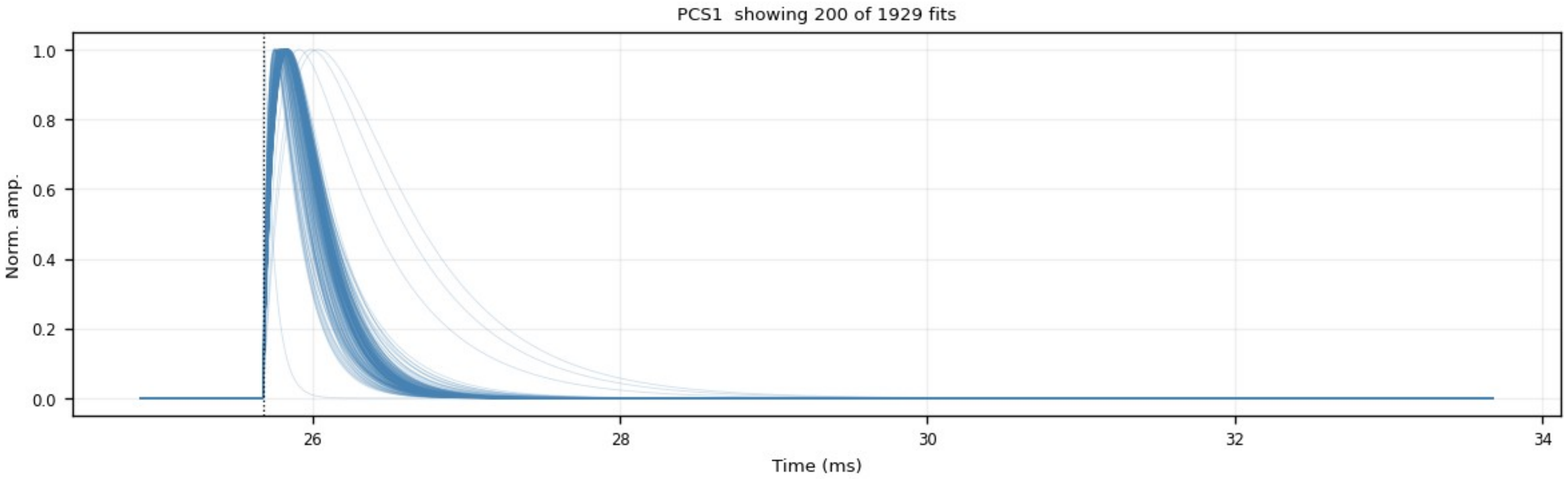
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms, the exact PCA input population: the clean fast-pulse bundle only.

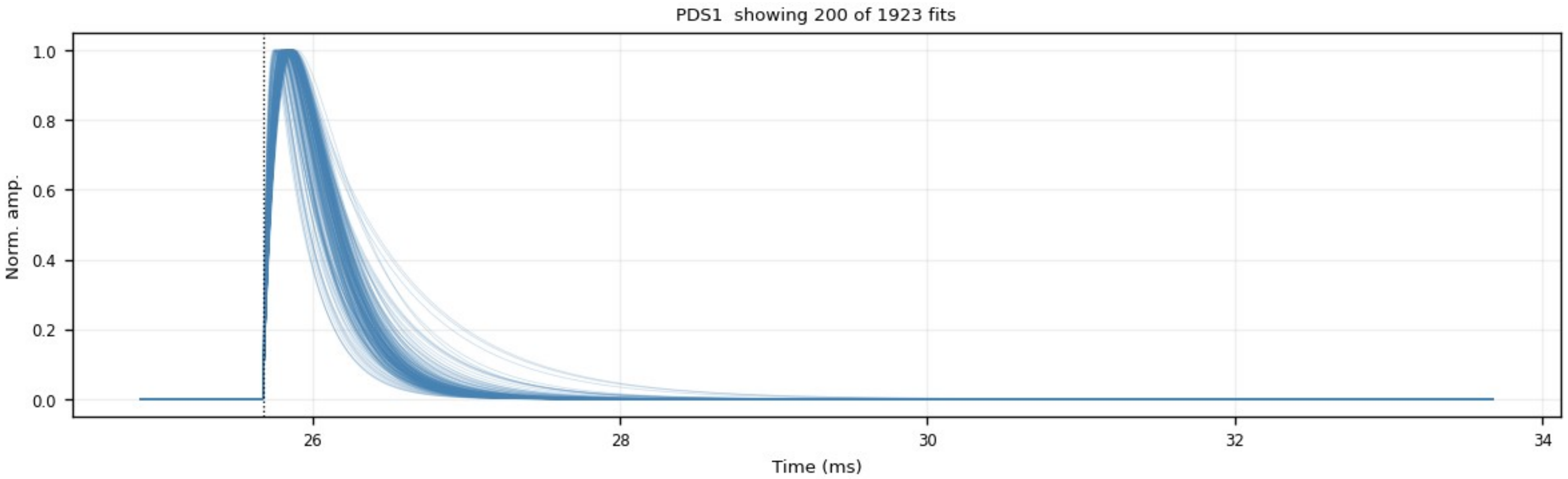


Backup · Z7 · fitted_curves_overlay: after both cuts (2/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 3–4 of 11, top to bottom

After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms, the exact PCA input population: the clean fast-pulse bundle only.

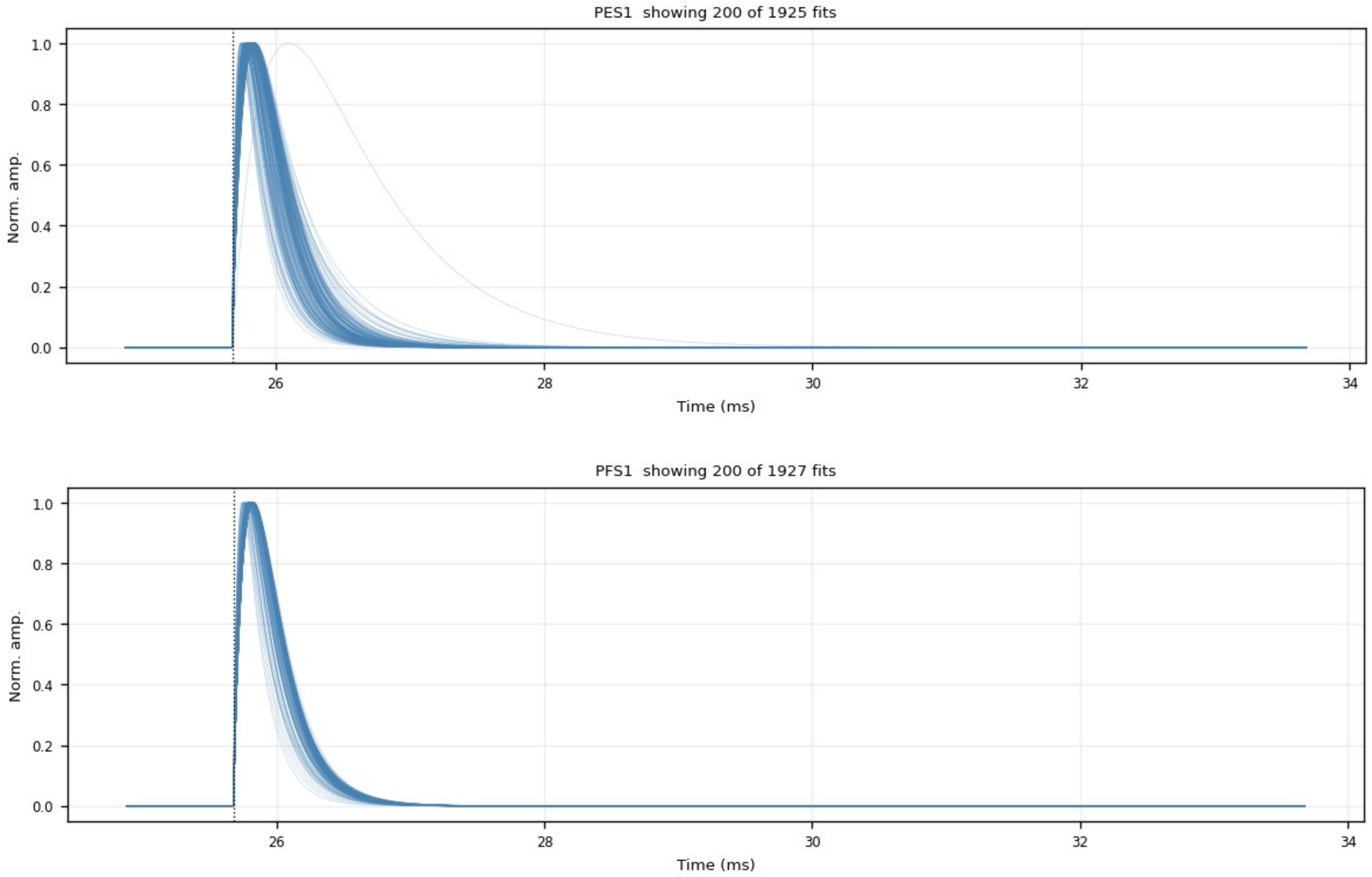




Backup · Z7 · fitted_curves_overlay: after both cuts (3/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 5–6 of 11, top to bottom

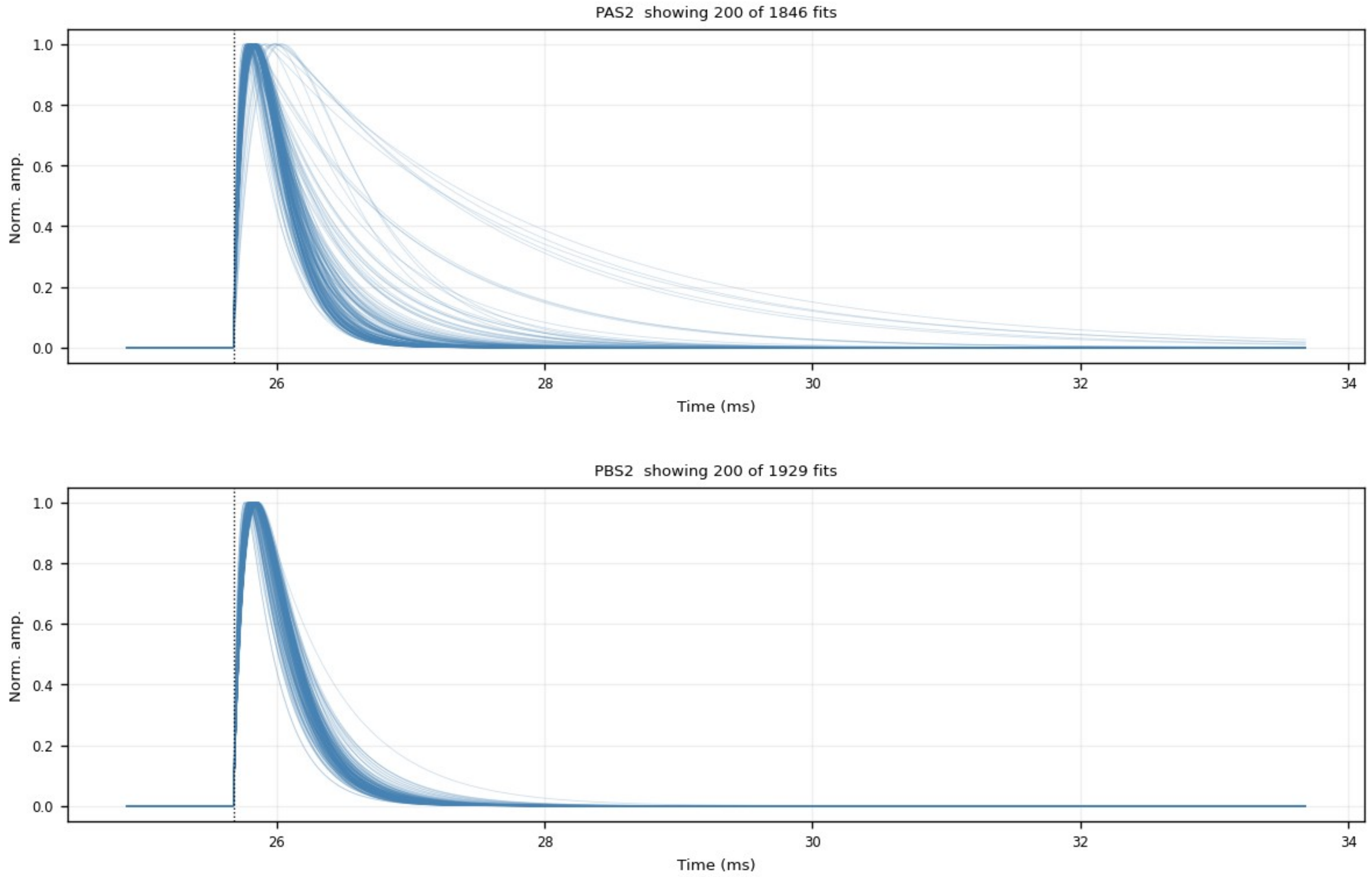
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · fitted_curves_overlay: after both cuts (4/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 7–8 of 11, top to bottom

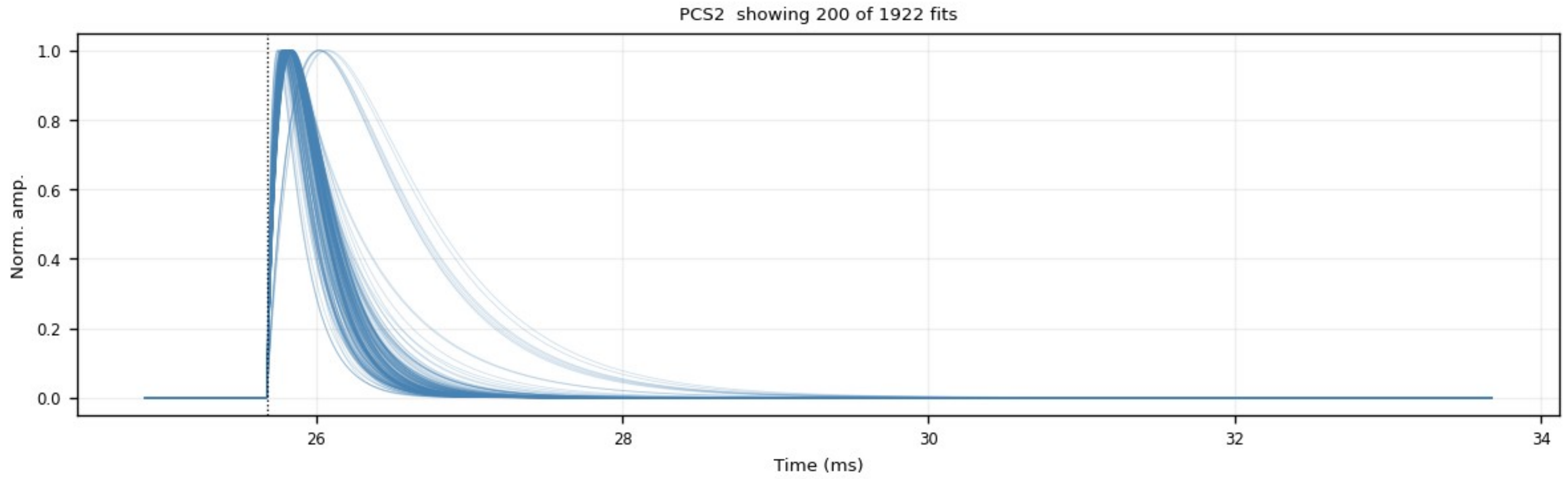
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms, the exact PCA input population: the clean fast-pulse bundle only.

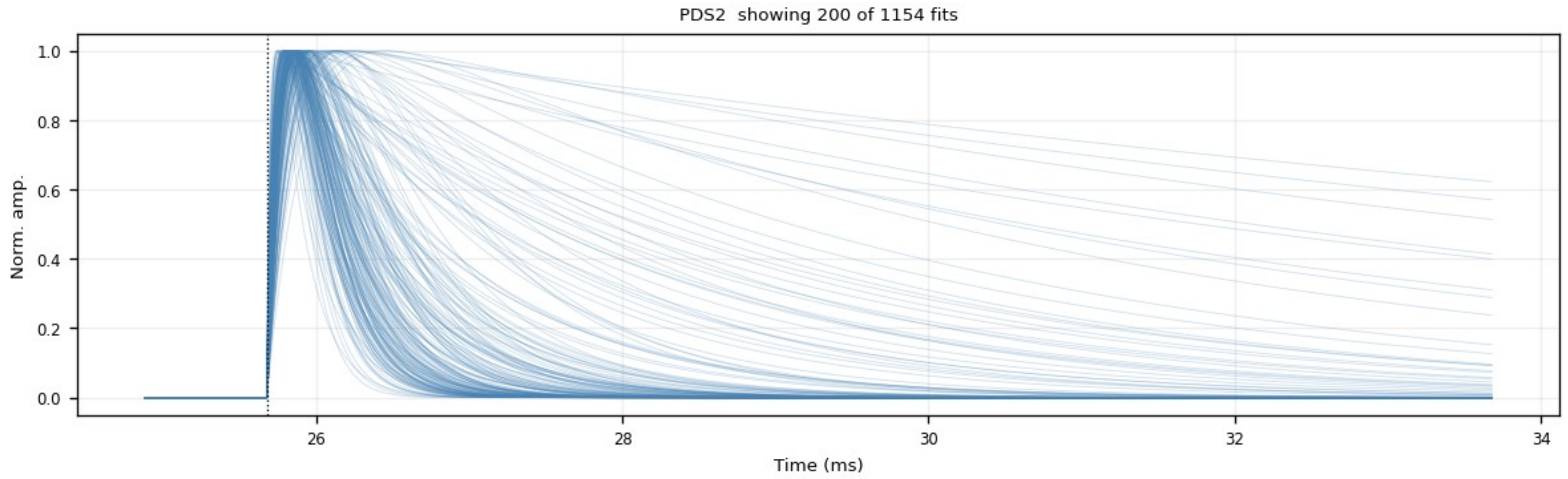


Backup · Z7 · fitted_curves_overlay: after both cuts (5/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 9–10 of 11, top to bottom

After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms, the exact PCA input population: the clean fast-pulse bundle only.

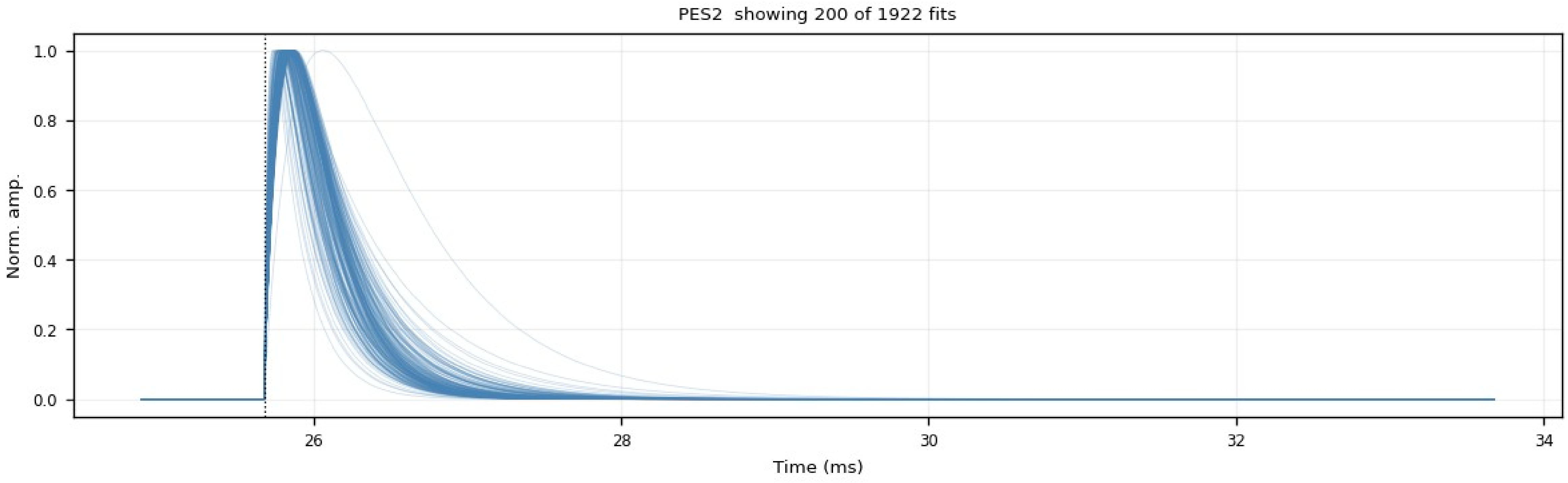




Backup · Z7 · fitted_curves_overlay: after both cuts (6/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 11–11 of 11, top to bottom

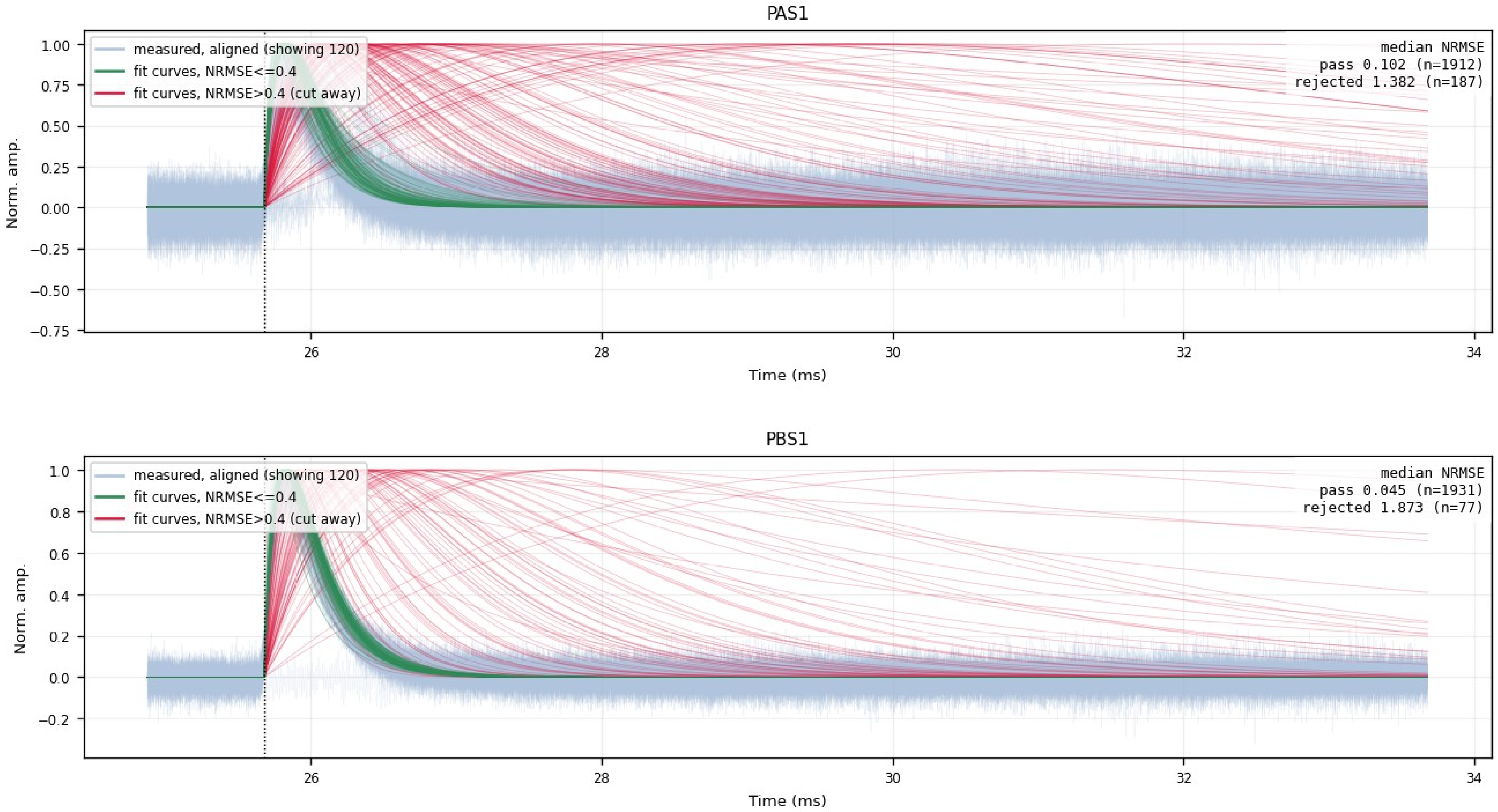
After NRMSE ≤ 0.4 AND τ_rise ≤ 0.3 ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (1/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

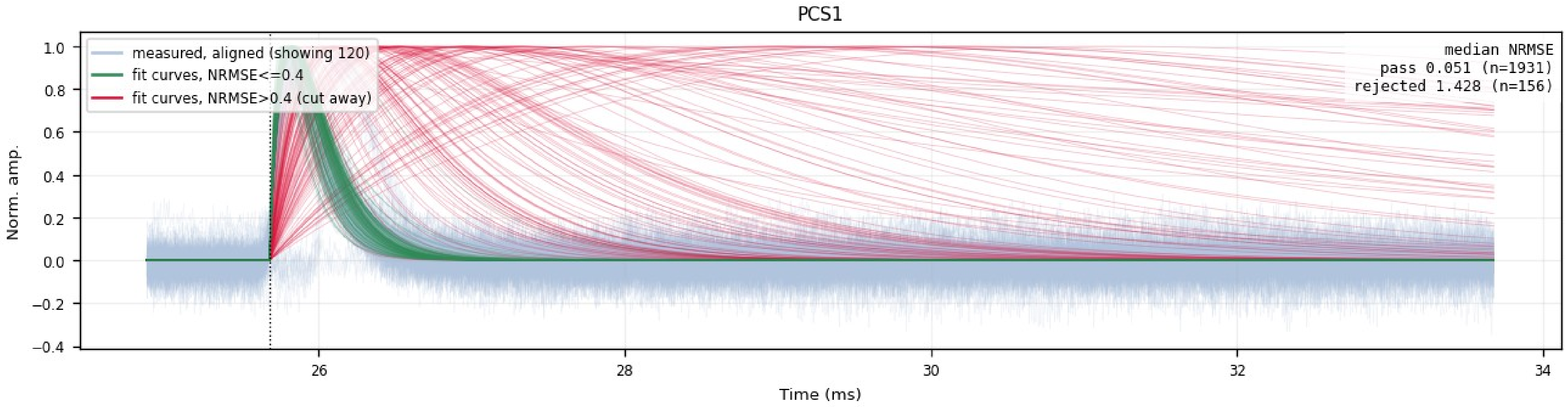
Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

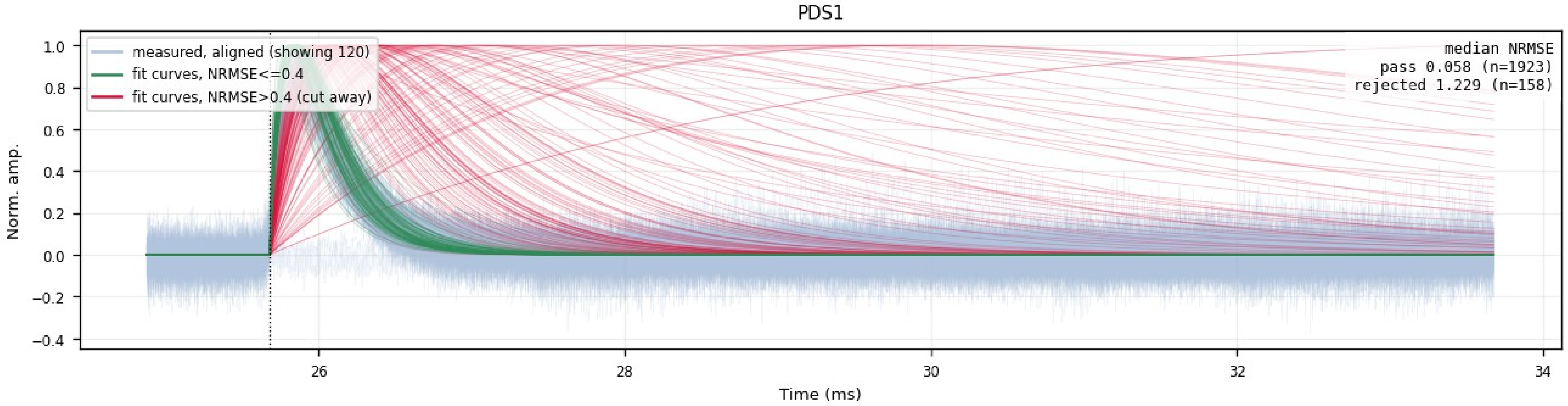


Backup · Z7 · overlay_fan_cut: kept vs rejected fits (2/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

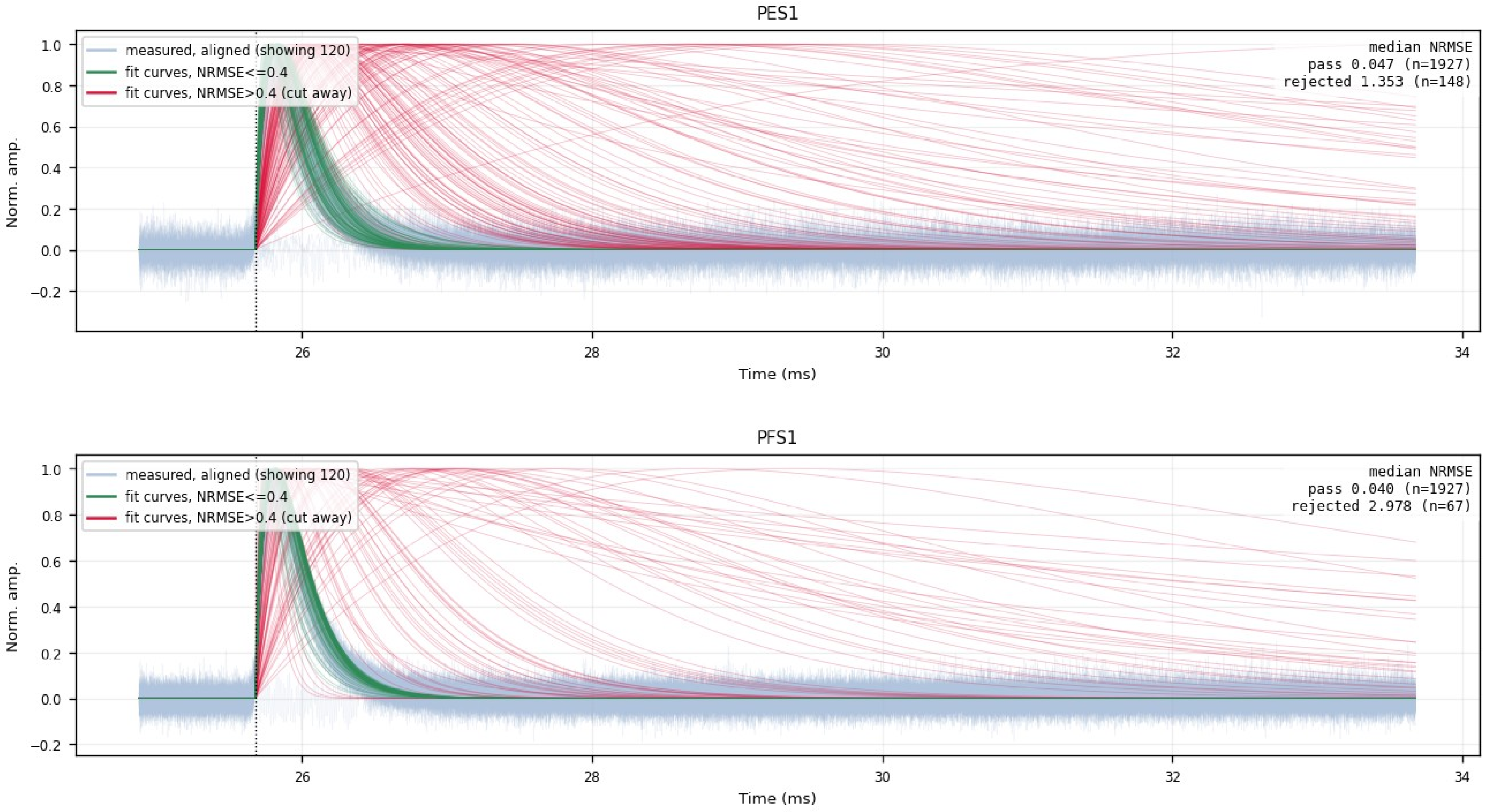




Backup · Z7 · overlay_fan_cut: kept vs rejected fits (3/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

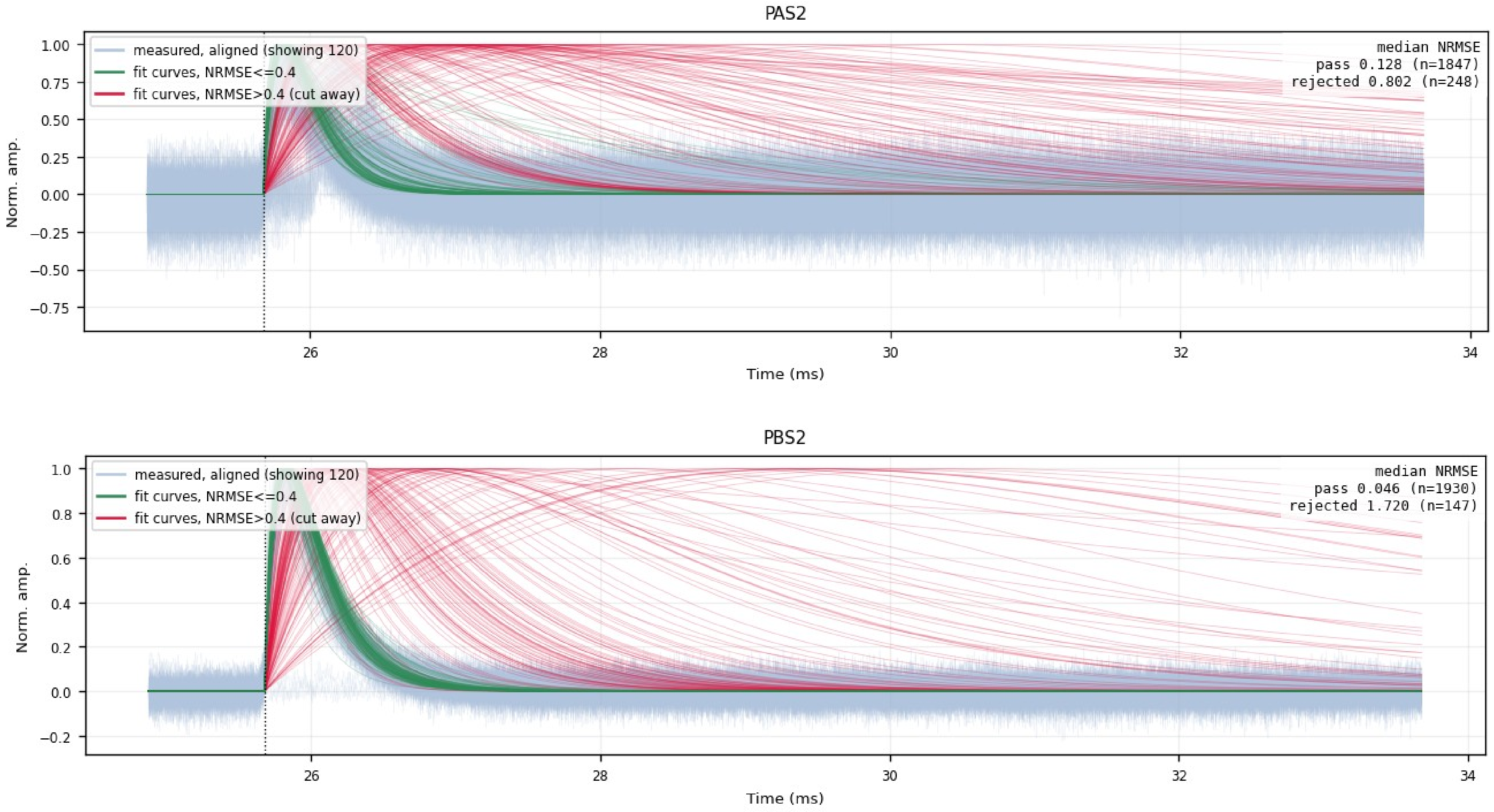
Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (4/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

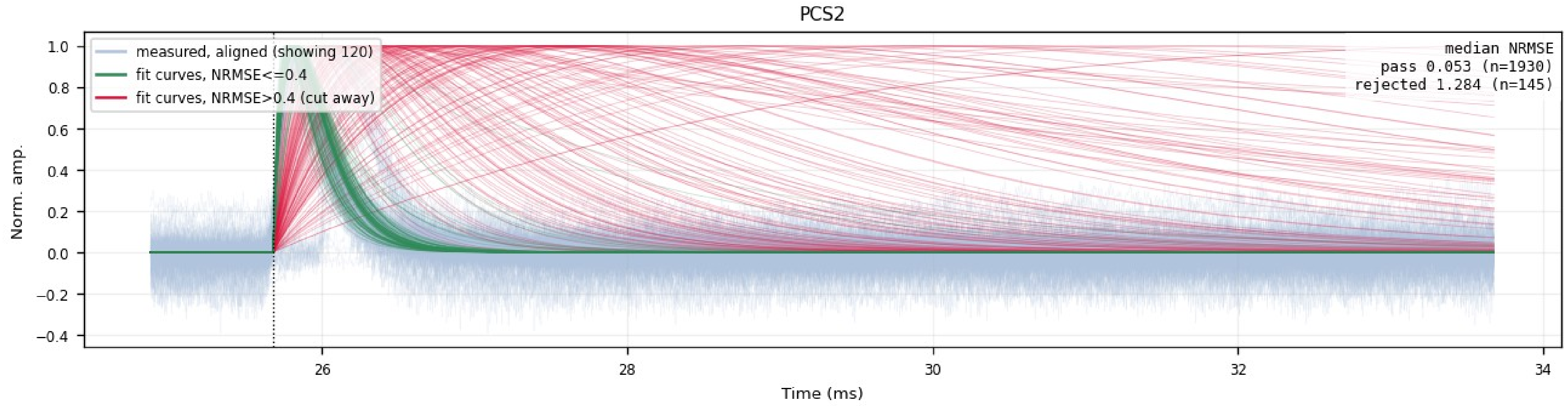
Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

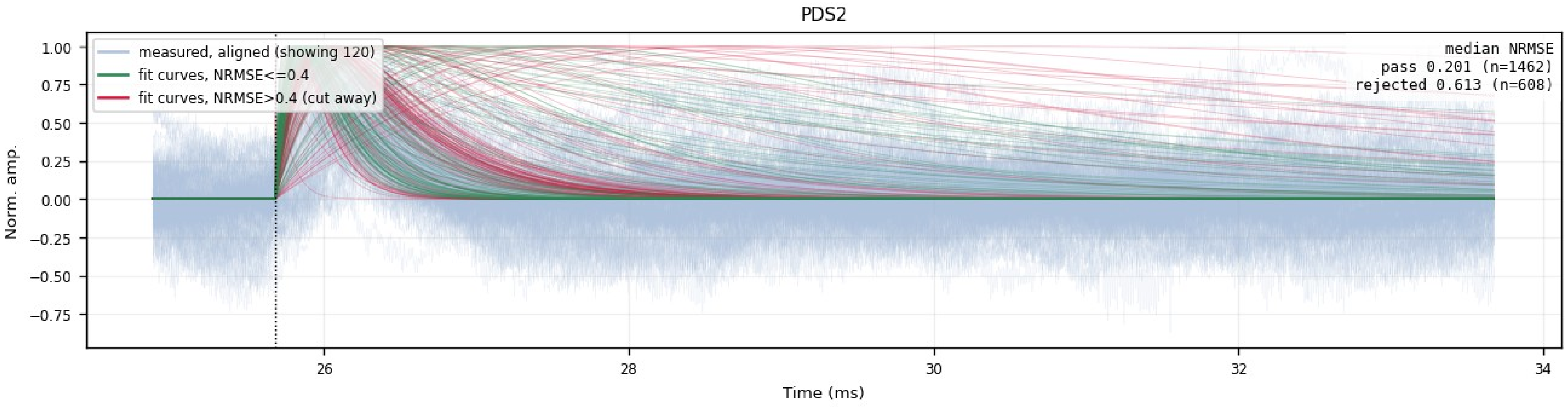


Backup · Z7 · overlay_fan_cut: kept vs rejected fits (5/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.

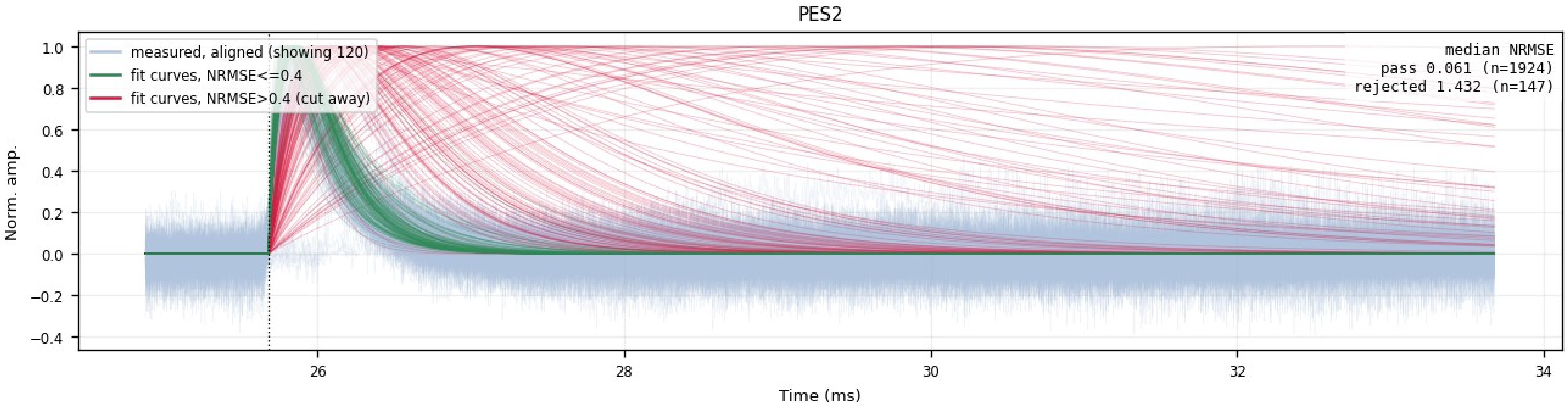




Backup · Z7 · overlay_fan_cut: kept vs rejected fits (6/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

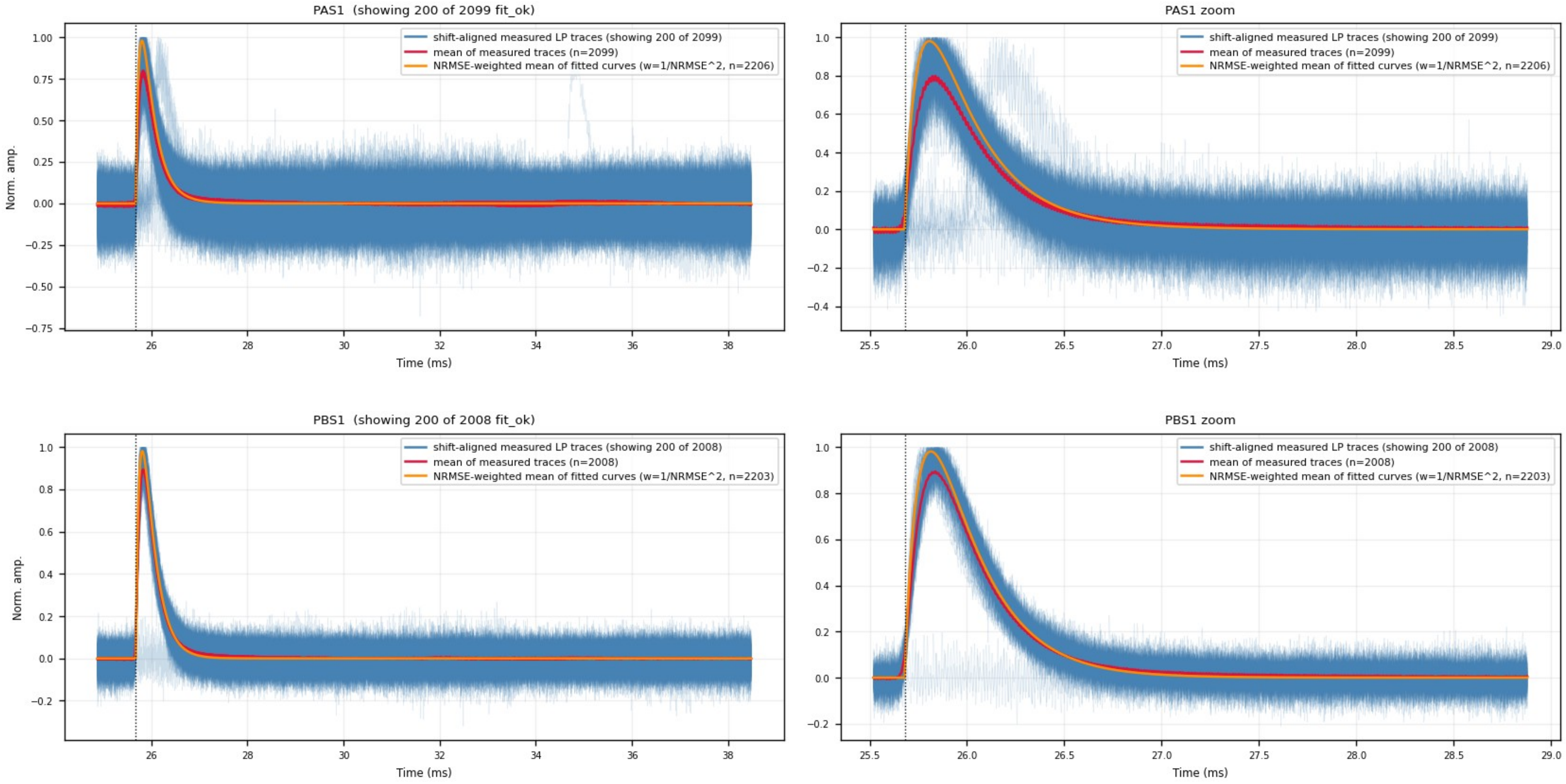
Gray-blue = aligned measured traces; green = fits with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · aligned_overlay: aligned measured traces (1/6)

figure: zip7_lp_aligned_overlay.png, channel panels 1–2 of 11, top to bottom

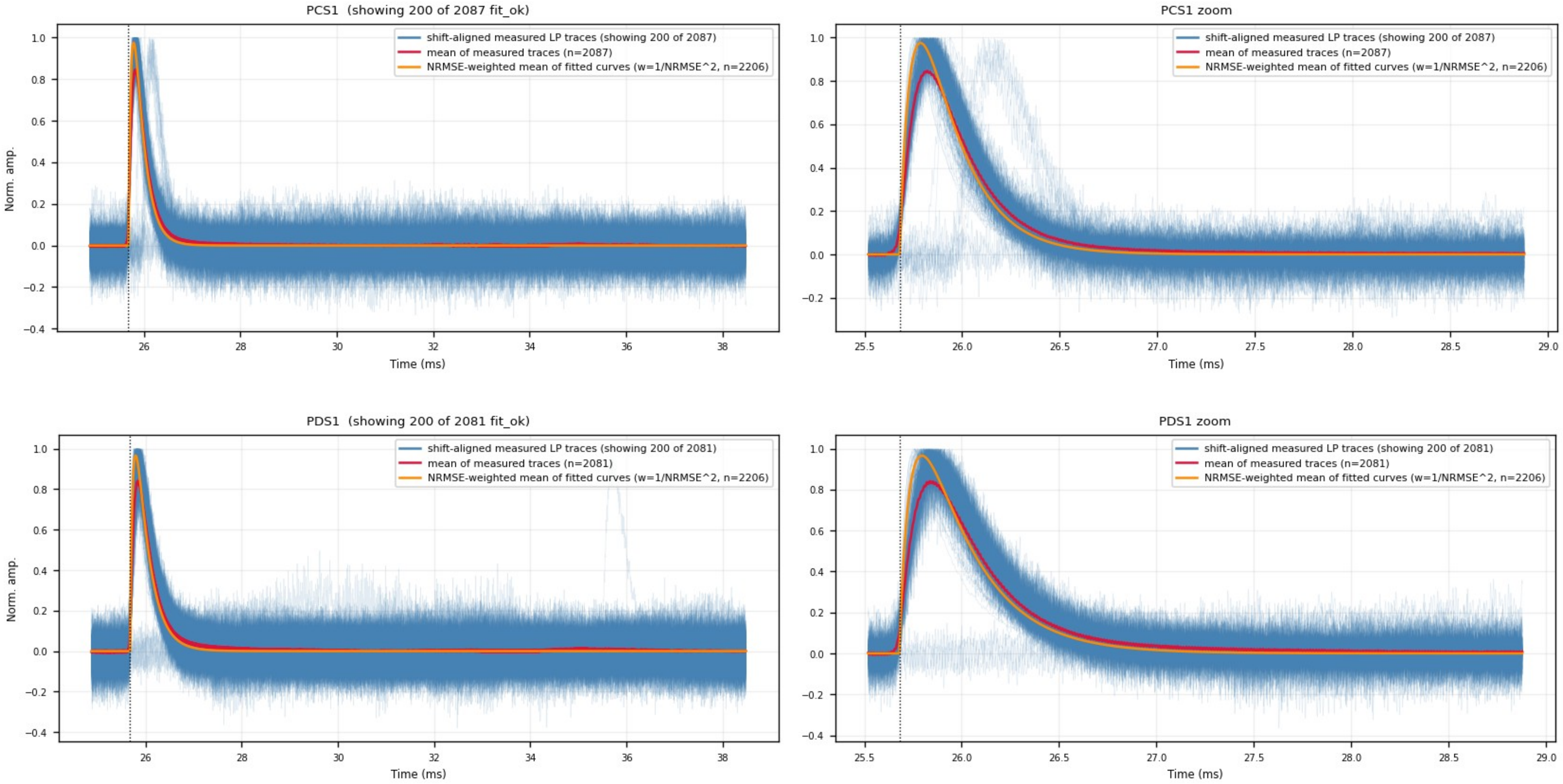
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (2/6)

figure: zip7_lp_aligned_overlay.png, channel panels 3–4 of 11, top to bottom

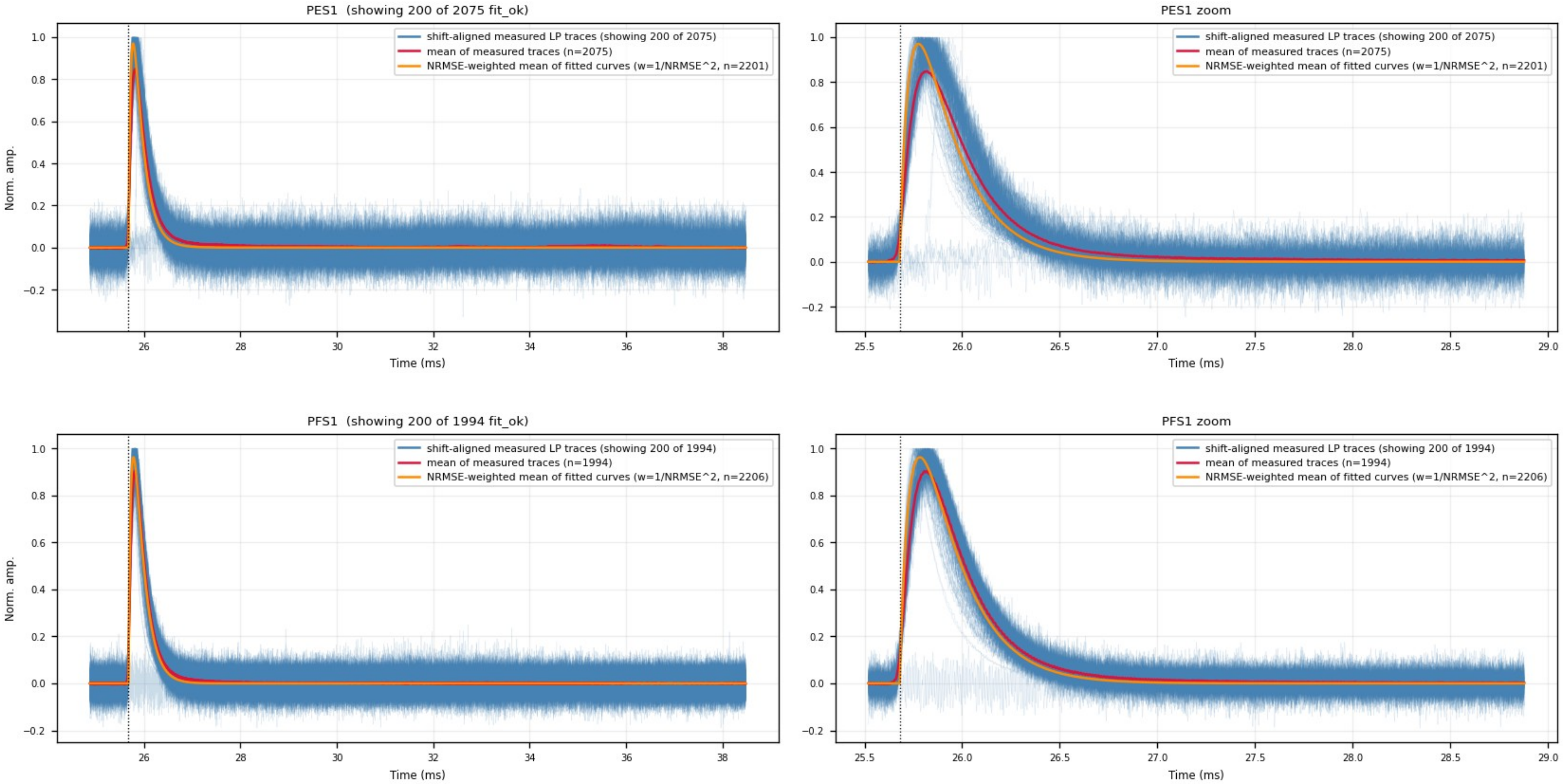
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (3/6)

figure: zip7_lp_aligned_overlay.png, channel panels 5–6 of 11, top to bottom

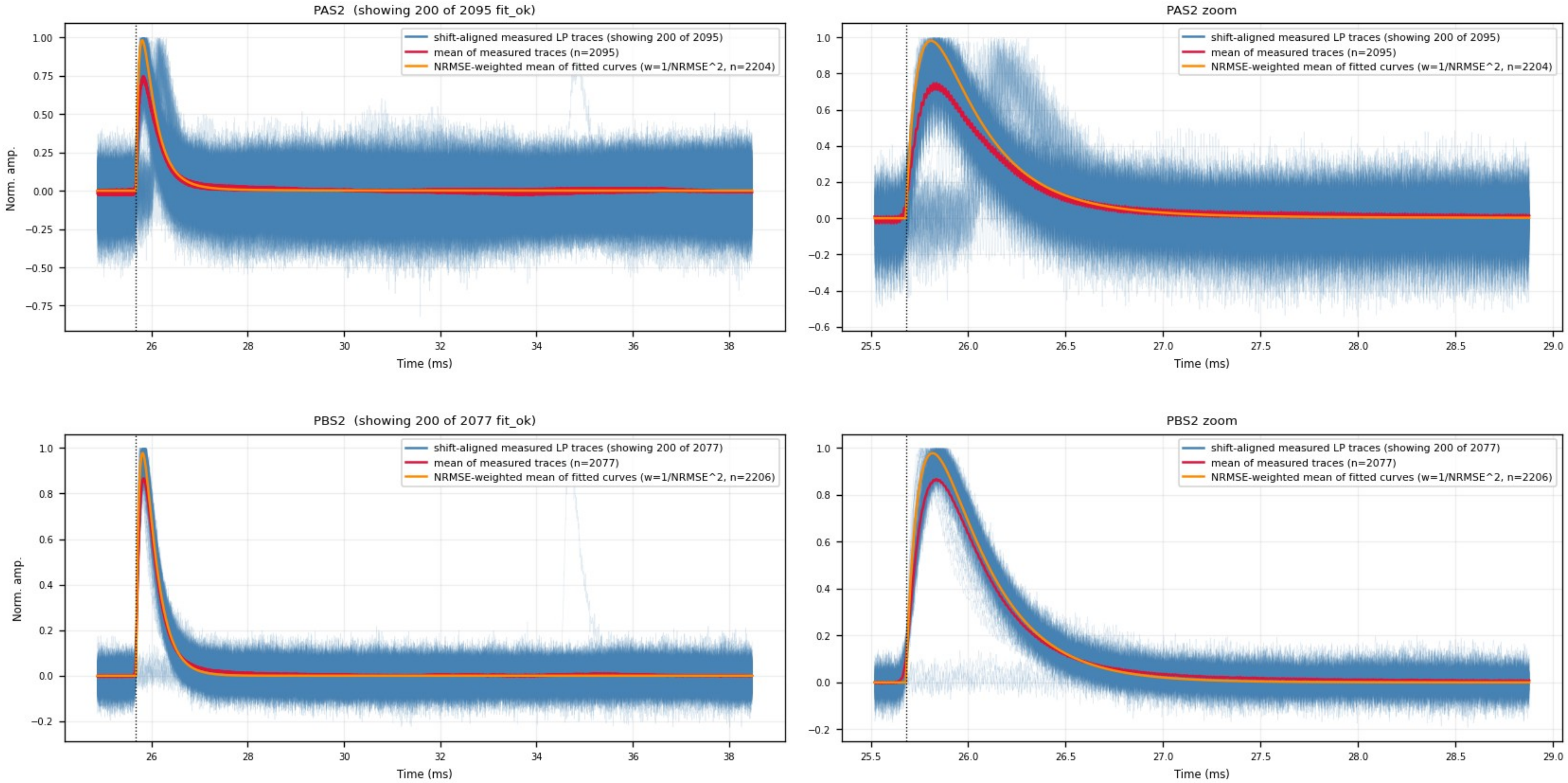
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (4/6)

figure: zip7_lp_aligned_overlay.png, channel panels 7–8 of 11, top to bottom

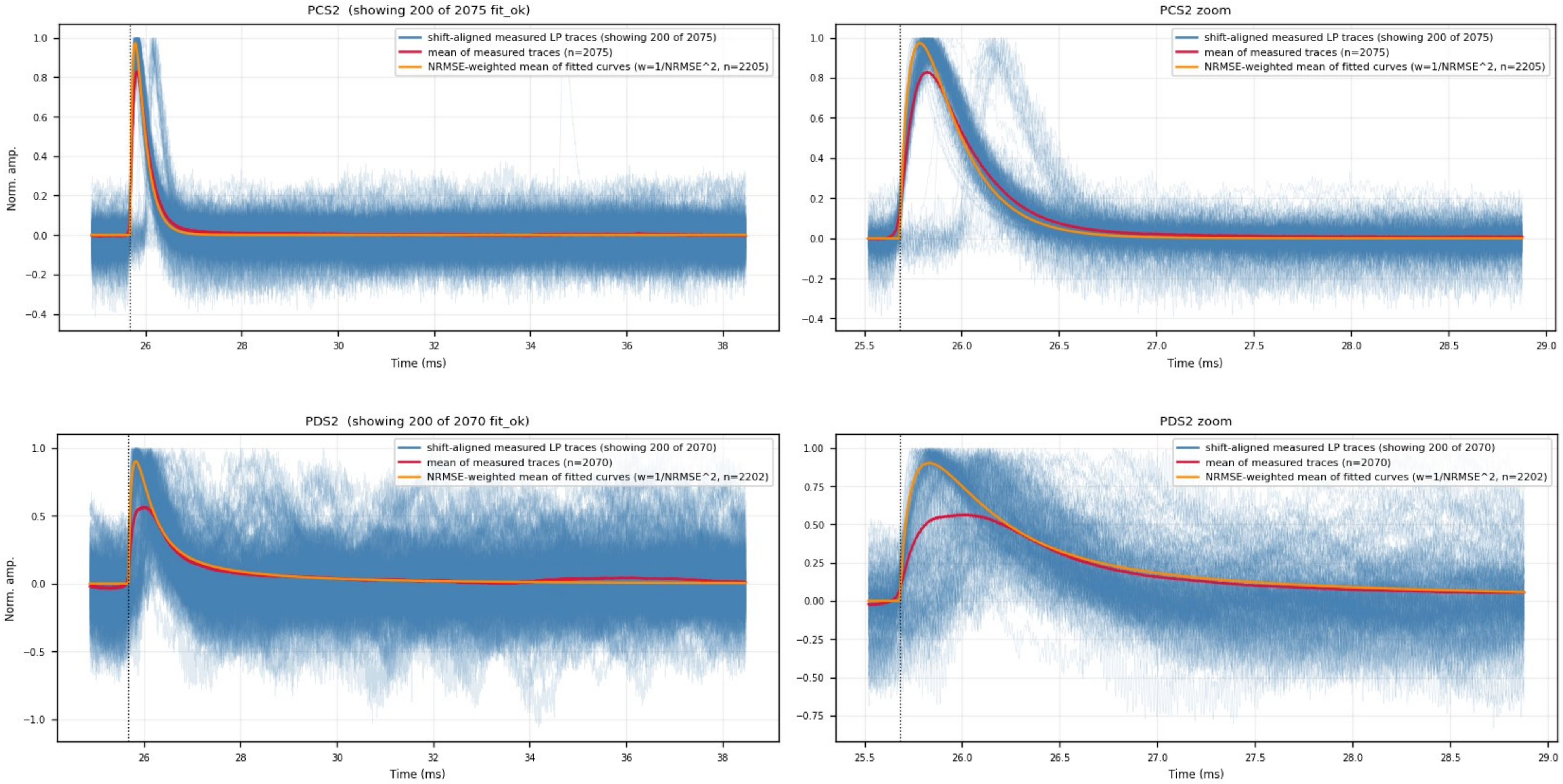
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (5/6)

figure: zip7_lp_aligned_overlay.png, channel panels 9–10 of 11, top to bottom

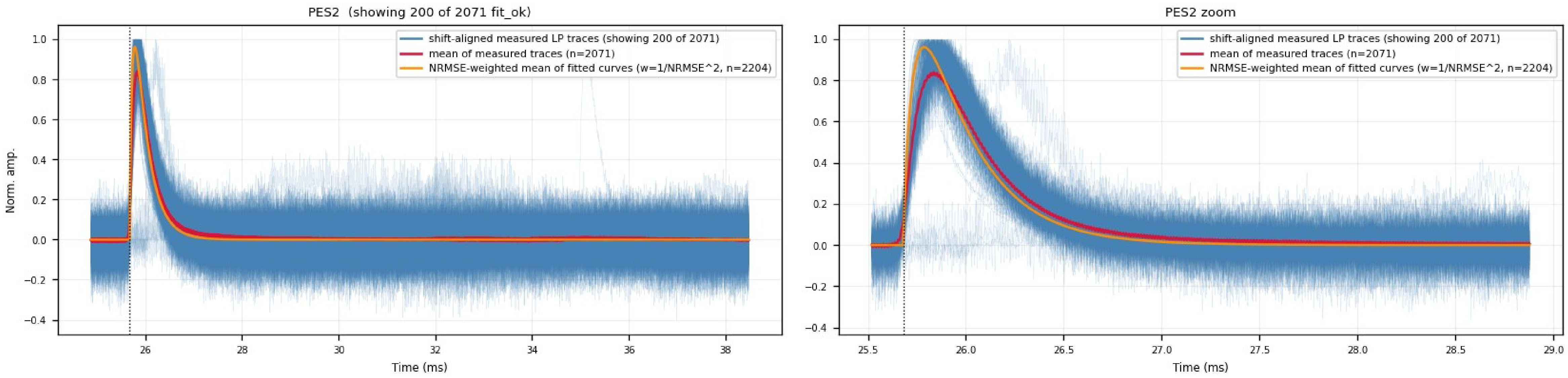
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (6/6)

figure: zip7_lp_aligned_overlay.png, channel panels 11–11 of 11, top to bottom

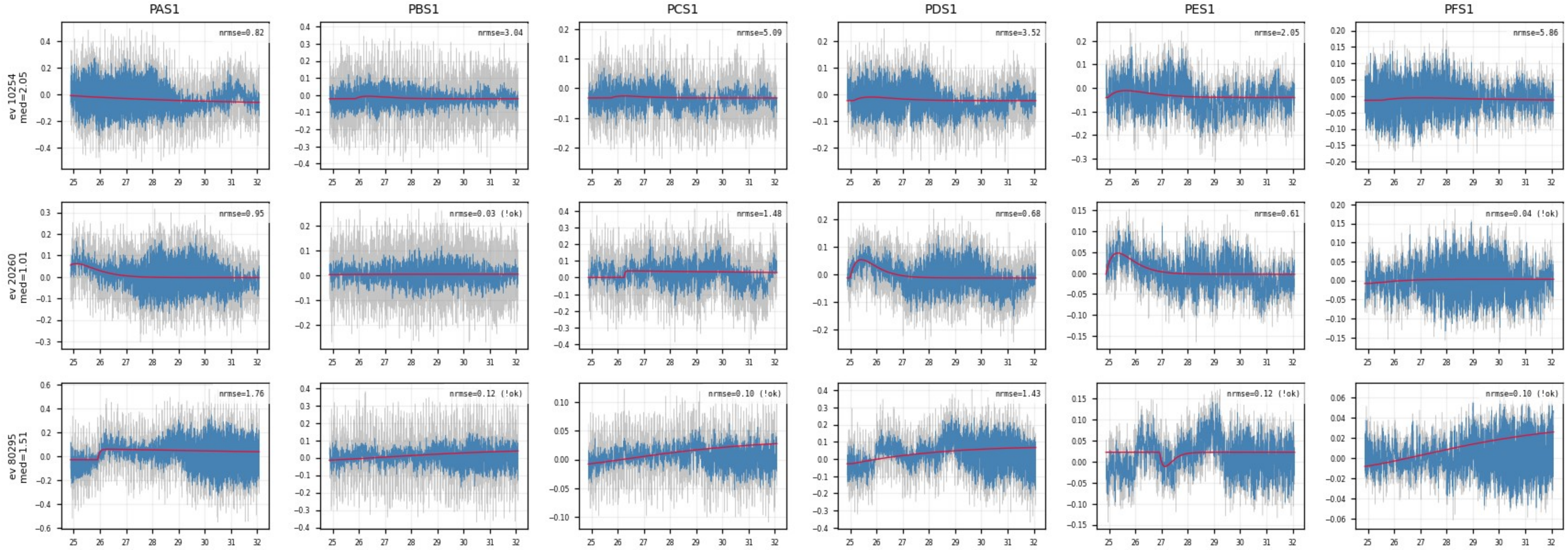
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (1/10)

figure: zip7_slow_rise_events.png, event rows 1–3 of 15, left half of the channels

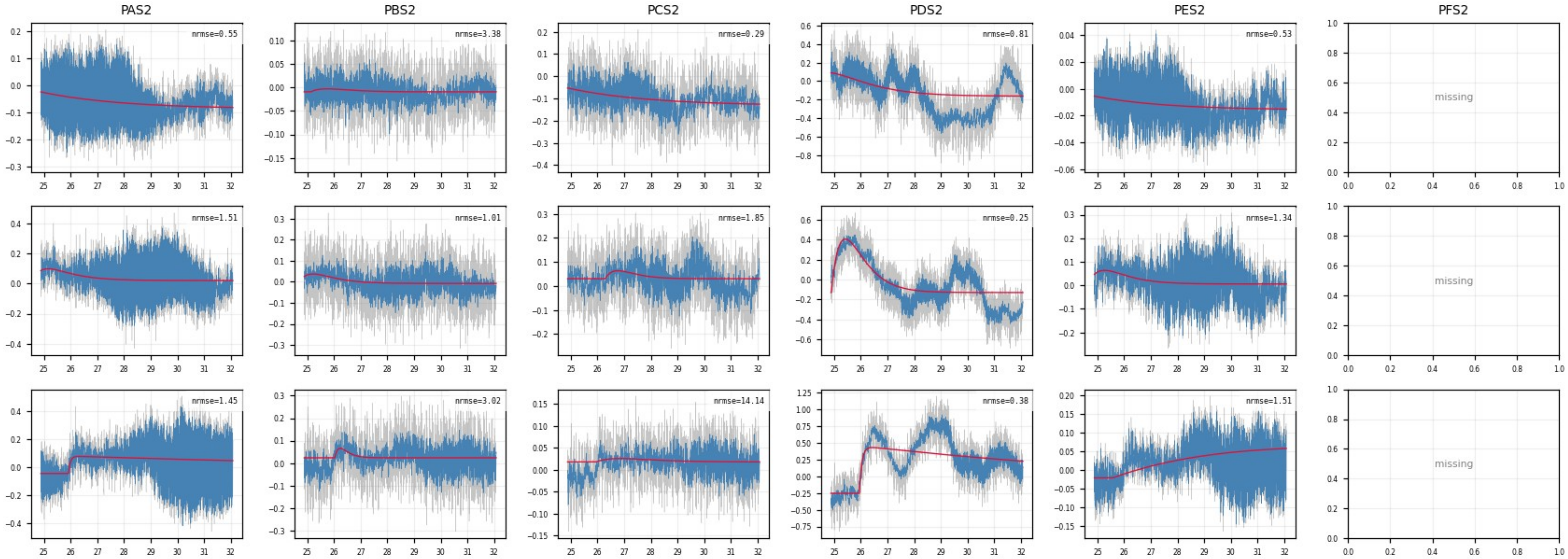
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (2/10)

figure: zip7_slow_rise_events.png, event rows 1–3 of 15, right half of the channels

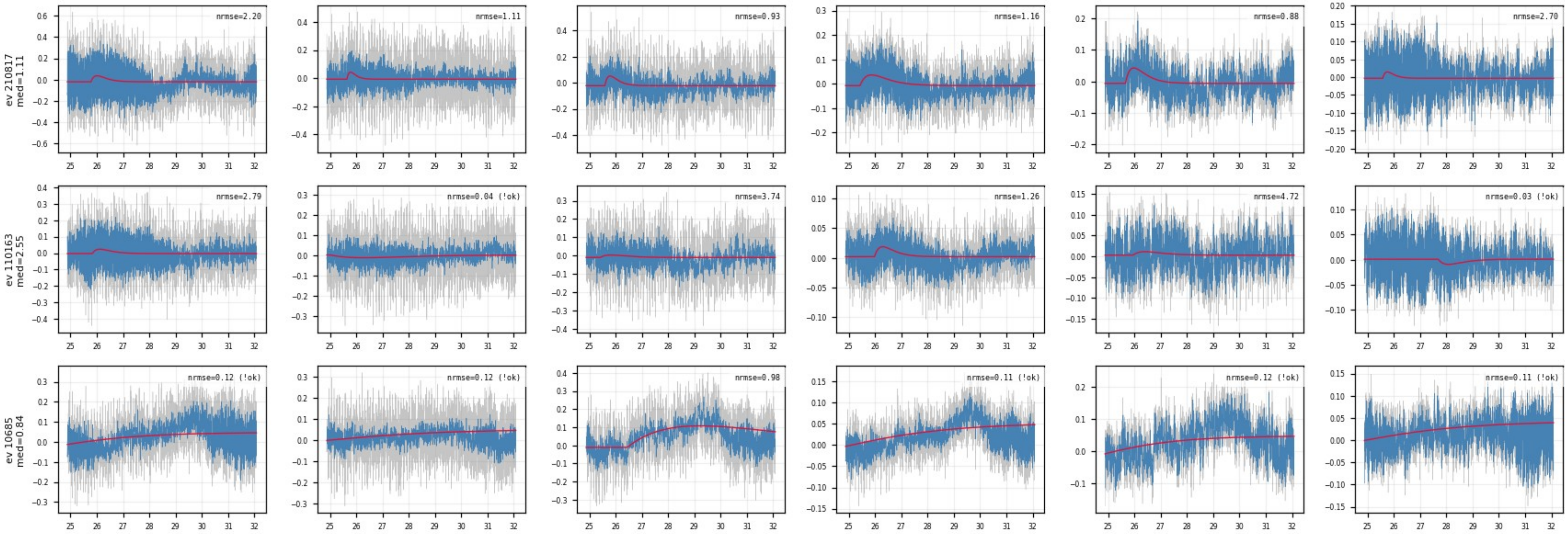
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (3/10)

figure: zip7_slow_rise_events.png, event rows 4–6 of 15, left half of the channels

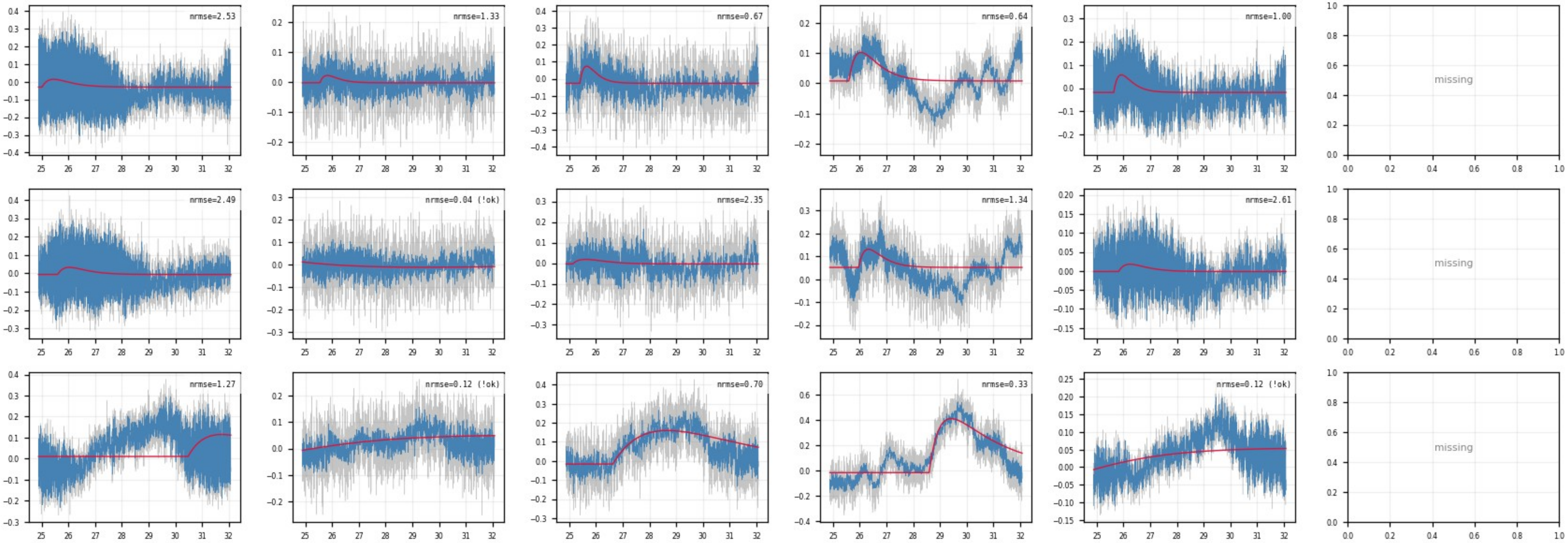
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (4/10)

figure: zip7_slow_rise_events.png, event rows 4–6 of 15, right half of the channels

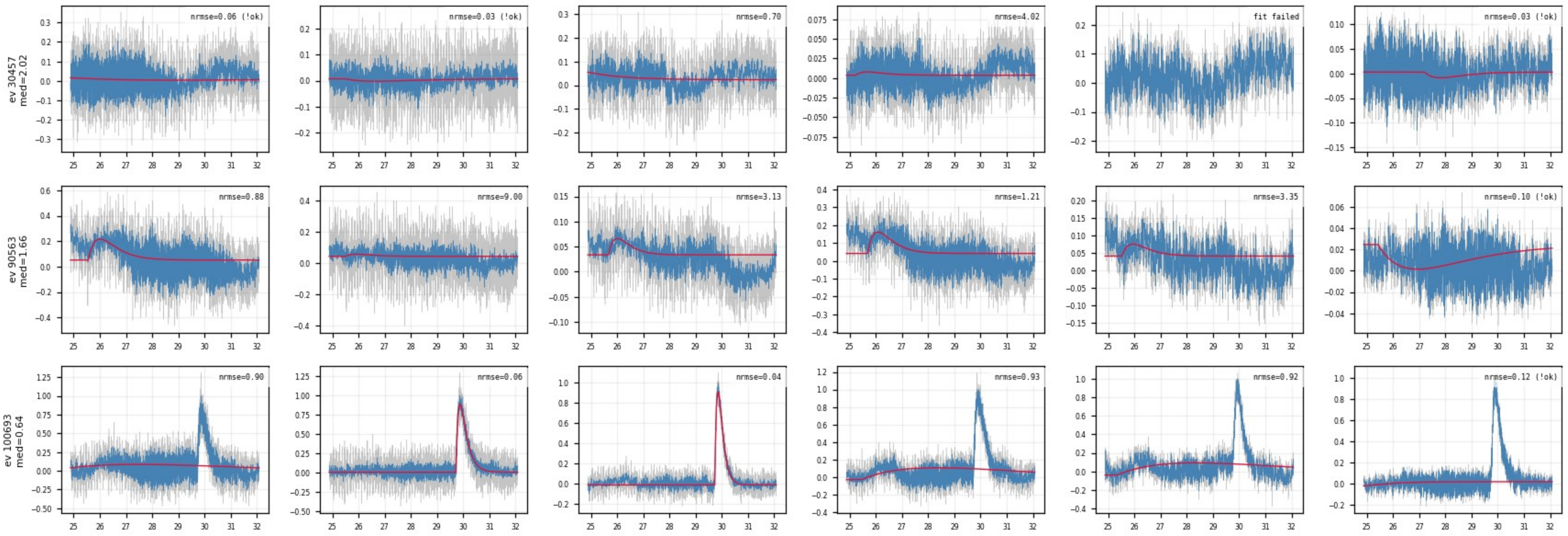
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (5/10)

figure: zip7_slow_rise_events.png, event rows 7–9 of 15, left half of the channels

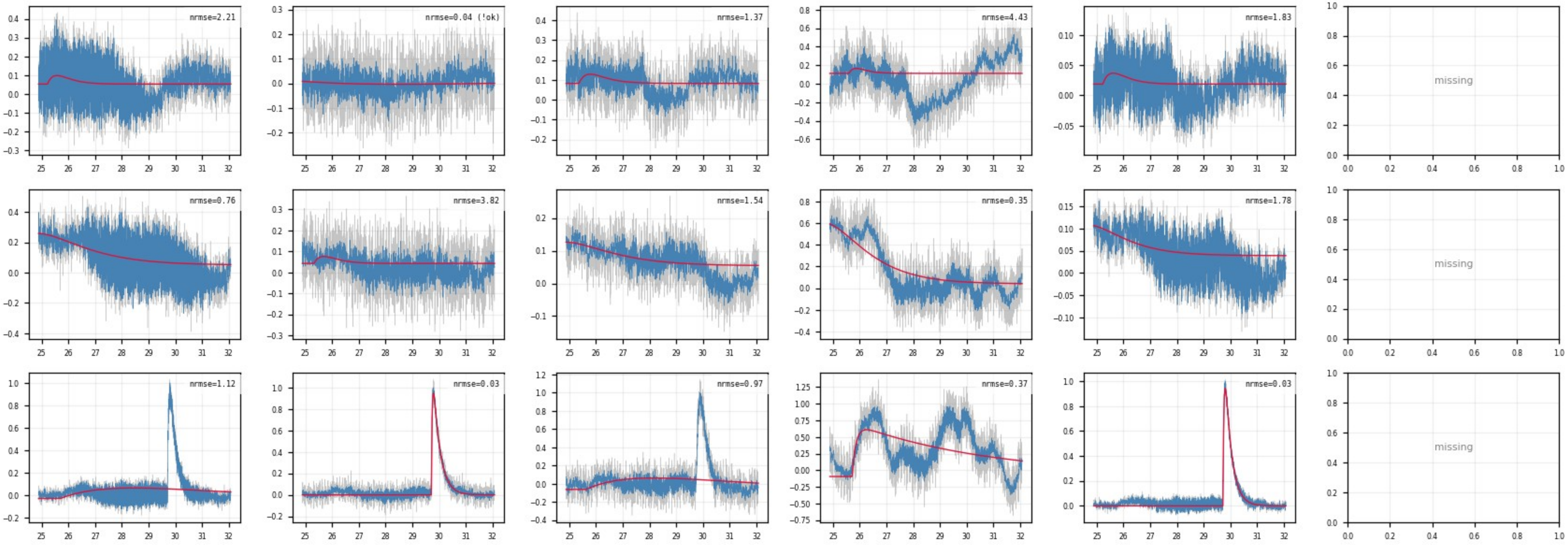
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (6/10)

figure: zip7_slow_rise_events.png, event rows 7–9 of 15, right half of the channels

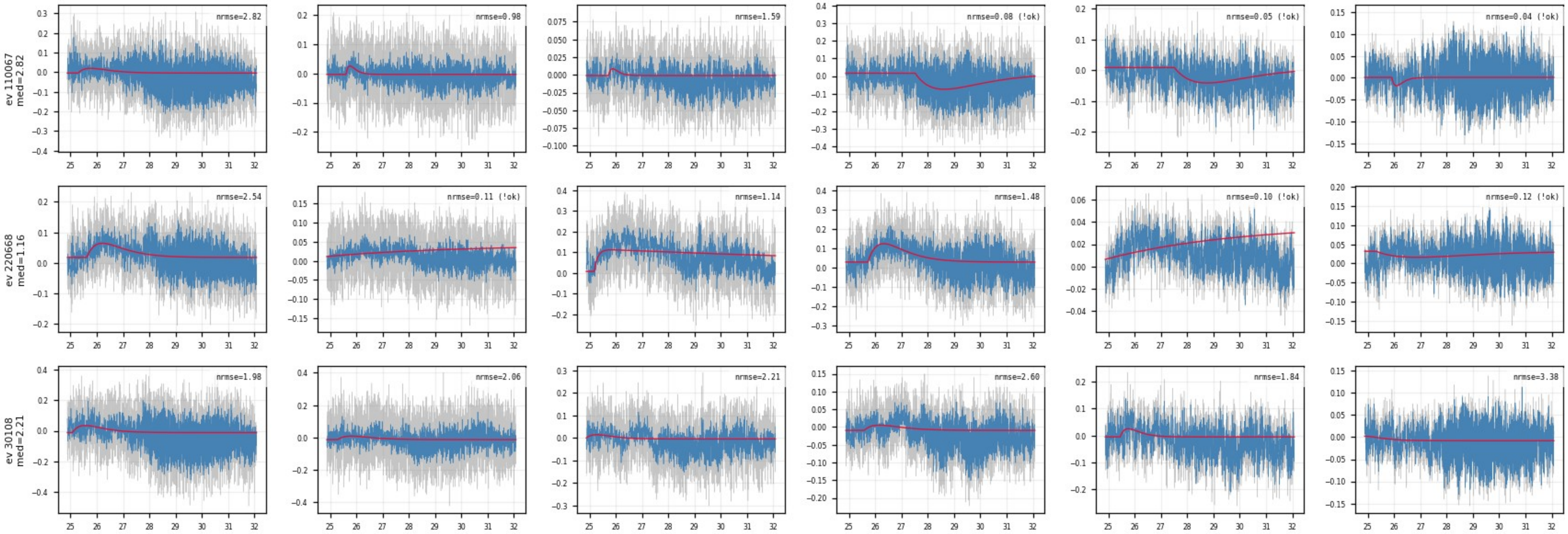
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (7/10)

figure: zip7_slow_rise_events.png, event rows 10–12 of 15, left half of the channels

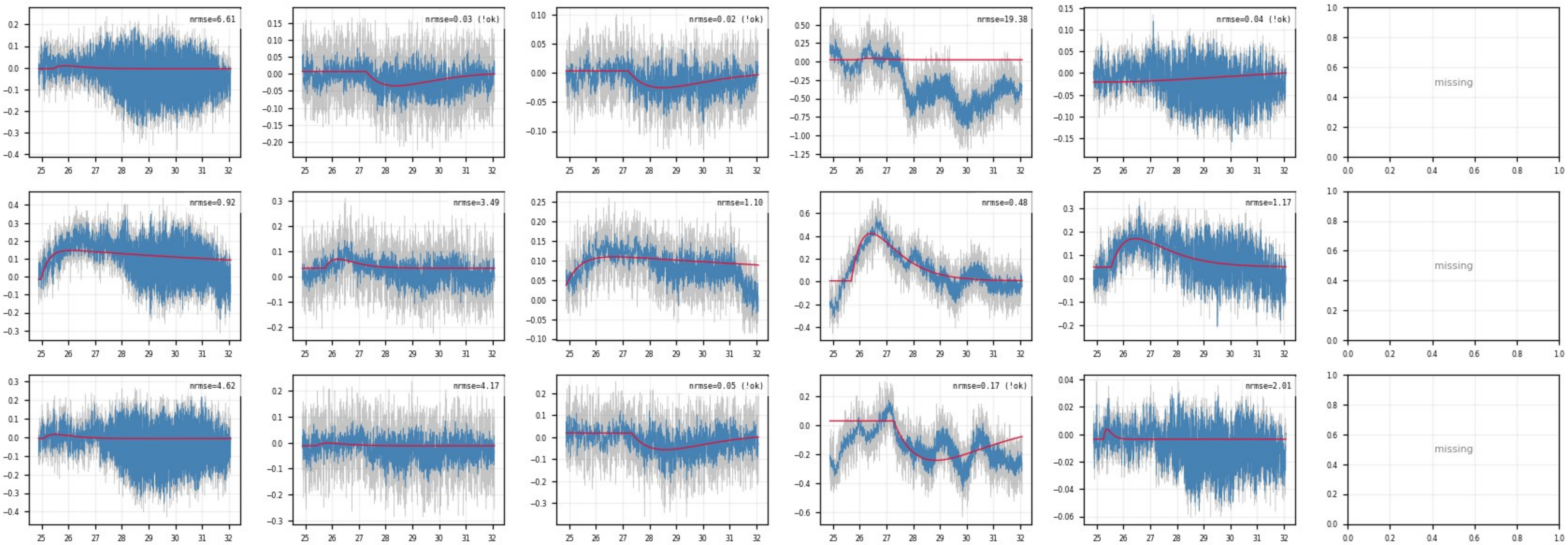
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (8/10)

figure: zip7_slow_rise_events.png, event rows 10–12 of 15, right half of the channels

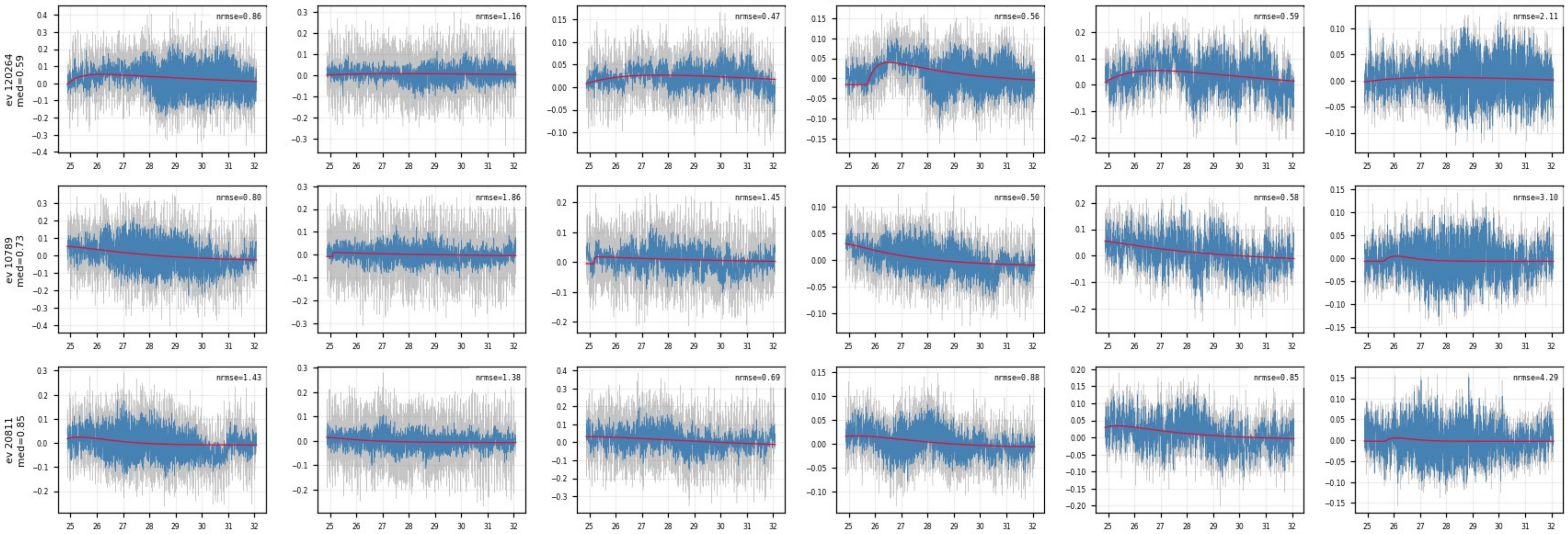
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (9/10)

figure: zip7_slow_rise_events.png, event rows 13–15 of 15, left half of the channels

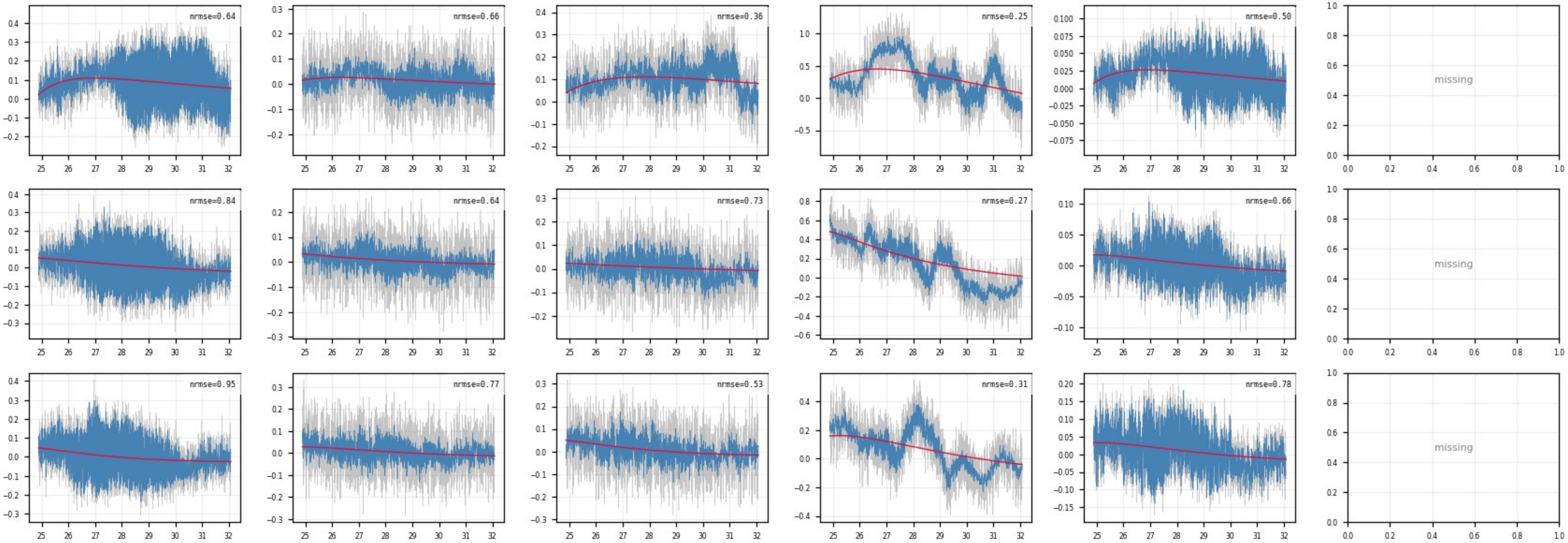
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (10/10)

figure: zip7_slow_rise_events.png, event rows 13–15 of 15, right half of the channels

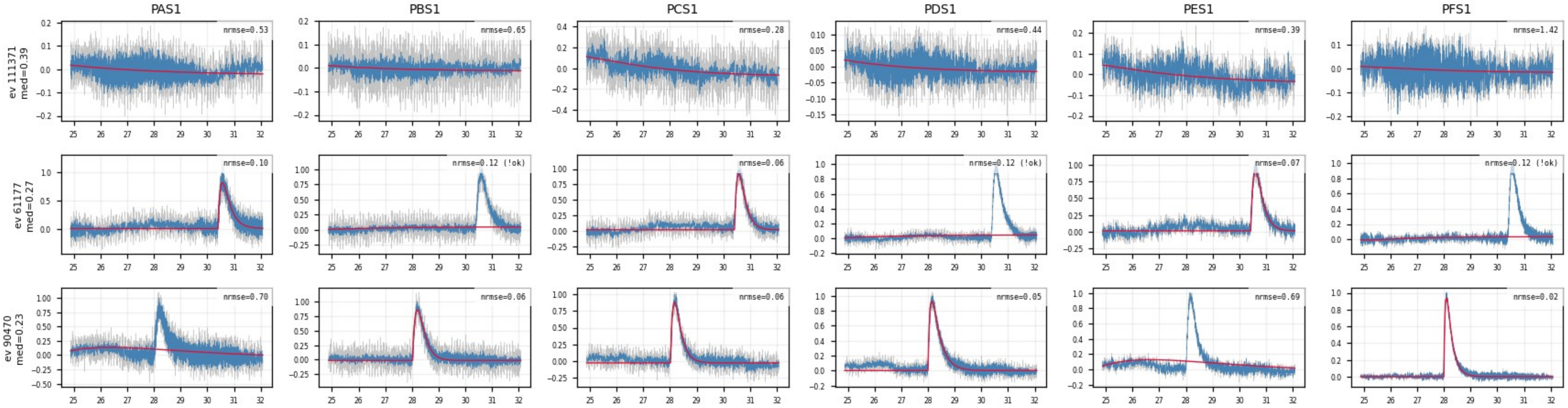
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · shadow_events: well-fit slow-rise events (1/2)

figure: zip7_shadow_events.png, event rows 1–3 of 3, left half of the channels

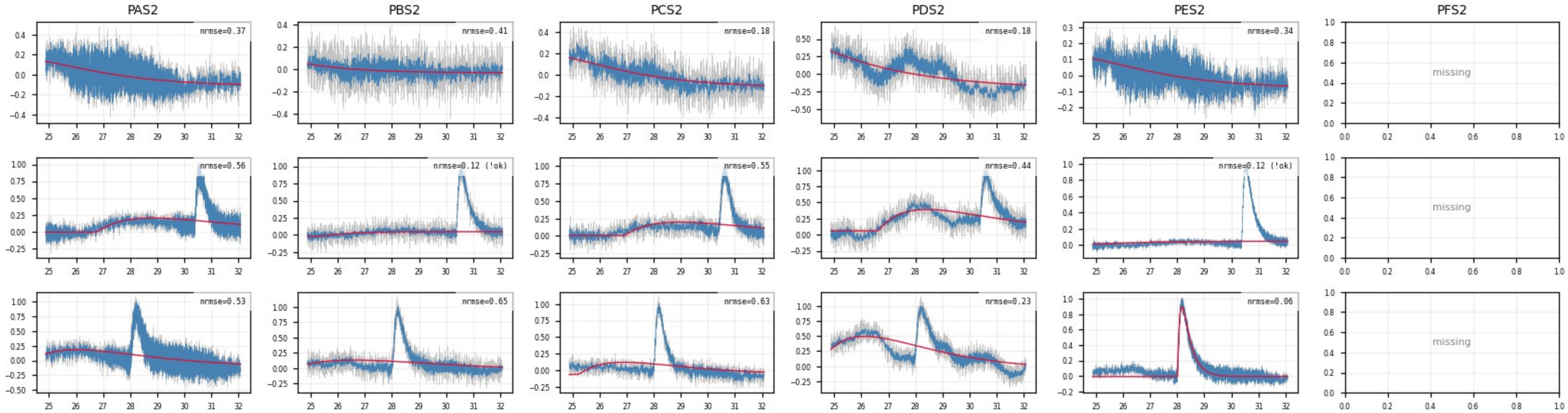
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses; kept, and exactly what the NxM multi-templates are for.



Backup · Z7 · shadow_events: well-fit slow-rise events (2/2)

figure: zip7_shadow_events.png, event rows 1–3 of 3, right half of the channels

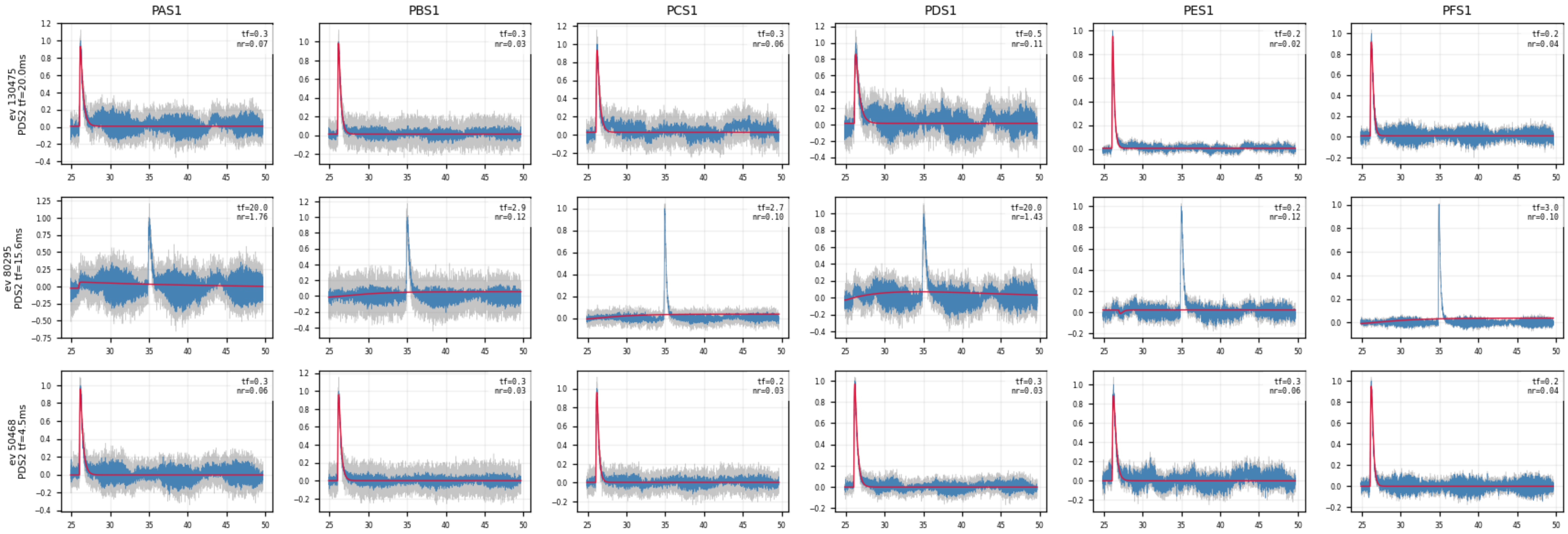
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses; kept, and exactly what the NxM multi-templates are for.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (1/8)

figure: zip7_PDS2_slow_fall.png, event rows 1–3 of 10, left half of the channels

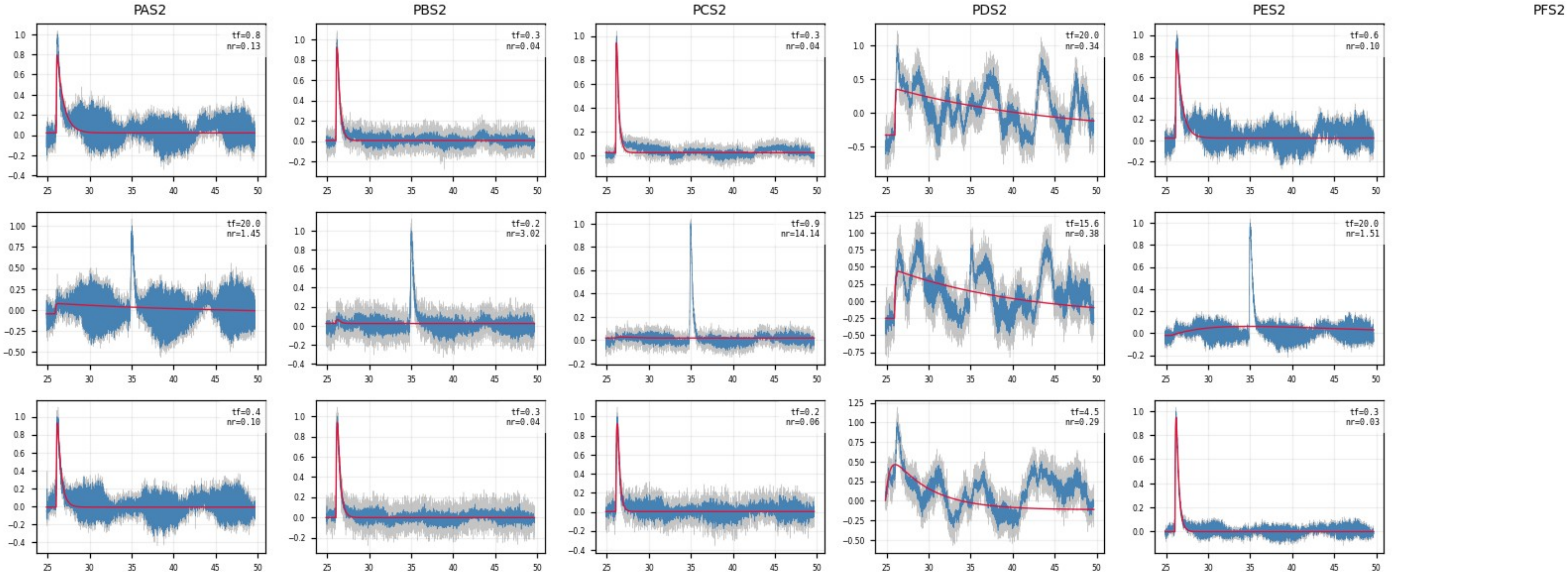
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (2/8)

figure: zip7_PDS2_slow_fall.png, event rows 1–3 of 10, right half of the channels

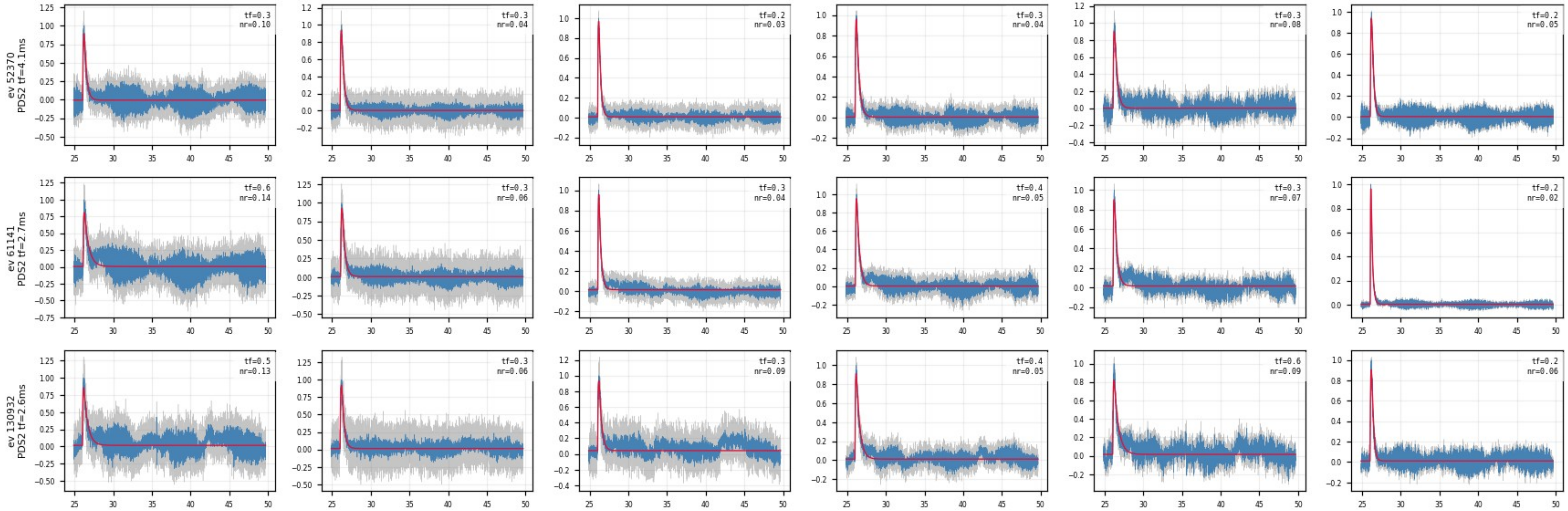
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (3/8)

figure: zip7_PDS2_slow_fall.png, event rows 4–6 of 10, left half of the channels

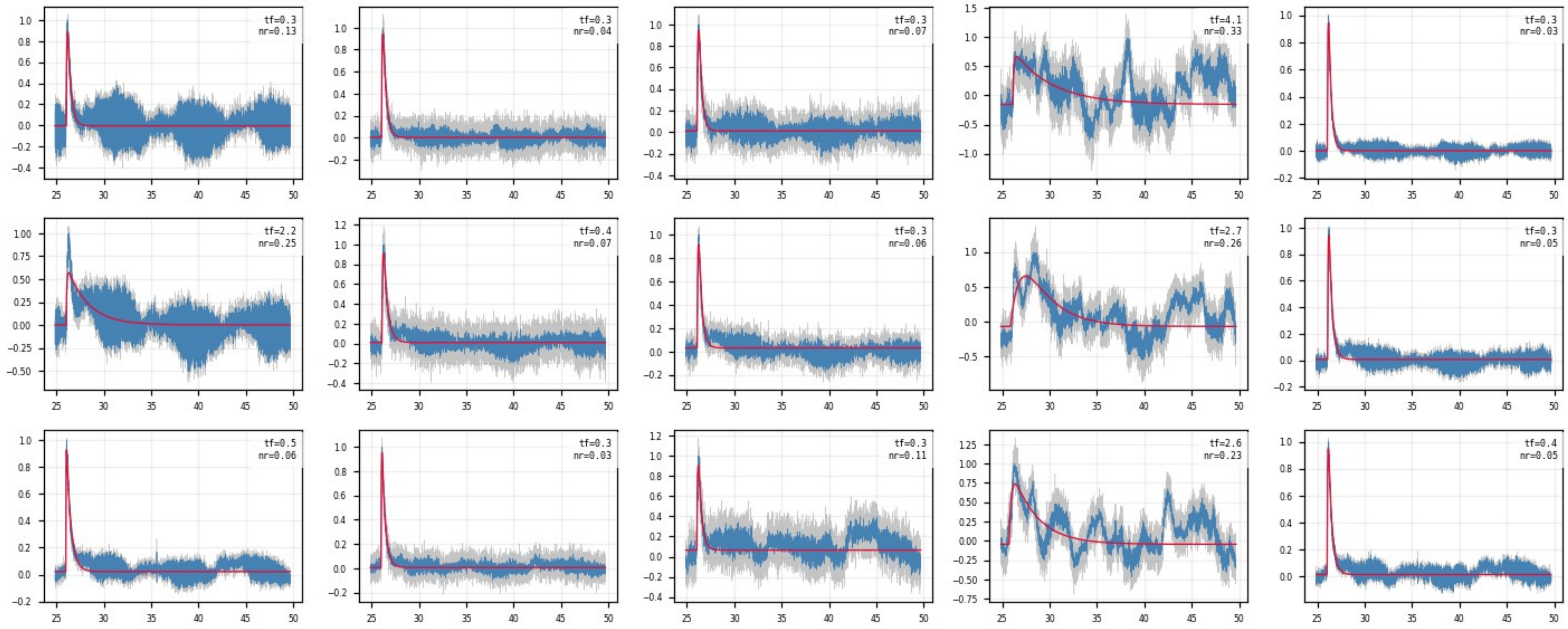
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (4/8)

figure: zip7_PDS2_slow_fall.png, event rows 4–6 of 10, right half of the channels

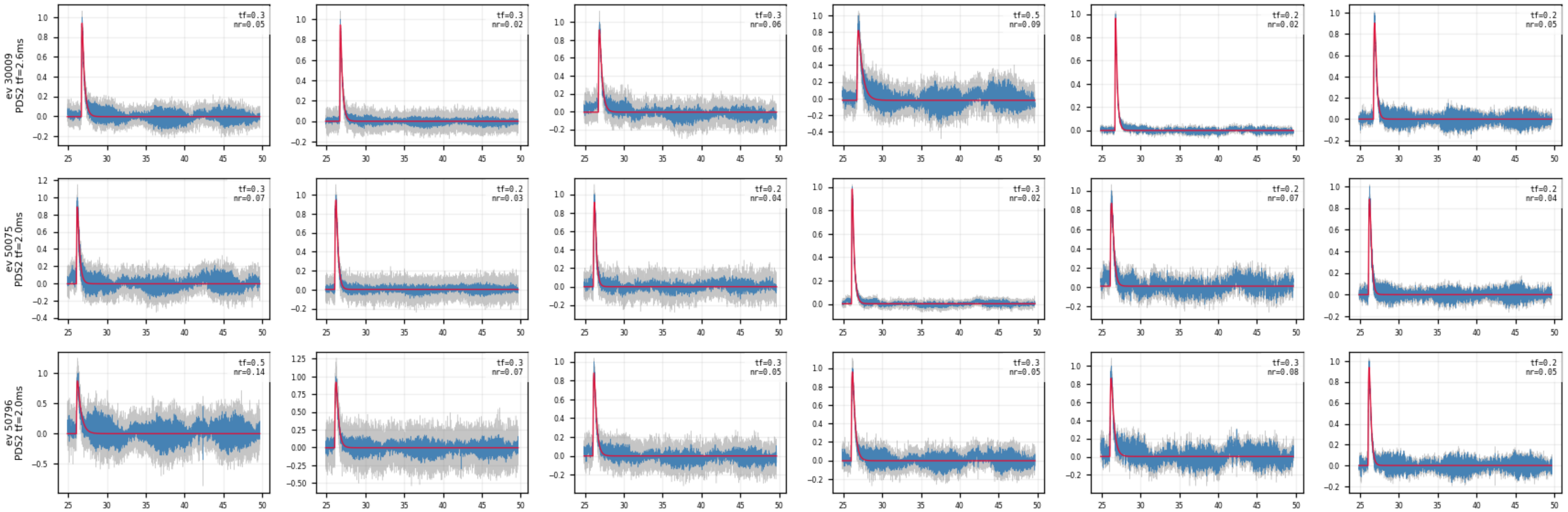
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (5/8)

figure: zip7_PDS2_slow_fall.png, event rows 7–9 of 10, left half of the channels

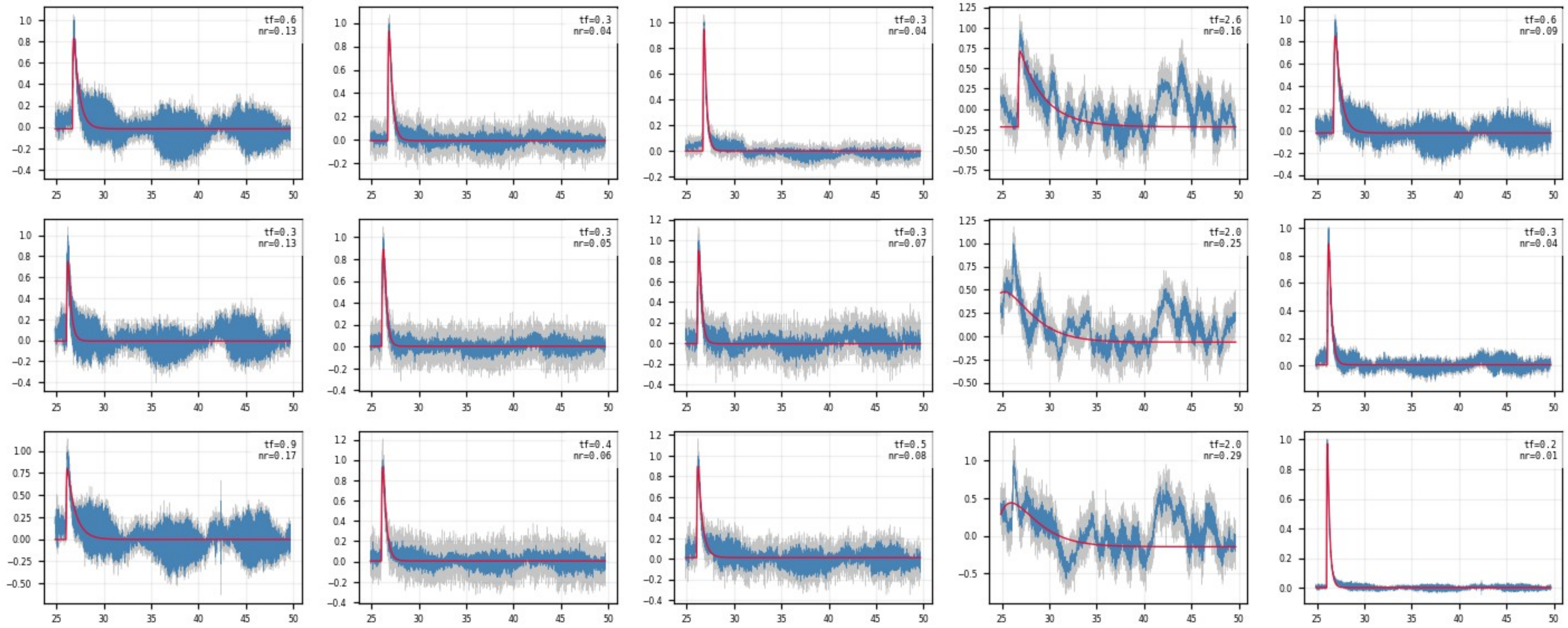
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (6/8)

figure: zip7_PDS2_slow_fall.png, event rows 7–9 of 10, right half of the channels

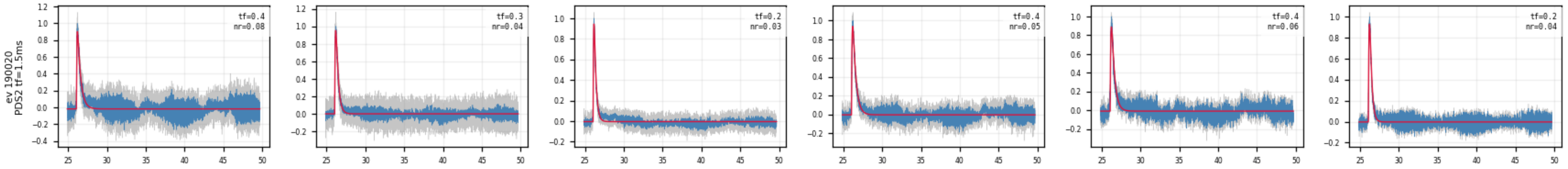
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (7/8)

figure: zip7_PDS2_slow_fall.png, event rows 10–10 of 10, left half of the channels

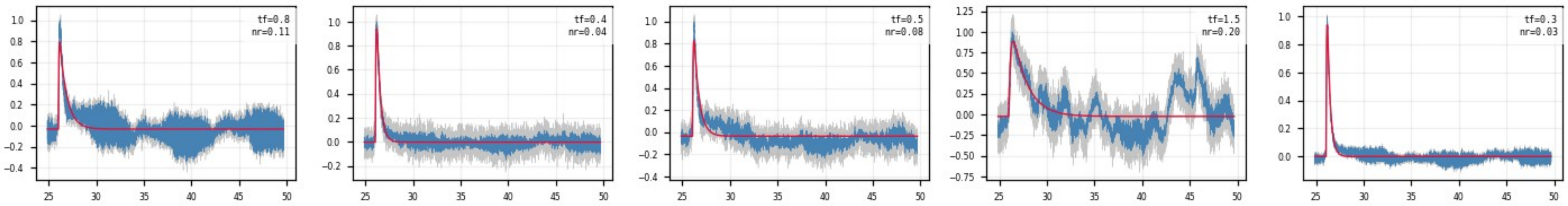
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long-τ_fall study (PDS2) (8/8)

figure: zip7_PDS2_slow_fall.png, event rows 10–10 of 10, right half of the channels

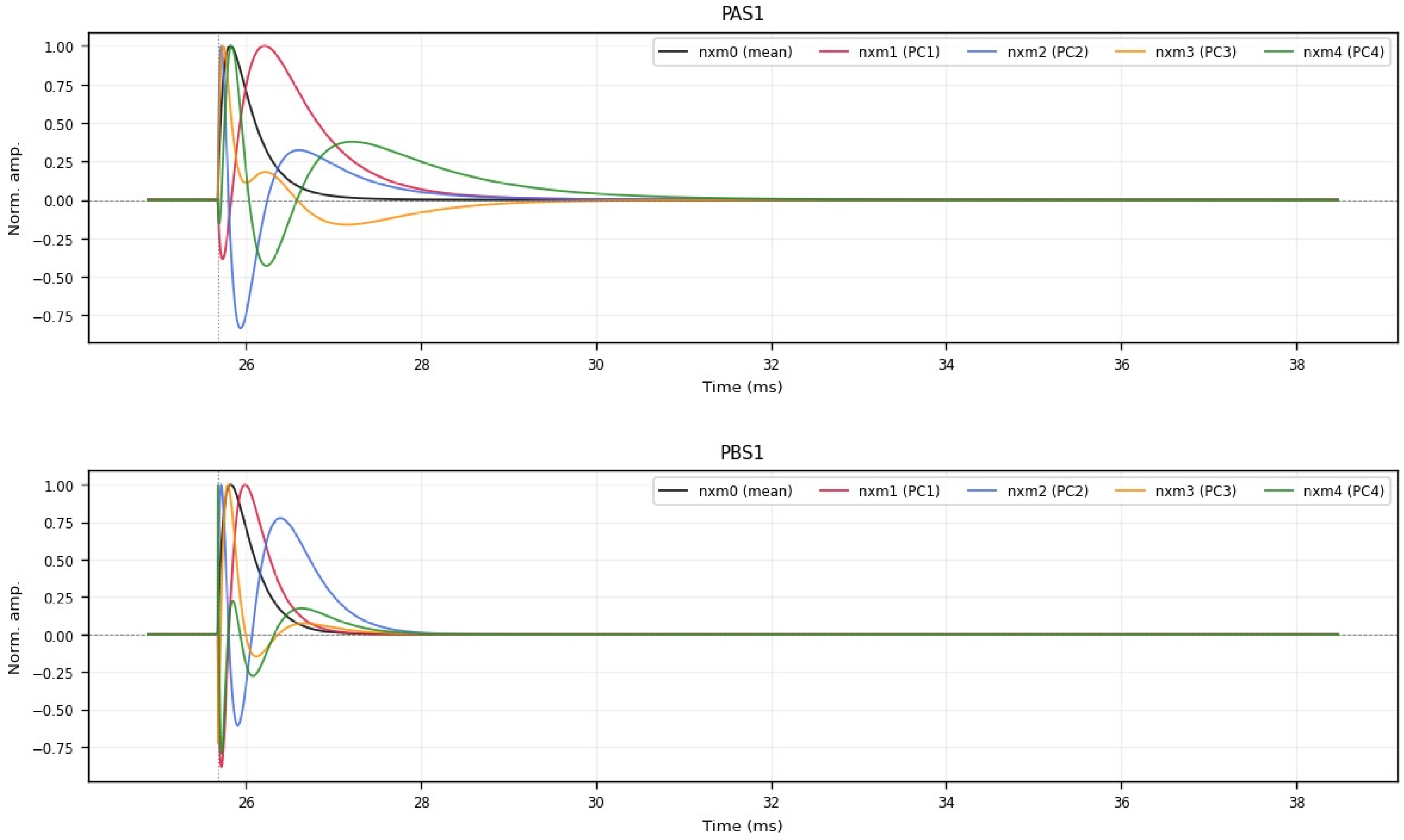
10 random events with τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (1/6)

figure: zip7_pca_templates.png, channel panels 1–2 of 11, top to bottom

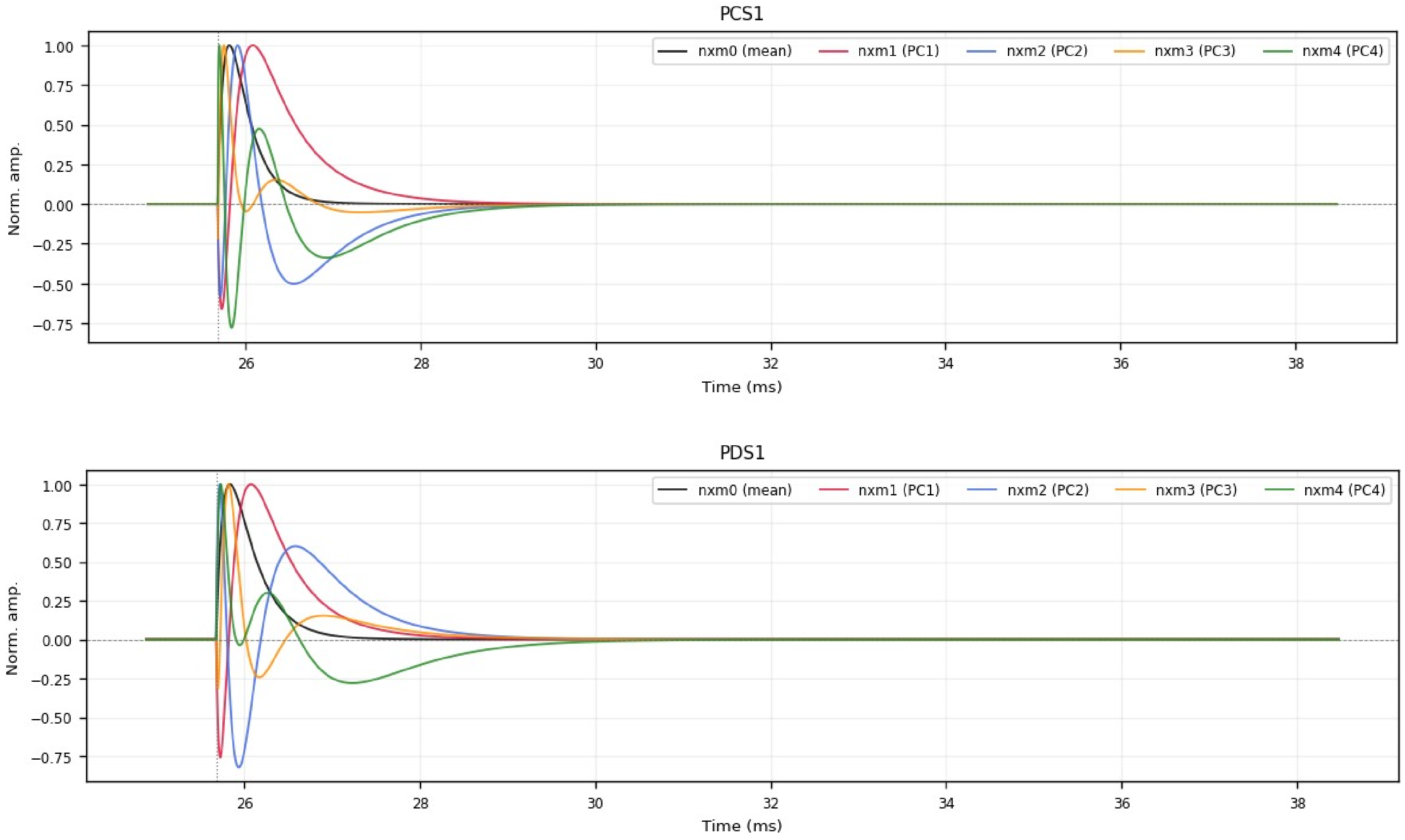
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (2/6)

figure: zip7_pca_templates.png, channel panels 3–4 of 11, top to bottom

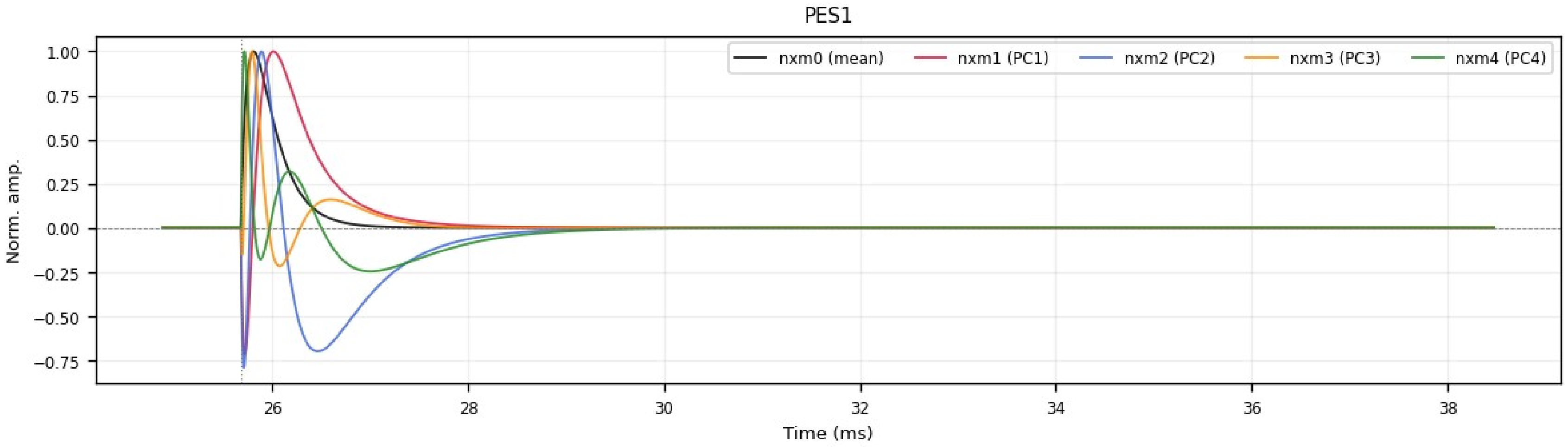
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance.
PDS2 replaced by the PDS1 template in the delivered files.

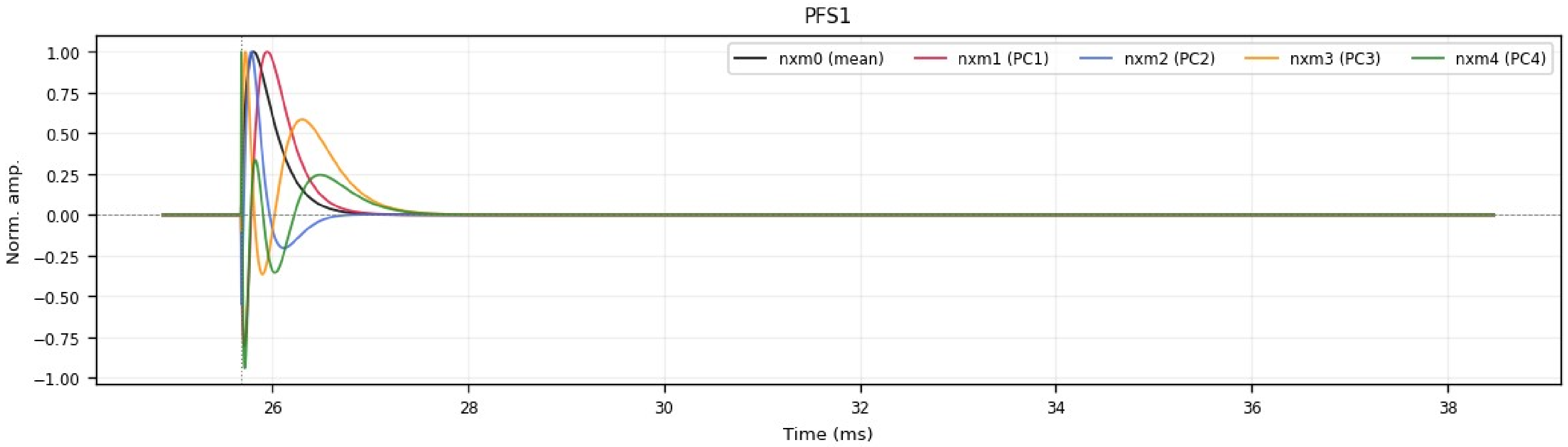


Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (3/6)

figure: zip7_pca_templates.png, channel panels 5–6 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance.
PDS2 replaced by the PDS1 template in the delivered files.

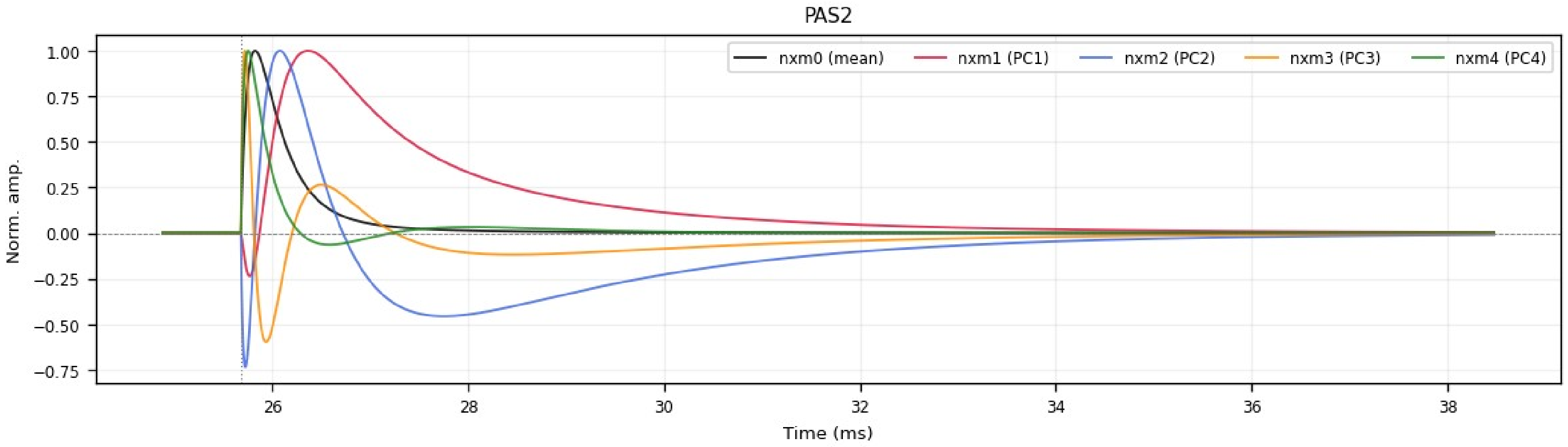


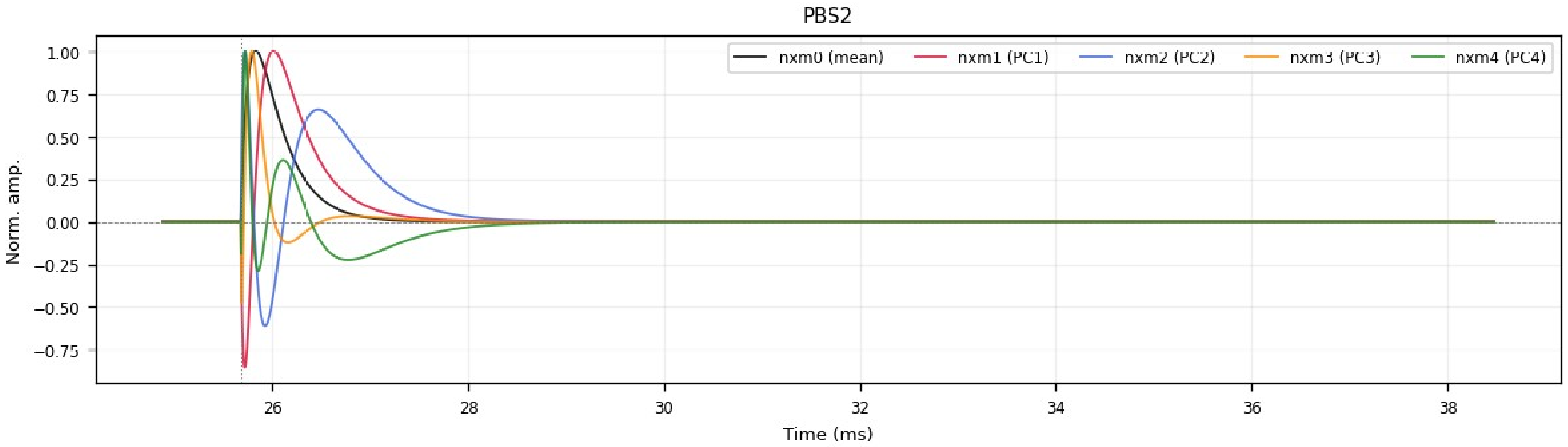


Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (4/6)

figure: zip7_pca_templates.png, channel panels 7–8 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.

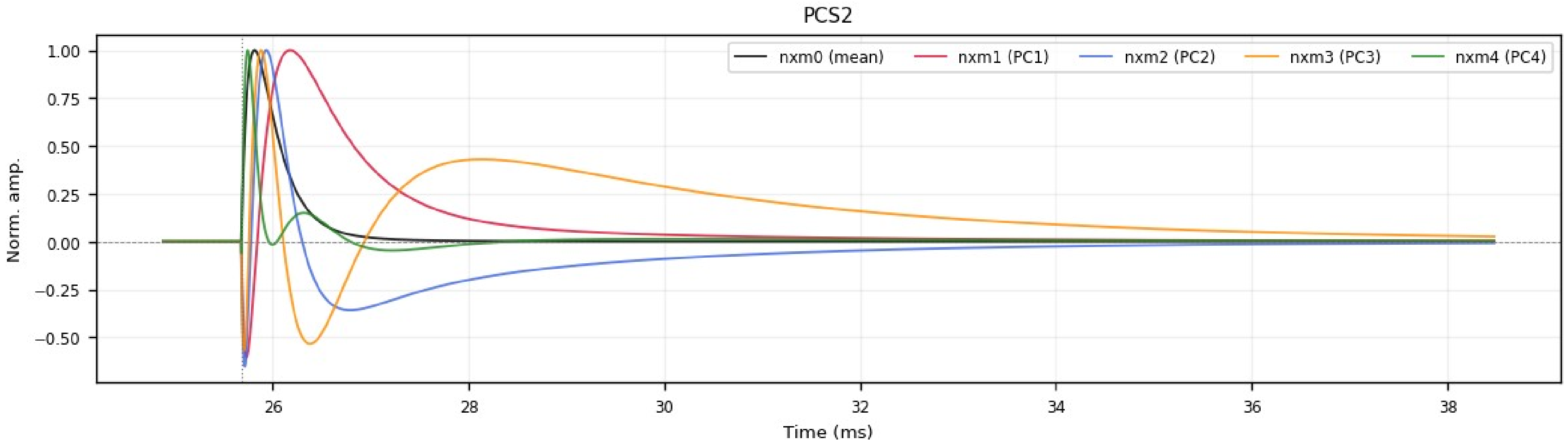


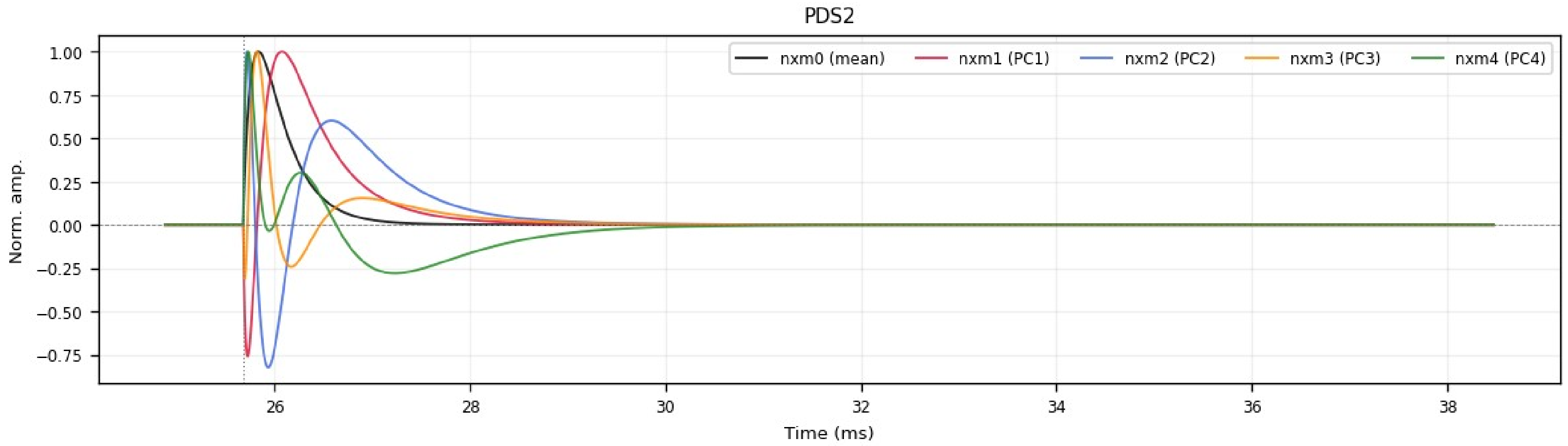


Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (5/6)

figure: zip7_pca_templates.png, channel panels 9–10 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.

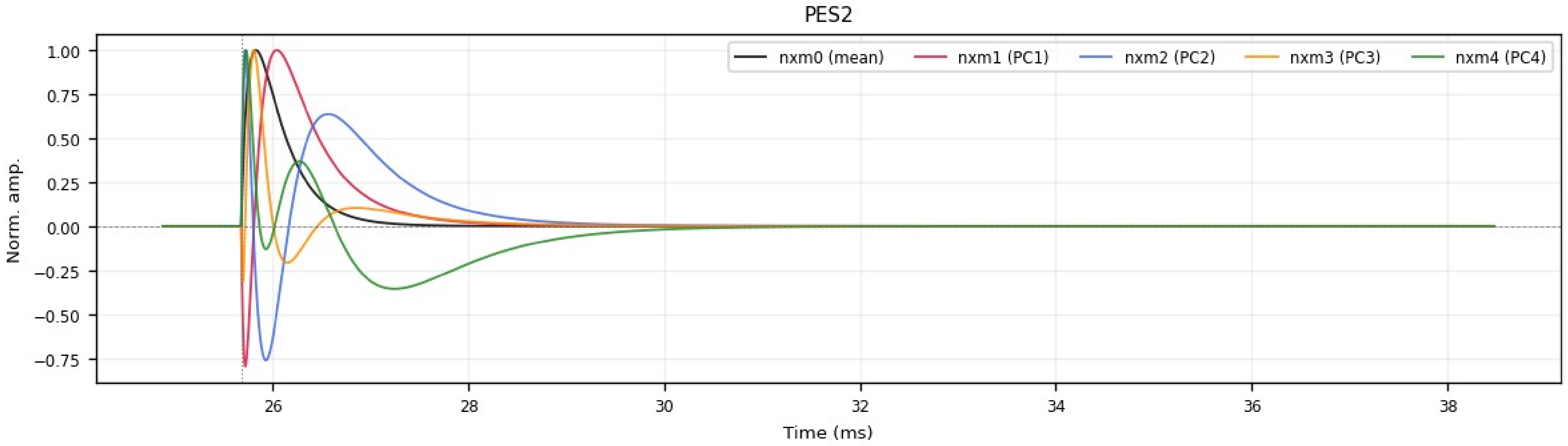




Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (6/6)

figure: zip7_pca_templates.png, channel panels 11–11 of 11, top to bottom

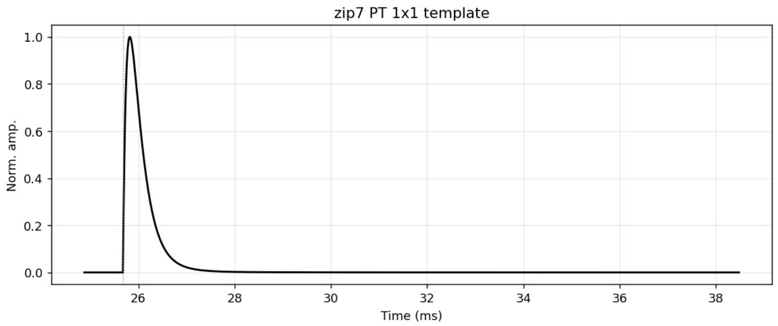
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



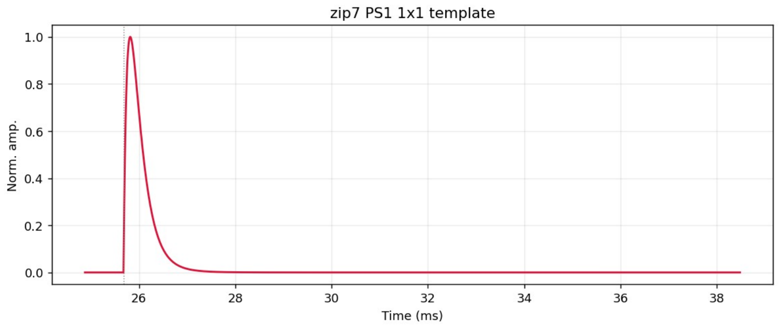
Backup · Z7 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip7_*.png

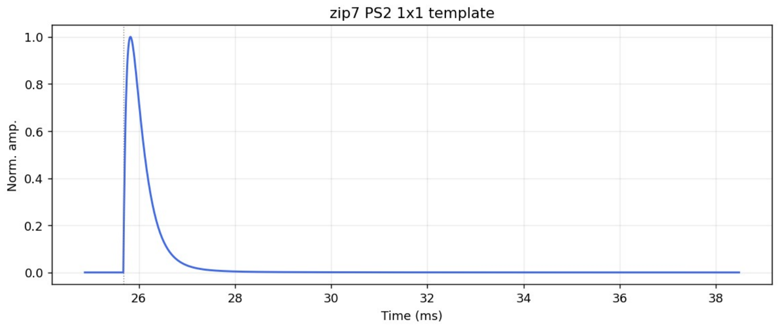
Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates: PT = all channels, PS1 / PS2 = side-1 / side-2 only.



Z7 PT



Z7 PS1



Z7 PS2